MARTECH BOILER
The Energy Solutions

BẢNG KIỂM SOÁT VẬT TƯ

Số KS/ Doc.No: BOM.BB.TP2887.0100

Trang/ Page:

Dự án/ *Project* : **TP2887**

Nơi đến/ *Destination Place* :

Tên hạng mục/ *Name of item* : **BỒN NƯỚC 20M3, CÓ BỘ KHỬ KHÍ**

Mã Thành Phẩm/ *Finished Products Code* : **TP2887**

Ngày ban hành/ Issue Date: 30/01/2026

Lần Ban Hành /Rev: 00

STT	Mã VT	Loại VT	Qui cách	Đv	SI Thiết Kế				SI Đề Nghị (ERP)				KH Trình Dự Toán (ERP)			Ghi chú
					Σ1 =A+B+C	Lần 1 (A)	Lần 2 (B)	Lần 3 (C)	Σ2 =a+b+c	Đợt 1 (a)	Đợt 2 (b)	Đợt 3 (c)	Đợt 1	Đợt 2	Đợt 3	
1	VC010860	Thép tấm ASTM A240 304, 2x1219 (Có lớp film)		m	7.0	7.00			7.0	7.00						
2	VC009918	Thép tấm ASTM A240 304, 3x1500x6000mm		m2	4.5	4.50			4.5	4.50						
3	VC020680	Thép tấm JIS G3101 SS400, 4x2000x6000mm		m2	4.2	4.20			4.2	4.20						
4	VC020679	Thép tấm JIS G3101 SS400, 5x2000x6000mm		m2	51.5	51.50			51.5	51.50						
5	VC009921	Thép tấm ASTM A240 304, 6x1500x6000mm		m2	0.5	0.50			0.5	0.50						
6	VC020677	Thép tấm JIS G3101 SS400, 6x2000x6000mm		m2	25.6	25.60			25.6	25.60						
7	VC020676	Thép tấm JIS G3101 SS400, 8x2000x6000mm		m2	0.8	0.80			0.8	0.80						
8	VC020675	Thép tấm JIS G3101 SS400, 10x2000x6000mm		m2	4.5	4.50			4.5	4.50						
9	VC009924	Thép tấm ASTM A240 304, 10x1500x6000mm		m2	0.5	0.50			0.5	0.50						
10	VC020674	Thép tấm JIS G3101 SS400, 12x2000x6000mm		m2	1.0	1.00			1.0	1.00						
11	VC000879	Thép tấm ASTM A36, 16x2000x6000mm		m2	0.2	0.20			0.2	0.20						
12	VC000881	Thép tấm ASTM A36, 20x2000x6000mm		m2	0.1	0.10			0.1	0.10						
13	VC009929	Thép tấm ASTM A240 304, 20x1500x6000mm		m2	0.1	0.05			0.1	0.05						
14	VC001053	Ống đúc ASTM A312 TP304, Ø21.3x2.77x6000mm		m	0.4	0.40			0.4	0.40						
15	VC001022	Ống đúc ASTM A106 Gr.B, Ø26.7x2.87x6000 mm		m	1.6	1.60			1.6	1.60						
16	VC000917	Ống hàn JIS G3131 SPHC, Ø26.7x1.8x6000mm		m	2.0	2.00			2.0	2.00						
17	VC003790	Ống đúc ASME SA-106 Gr.B, Ø33.4x3.38x11800mm		m	1.4	1.40			1.4	1.40						
18	VC001024	Ống đúc ASTM A106 Gr.B, Ø42.2x3.56x6000 mm		m	0.4	0.40			0.4	0.40						
19	VC000920	Ống hàn JIS G3131 SPHC, Ø42.2x1.8x6000mm		m	4.2	4.20			4.2	4.20						
20	VC017706	Ống đúc DIN EN 10216-2 P235GH-TC1, Ø60.3x3.91x11800mm		m	0.4	0.40			0.4	0.40						

21	VC001059	Ống đúc ASTM A312 TP304, Ø76.1x5.16x6000mm		m	0.1	0.10			0.1	0.10						
22	VC001028	Ống đúc ASTM A106 Gr.B, Ø88.9x5.49x6000 mm		m	0.8	0.80			0.8	0.80						
23	VC001029	Ống đúc ASTM A106 Gr.B, Ø114.3x6.02x6000mm		m	8.2	8.20			8.2	8.20						
24	VC018920	Ống đúc DIN EN 10216-2 P235GH-TC1, Ø219.1x8.18x11800mm		m	7.0	7.00			7.0	7.00						
25	VC001036	Ống đúc ASTM A106 Gr.B, Ø323.9x10.31x6000mm		m	0.4	0.40			0.4	0.40						
26	VC001172	Thép hình H125x125x6.5x9x12000mm/SS400		m	3.0	3.00			3.0	3.00						
27	VC010521	Thép hình V63x63x6x6000mm CT38/SS400/AGS 400		m	3.0	3.00			3.0	3.00						
28	VC010520	Thép hình V50x50x5x6000 mm, SUS304		m	10.2	10.20			10.2	10.20						
29	VC001270	Thép hình V30x30x3x6000mm, SUS304		m	2.5	2.50			2.5	2.50						
30	VC014410	Ty ren M12, SUS304		m	0.4	0.40			0.4	0.40						
31	VC014590	Ty ren M14, SUS304		m	1.8	1.80			1.8	1.80						
32	VC013742	Láp JIS G4051 S45C Ø16x6000mm		m	0.8	0.80			0.8	0.80						
33	VC013734	Láp ASTM A276 304/304L, Ø10x6000mm		m	1.0	1.00			1.0	1.00						
34	VC011158	Láp SUS304 Ø14x6000mm		m	0.5	0.50			0.5	0.50						
35	VC019285	Mặt bích Type 01.B1 A516, DN20-PN16		Cái	4.0	4.00			4.0	4.00						
36	VC019273	Mặt bích Type 01.B1 ASTM A516, DN25-PN16		Cái	3.0	3.00			3.0	3.00						
37	VC019276	Mặt bích Type 01.B1 A516, DN32-PN16		Cái	1.0	1.00			1.0	1.00						
38	VC019279	Mặt bích Type 01.B1 ASTM A516, DN50-PN16		Cái	1.0	1.00			1.0	1.00						
39	VC019281	Mặt bích Type 01.B1 ASTM A516, DN80-PN16		Cái	1.0	1.00			1.0	1.00						
40	VC019267	Mặt bích Type 01.B1 ASTM A516, DN100-PN16		Cái	5.0	5.00			5.0	5.00						
41	VC019283	Mặt bích Type 05.B1 ASTM A516, DN100-PN16		Cái	1.0	1.00			1.0	1.00						
42	VC019272	Mặt bích Type 01.B1 ASTM A516, DN200-PN16		Cái	3.0	3.00			3.0	3.00						
43	VC019275	Mặt bích Type 01.B1 A516, DN300-PN16		Cái	1.0	1.00			1.0	1.00						
44	VC000274	Mặt bích Type 01 CT3/SS400, DN400 PN16		Cái	1.0	1.00			1.0	1.00						
45	VC005492	Mặt bích Type 05 CT3/SS400, DN400 PN16		Cái	1.0	1.00			1.0	1.00						
46	VC020000	Mặt bích Type 01.B1 A516, DN450-PN16		Cái	1.0	1.00			1.0	1.00						
47	VC019295	Mặt bích Type 05.B1 A516, DN450-PN16		Cái	1.0	1.00			1.0	1.00						
48	VC000575	Roong mặt bích SW DN20 PN10/16/25/40, 316 FG, 316 IR, CS OR 4.5mmT		Cái	4.0	4.00			4.0	4.00						PIPING CÁP
49	VC000578	Roong mặt bích SW DN25 PN10/16/25/40, 316 FG, 316 IR, CS OR 4.5mmT		Cái	3.0	3.00			3.0	3.00						PIPING CÁP

50	VC005343	Roong mặt bích SW DN32 PN10/16/25/40, 316L FG, 316L IR, CS OR 4.5mmT		Cái	1.0	1.00			1.0	1.00						PIPING CÁP
51	VC000585	Roong mặt bích SW DN50 PN10/16/25/40, 316 FG, 316 IR, CS OR 4.5mmT		Cái	1.0	1.00			1.0	1.00						PIPING CÁP
52	VC005349	Roong mặt bích SW DN80 PN10/16/25/40, 316L FG, 316L IR, CS OR 4.5mmT		Cái	1.0	1.00			1.0	1.00						PIPING CÁP
53	VC000594	Roong mặt bích SW DN100 PN10/16, 316L FG, 316L IR, CS OR 4.5mmT		Cái	2.0	2.00			2.0	2.00						PIPING CÁP
54	VC000594	Roong mặt bích SW DN100 PN10/16, 316L FG, 316L IR, CS OR 4.5mmT		Cái	1.0	1.00			1.0	1.00						SITE
55	VC000594	Roong mặt bích SW DN100 PN10/16, 316L FG, 316L IR, CS OR 4.5mmT		Cái	1.0	1.00			1.0	1.00						
56	VC000602	Roong mặt bích SW DN200 PN16, 316 FG, 316 IR, CS OR 4.5mmT		Cái	2.0	2.00			2.0	2.00						SITE
57	VC000608	Roong mặt bích SW DN300 PN16, 316 FG, 316 IR, CS OR 4.5mmT		Cái	1.0	1.00			1.0	1.00						PIPING CÁP
58	VC000609	Roong mặt bích SW DN400 PN16, 316 FG, 316 IR, CS OR 4.5mmT		Cái	1.0	1.00			1.0	1.00						
59	VC000611	Roong mặt bích SW DN450 PN16, 316L FG, 316L IR, CS OR 4.5mmT		Cái	1.0	1.00			1.0	1.00						
60	VC001413	Bulong M12x60mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm		Bộ	28.0	28.00			28.0	28.00						PIPING CÁP
61	VC001408	Bulong M16x60mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm		Bộ	4.0	4.00			4.0	4.00						PIPING CÁP
62	VC001389	Bulong M16x70mm + tán cường lực + lông đền/35K 8.8 mạ kẽm		Bộ	28.0	28.00			28.0	28.00						PIPING CÁP
63	VC001389	Bulong M16x70mm + tán cường lực + lông đền/35K 8.8 mạ kẽm		Bộ	8.0	8.00			8.0	8.00						SITE
64	VC001389	Bulong M16x70mm + tán cường lực + lông đền/35K 8.8 mạ kẽm		Bộ	8.0	8.00			8.0	8.00						
65	VC001446	Bulong M20x80mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm		Bộ	12.0	12.00			12.0	12.00						PIPING CÁP
66	VC001446	Bulong M20x80mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm		Bộ	12.0	12.00			12.0	12.00						SITE
67	VC001461	Bulong M24x100mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm		Bộ	12.0	12.00			12.0	12.00						PIPING CÁP
68	VC001470	Bulong M27x110mm + tán cường lực + lông đền phe, 35K 8.8 mạ kẽm		Bộ	16.0	16.00			16.0	16.00						
69	VC001471	Bulong M27x130mm + tán cường lực + lông đền phe, 35K 8.8 mạ kẽm		Bộ	20.0	20.00			20.0	20.00						
70	VC001388	Bulong M16x50mm + tán cường lực + lông đền/35K 8.8 mạ kẽm		Bộ	10.0	10.00			10.0	10.00						SITE
71	VC001342	Bulong inox M12x40mm + tán + lông đền, SUS304		Bộ	2.0	2.00			2.0	2.00						
72	VC001343	Bulong inox M12x50mm + tán + lông đền, SUS304		Bộ	10.0	10.00			10.0	10.00						
73	VC013397	Bulong inox M10x50mm + tán + lông đền, SUS304		Bộ	8.0	8.00			8.0	8.00						
74	VC014144	Tán M12, SUS304		Con	2.0	2.00			2.0	2.00						
75	VC001385	Bulong M12x50mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm		Bộ	4.0	4.00			4.0	4.00						
76	VC001406	Bulong M10x50mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm		Bộ	25.0	25.00			25.0	25.00						
77	VC013548	Tán M14, SUS304		Bộ	12.0	12.00			12.0	12.00						
78	VC006409	Măng xông ren suốt A-105, G 1/2 x 30mm, Sch40		Cái	8.0	8.00			8.0	8.00						

79	VC002968	Van cầu lắp bích DN25-PN16, KSB		Cái	1.0	1.00			1.0	1.00						PIPING CÁP
80	VC019289	Mặt bích Type 05.B1 ASTM A516, DN25-PN16		Cái	1.0	1.00			1.0	1.00						
81	VC000578	Roong mặt bích SW DN25 PN10/16/25/40, 316 FG, 316 IR, CS OR 4.5mmT		Cái	1.0	1.00			1.0	1.00						PIPING CÁP
82	VC001413	Bulong M12x60mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm		Cái	4.0	4.00			4.0	4.00						PIPING CÁP
83	VC015730	Lò xo nén Ø3xØ30, p= 9.5mm, L= 40mm, SU304		Cái	1.0	1.00			1.0	1.00						
84	VC000492	Tê đúc A-234 WPB, DN20 Sch40 BW		Cái	1.0	1.00			1.0	1.00						
85	VC006464	Siphons SUS 304, 1/2 NPT		Cái	1.0	1.00			1.0	1.00						
86	VC006299	Cà rá đồng, 1/2 NPT x 1/4 NPT		Cái	1.0	1.00			1.0	1.00						
87	VC000787	Teplong cuộn		Cuộn	2.0	2.00			2.0	2.00						
88	VC000787	Teplong cuộn		Cuộn	2.0	2.00			2.0	2.00						PIPING CÁP
89	VC003540	Cao su tấm, dày 3mm		m2	0.1	0.10			0.1	0.10						
90	VC006408	Măng xông ren suốt A-105, M18x25		Cái	4.0	4.00			4.0	4.00						
91	VC003051	Van kính thủy sáng lắp ren Ø16		Bộ	2.0	2.00			2.0	2.00						
92	VC002811	Kính thủy bồn nước Ø16		Bộ	1.0	1.00			1.0	1.00						
93	VC004182	Bulong M6x20mm + tán + long đền, 35K 8.8 mạ kẽm		Bộ	4.0	4.00			4.0	4.00						
94	VC000382	Bầu giảm đúc đồng tâm A-234 WPB, DN100 x 25 Sch40 BW		Cái	1.0	1.00			1.0	1.00						
95	VC015735	Bảng tên inox, nền inox, chữ đen, 0.5x150x100mm		Cái	1.0	1.00			1.0	1.00						
96	VC009342	Vít đầu dù đuôi cá M4.2x12mm		Con	4.0	4.00			4.0	4.00						
97	VC000380	Bầu giảm đúc đồng tâm A-234 WPB, DN80 x 65 Sch40 BW		Cái	1.0	1.00			1.0	1.00						
98	VC006377	Cùm ống chữ U + tán + long đền, Ø219 x M8, CT3		Cái	3.0	3.00			3.0	3.00						
99	VC006376	Cùm ống chữ U + tán + long đền, Ø114 x M8, CT3		Cái	3.0	3.00			3.0	3.00						
100	VC000077	Co đúc 90º LR A-234 WPB, DN200 Sch40 BW		Cái	1.0	1.00			1.0	1.00						
101	VC005918	Co đúc 90º LR A-234 WPB, DN100 Sch40 BW		Cái	1.0	1.00			1.0	1.00						
102	VC000110	Co đúc 45º LR A-234 WPB, DN100 Sch40 BW		Cái	1.0	1.00			1.0	1.00						
103	VC001133	Bảo ôn vàng 6 tấm 600x1200mm, dày 50mm, tỉ trọng 100 kg/m3		Kiện	30.0	30.00			30.0	30.00						BẢO ÔN 2 LỚP
104	VC020568	Tôn cuộn G550 AZ050, 0.50x1200mm, mạ màu trắng sữa BWL01		m	65.0	65.00			65.0	65.00						
105	VC009342	Vít đầu dù đuôi cá M4.2x12mm		Con	1800.0	1800.00			1800.0	1800.00						
106	VC010565	Kẽm buộc bồi sắt 1mm		Kg	10.0	10.00			10.0	10.00						

Người lập	Người duyệt	Sản xuất chung	QC	Ban QLNM

MARTECH BOILER The Energy Solutions	**BIÊN BẢN NGHIỆM THU BÁN THÀNH PHẨM-THÀNH PHẨM**	Số KS/ Doc.No: COA.BB.TP2887.00 Trang/ Page: 01/01

Dự án/ Project: TP2887	Khối lượng (Kg): 3800	Thiết bị	Xử lý bề mặt	Sơn lót	Sơn phủ/ Tẩy	Màu	Code
Nơi đến/ *Destination Place* :	DT mặt trong (m2):						
Tên hạng mục/ *Name of item* : BỒN NƯỚC 20M3 + BỘ KHỬ KHÍ	DT mặt ngoài (m2): 80						
Mã Thành Phẩm/ *Finished Products Code* :TP2887	DT sơn SP nhỏ (m2):						
Ngày ban hành/ Issue Date: 30/01/2026	DT sơn SP lớn (m2):						
Lần Ban Hành /Rev: 00							

No.	**Mô tả hàng hóa** ***Description of Goods Specification***	**S.lượng** ***Quantity***	**Đ/vị** ***Unit***	**Thẻ nhận dạng/** ***Tag No.***	**Kích thước/ *Dimension***			**T/lượng TK (kg)** ***Design Weight***	**T/lượng thực (kg)** ***Net Weight***	**Tổ trưởng SX/** ***Mnf. Team Leader***	**QLSX/ *Mnf. Sup.***	**QC SX/** ***Mnf. QC***	**Tổ trưởng /** ***Painting Team Leader***	**QLSX Sơn/** ***Painting Sup.***	**QC Sơn/** ***Painting QC***	**QLSX/** ***Mnf. Sup.***
					L(mm)	**W (mm)**	**H (mm)**									
1	Bồn nước	1	Cụm	XXXXXX-2887-01	7192	2473	2545	3000								
2	Bình khử khí	1	Cụm	XXXXXX-2887-02	1668	1158	2536	800								
3	Cụm PT&PG	1	Cụm	XXXXXX-2887-03	N/A	N/A	N/A	N/A								
4	Cụm kính thủy	1	Cụm	XXXXXX-2887-04	N/A	N/A	N/A	N/A								

MARTECH BOILER The Energy Solutions	DANH SÁCH ĐÓNG KIỆN PACKAGING LIST	Số KS/ Doc.No: FPL.BB.TP2887.01 Trang/ Page: 01/01

Dự án/ *Project* : **TP2887**

Nơi đến/ *Destination Place* :

Tên hạng mục/ *Name of item* : BỒN NƯỚC 20M3 + BKK

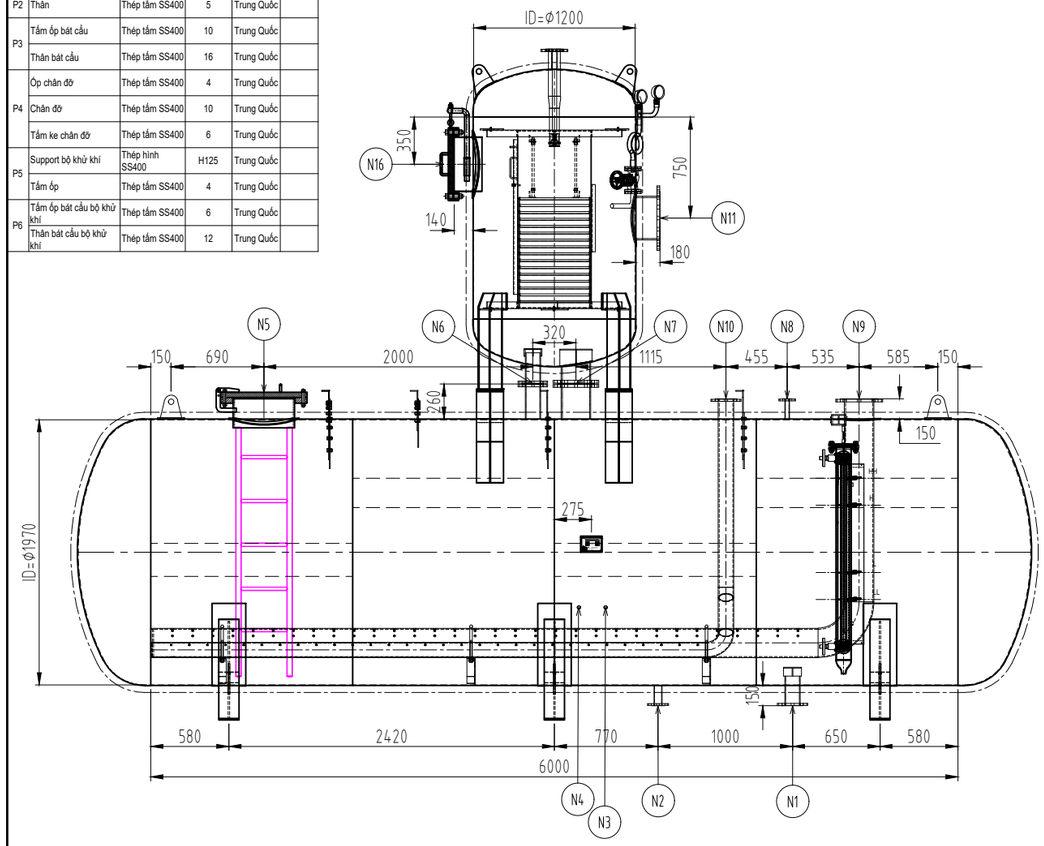
Ngày ban hành/ Issue Date: 30/01/2026

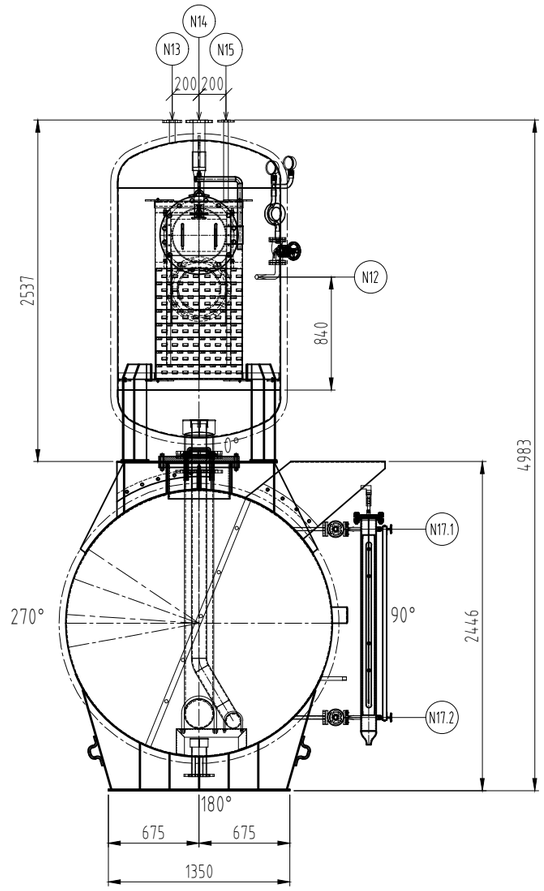
Lần Ban Hành /Rev: 00

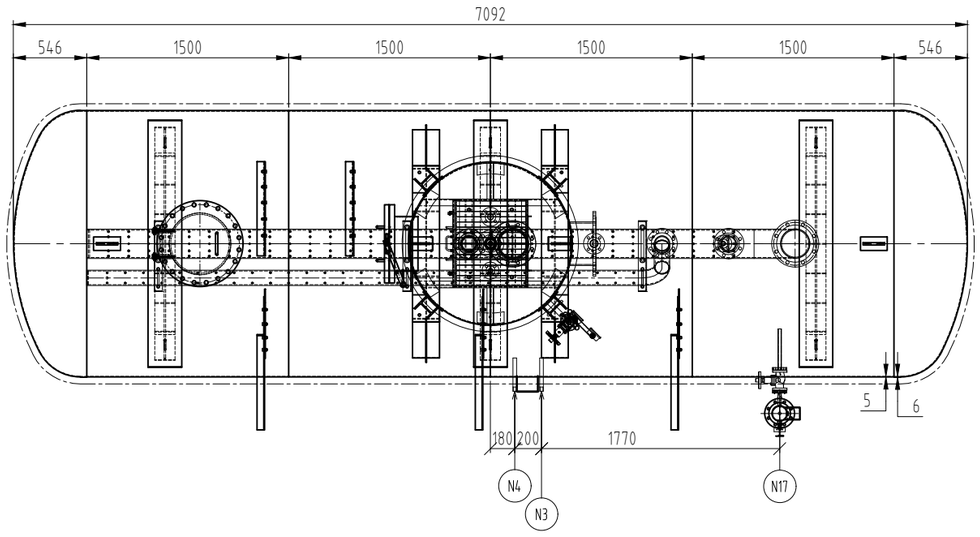
No.	Nhãn kiện hàng/Package No.	Mô tả hàng hóa/ Description of Goods Specification	S.lượng *Quantity*	Đ/vị *Unit*	Thẻ nhận dạng/ *Tag No.*	Kích thước/ *Dimension*			T/lượng TK (kg) *Design Weight*	T/lượng thực (kg) *Net Weight*	Mã VT / Material Code	Mã TP/ Finished Products Code	Ghi chú/ Remarks
						L(mm)	W (mm)	H (mm)					
01		Bồn nước	1	Cụm	XXXXXX-2887-01	7192	2473	2545	3000			TP2887	
02		Bình khử khí	1	Cụm	XXXXXX-2887-02	1668	1158	2536	800			TP2887	
03		Cụm PT&PG	1	Cụm	XXXXXX-2887-03	N/A	N/A	N/A	N/A			TP2887	
04		Cụm kính thủy	1	Cụm	XXXXXX-2887-04	N/A	N/A	N/A	N/A			TP2887	
05		Roong mặt bích SW DN100 PN10/16, 316L FG, 316L IR, CS OR 4.5mmT	1	Cái	N/A	N/A	N/A	N/A	N/A		VC000594	TP2887	
06		Roong mặt bích SW DN200 PN16, 316 FG, 316 IR, CS OR 4.5mmT	1	Cái	N/A	N/A	N/A	N/A	N/A		VC000602	TP2887	
07		Bulong M16x70mm + tán cường lực + lông đền/35K 8.8 mạ kẽm	8	Bộ	N/A	N/A	N/A	N/A	N/A		VC001389	TP2887	
08		Bulong M20x80mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	12	Bộ	N/A	N/A	N/A	N/A	N/A		VC001446	TP2887	
09		Bulong M16x50mm + tán cường lực + lông đền/35K 8.8 mạ kẽm	10	Bộ	N/A	N/A	N/A	N/A	N/A		VC001388	TP2887	
Người lập		Người Duyệt	Sản xuất Chung			QC					Ban QLNM		
Nguyễn Minh Vương													

LIST VẬT LIỆU

No.	Mô tả	Vật liệu	Chiều dày (mm)	Xuất xứ	Ghi chú
P1	Đầu bo	Thép tấm SS400	6	Trung Quốc	
P2	Thân	Thép tấm SS400	5	Trung Quốc	
P3	Tấm ốp bát cầu	Thép tấm SS400	10	Trung Quốc	
	Thân bát cầu	Thép tấm SS400	16	Trung Quốc	
P4	Ốp chân đỡ	Thép tấm SS400	4	Trung Quốc	
	Chân đỡ	Thép tấm SS400	10	Trung Quốc	
	Tấm ke chân đỡ	Thép tấm SS400	6	Trung Quốc	
P5	Support bộ khử khí	Thép hình SS400	H125	Trung Quốc	
	Tấm ốp	Thép tấm SS400	4	Trung Quốc	
P6	Tấm ốp bát cầu bộ khử khí	Thép tấm SS400	6	Trung Quốc	
	Thân bát cầu bộ khử khí	Thép tấm SS400	12	Trung Quốc	







THÔNG SỐ KỸ THUẬT BỒN NƯỚC:

- Kích thước: ID1970x7092
- Tiêu chuẩn thiết kế và chế tạo: ASME VIII Ed 2025
- ASTK: 2 bar.g
- AS thử nước: 3 bar.g
- AS làm việc max: 0.2 bar.g
- Nhiệt độ làm việc max: 105°C
- Vật liệu: SS400
- NDT: No Radiography
- Đầu bo dạng Ellipsoidal (bán kính lớn = 0.9ID, bán kính nhỏ =0.17ID)
- Độ dày:
  - -- Shell - 5 mm (t min sau khi bo >=4.8mm)
  - -- Head - 6 mm (t min sau khi bo >=5.1 mm)
- Thể tích: 20m^3
- Bảo ôn rockwool dày 100mm, tỷ trọng 100kg/m^3
- Bọc ngoài bằng tôn vỏ dày 0.5mm
- Khối lượng: 3000 kg
- Diện tích sơn: 65m2

THÔNG SỐ KỸ THUẬT BỒN KHỬ KHÍ:

- Kích thước: ID1200x2209
- Tiêu chuẩn thiết kế và chế tạo: ASME VIII Ed 2025
- ASTK: 2 bar.g
- AS thử nước: 3 bar.g
- AS làm việc max: 0.2 bar.g
- Nhiệt độ làm việc max: 105°C
- Vật liệu: Đầu bo và thân: SS400 - Kết cấu bên trong (ngoại trừ ống nước cấp bổ sung): SUS304
- NDT: No Radiography
- Đầu bo dạng Ellipsoidal (bán kính lớn = 0.9ID, bán kính nhỏ =0.17ID)
- Độ dày:
  - -- Shell - 5 mm(t min sau khi bo >=4.8mm)
  - -- Head - 6 mm (t min sau khi bo >=5.1 mm)
- Thể tích: 2.15 m^3
- Bảo ôn rockwool dày 100 mm, tỷ trọng 100kg/m^3
- Bọc ngoài bằng tôn vỏ dày 0.5mm
- Khối lượng: 800 kg
- Diện tích sơn: 15m2.

* Model TB: BỒN NƯỚC 20M3, CÓ BỘ KHỬ KHÍ
Serial No: (Z)-YY-XXXX-Q-NNNN-GG
+ Theo hướng dẫn đặt mã dự án, mã số chế tạo bản vẽ.
Mã hiệu: MB-TM-I01 - Ngày hiệu lực: 22/10/2022.

Ghi chú:

- Ngoài các kích thước được chỉ định rõ dung sai, thì các kích thước còn lại được tra theo tiêu chuẩn ISO 13920:2023

+ Table 1: cấp B với KT dài
+ Table 2: cấp B với KT góc
+ Table 3: cấp F với độ phẳng, thẳng và song song

BB-BO1: DANH SÁCH TÊN TB/CỤM CT/CT ĐI RỜI

STT	Mô tả	Thẻ nhận dạng (Tagname)	Kích thước: LxWxH (mm)			Khối lượng (Kg)	Số Lượng	Ghi chú
			Dài (L)	Rộng (W)	Cao (H)			
1	Bồn nước	XXXXXX-2887-01	7192	2473	2545	3000	1	
2	Bình khử khí	XXXXXX-2887-02	1668	1158	2536	800	1	
3	Cụm PT&PG	XXXXXX-2887-03	N/A	N/A	N/A	N/A	1	
4	Cụm kính thủy	XXXXXX-2887-04	N/A	N/A	N/A	N/A	1	

CONNECTION SIZE

No	Service	Q'ty	Nozzle neck			Flange			Remark
			Size	Sch.	Material	Type	Rating	Material	
N1	Ống nước đến cụm bơm	1	DN100	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N2	Ống xả đáy	1	DN50	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N3	Ống gắn nhiệt kế	1	DN20	40	SA/A106	Măng xông M21			-
N4	Ống gắn cảm biến nhiệt độ	1	DN20	40	SA/A106	Măng xông M21			-
N5	Cửa người chui	1	DN450	6T	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N6	Ống cân bằng hơi và nước	1	DN100	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N7	Ống xả xuống bồn nước	1	DN200	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N8	Ống gắn van xả áp chân không	1	DN25	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N9	Ống hơi cấp vào bồn nước	1	DN200	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N10	Ống nước hồi cấp vào bồn	1	DN100	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N11	Ống cấp hơi gia nhiệt bộ khử khí	1	DN300	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N12	Ống gắn TT, TG	1	DN25	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N13	Ống gắn van an toàn	1	DN32	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N14	Ống cấp nước bổ sung	1	DN80	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N15	Ống xả khí	1	DN25	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N16	Cửa vệ sinh bộ khử khí	1	DN400	6T	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N17	Ống gắn kính thủy tối	2	DN20	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

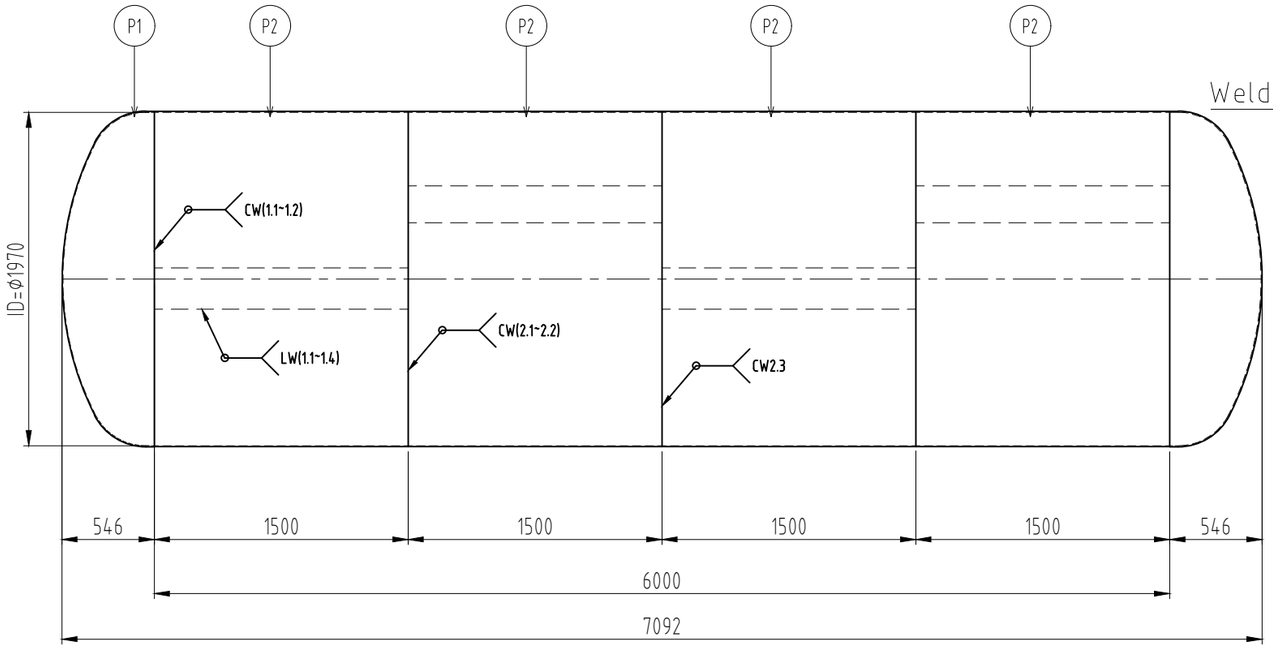
MANUFACTURER:
MARTECH BOILER The energy Solutions
MARTECH JOINT STOCK COMPANY

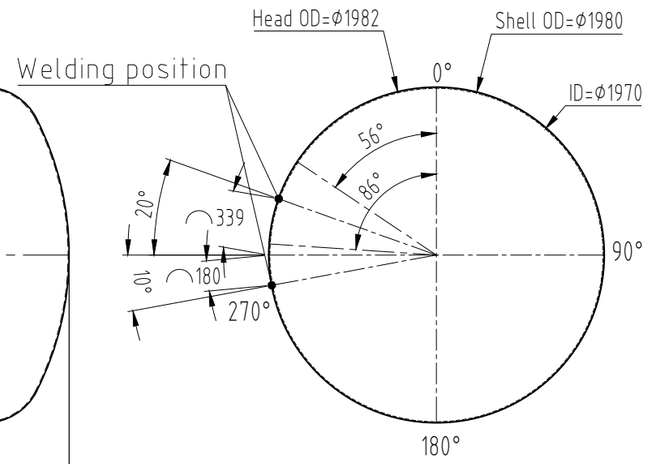
PROJECT JOB CODE: TP2887

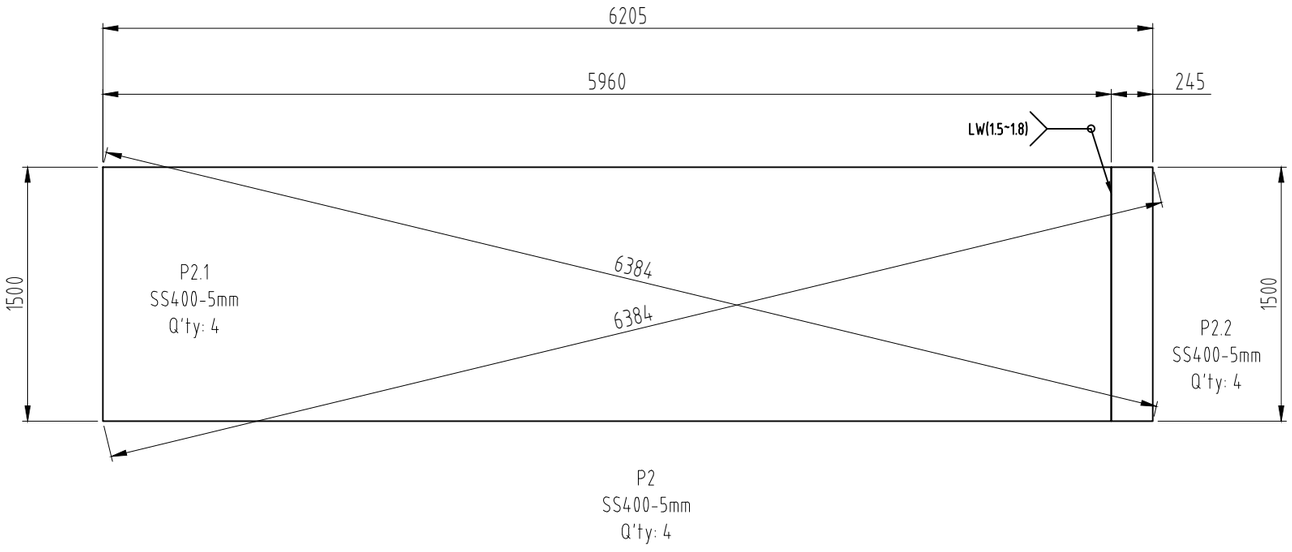
TITLE: GENERAL DRAWING
DEAERATOR AND WATER TANK

DRAWING NO.: 2887.0100.MGAD.01

Scale	Size	Rev.
1:1	A3	00







WELD PREPARATION		
No.	Sym.	Val.
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2	t2	6
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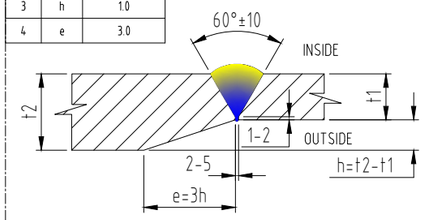
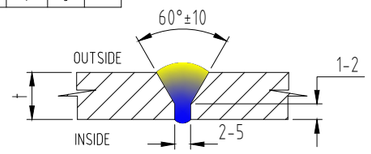
4	e	3.0

Detail: CW(1.1~1.2):
NDT: MT/PT 100%



WELD PREPARATION		
No.	Sym.	Val.
1	t	5

Detail: CW(2.1~2.2); LW(1.1~1.8)
NDT: MT/PT 100%



WELD PREPARATION		
No.	Sym.	Val.
1	t	5

Detail:CW(2.3)
NDT: MT/PT 100%

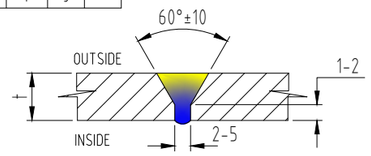


PLATE LIST

No.	Symbol	Material Decripsion		Specifications (mm)			Q'ty	Area (m2)	Mass (kg)	Pain are (m2)	Note
		Type of material	Material	Thk.	Length	Width					
1	P2.1	Plate steel	SS400	5	5960	1500	4	8.9	1403.6	71.5	
2	P2.2	Plate steel	SS400	5	245	1500	4	0.4	57.7	2.9	
TOTAL							8.0	8.9	1461.3	74.5	

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

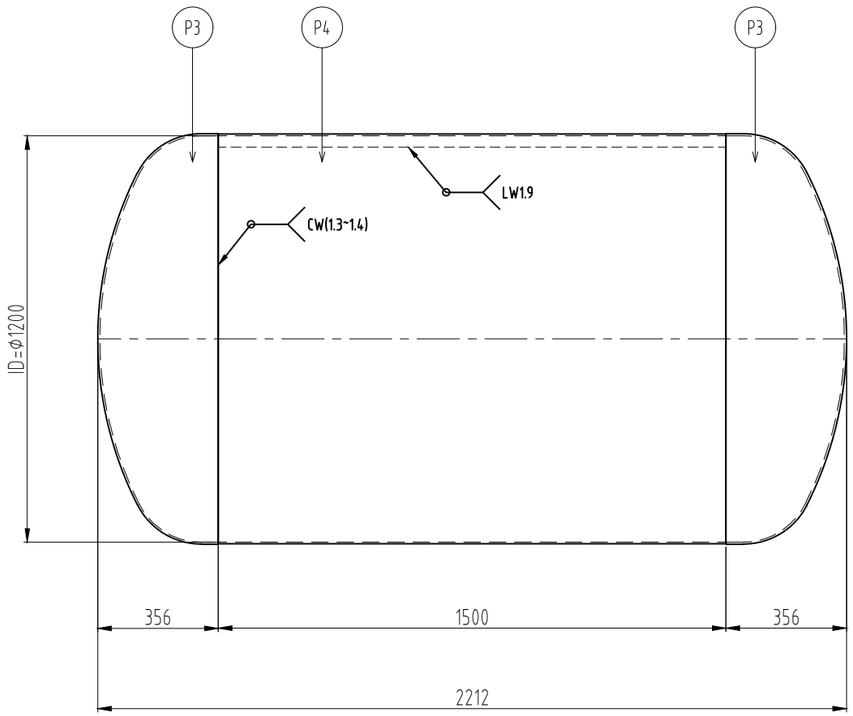
MANUFACTURER:
MARTECH BOILER The Energy Solutions
MARTECH JOINT STOCK COMPANY

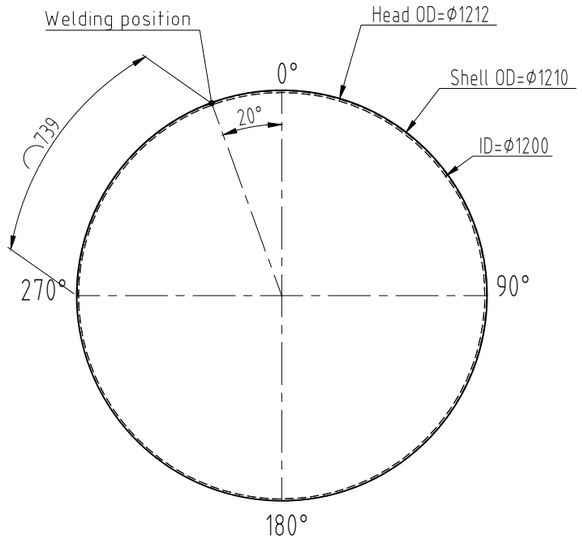
PROJECT JOB CODE: TP2887

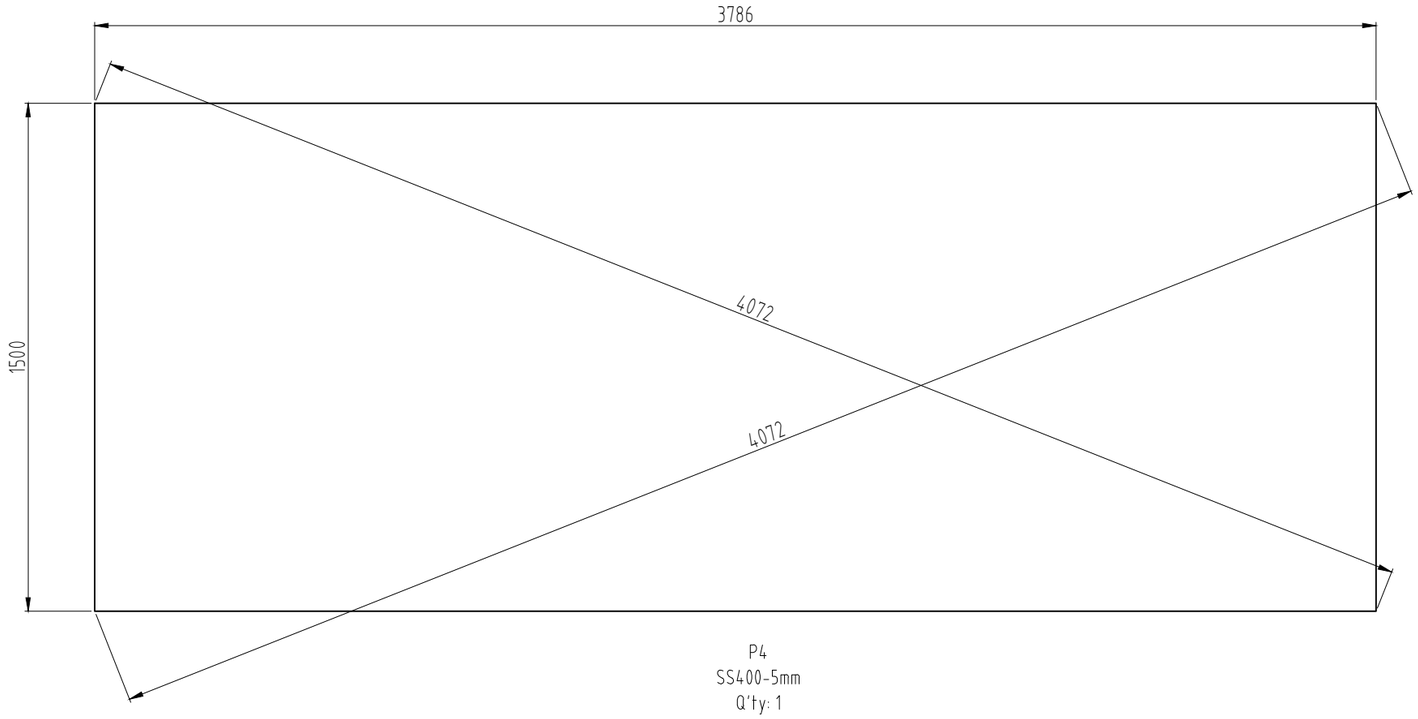
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.01







WELD PREPARATION		
No.	Sym.	Val.
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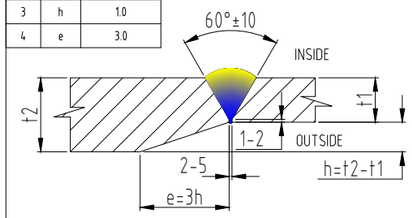
2	t2	6
3	h	1.0
4	e	3.0

Detail: CW(1.3~1.4):
NDT: MT/PT 100%



WELD PREPARATION		
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Detail: LW1.9
NDT: MT/PT 100%

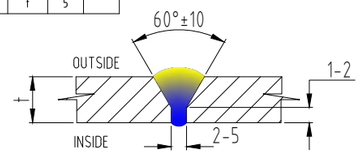


PLATE LIST

No.	Symbol	Material Decripsion		Specifications (mm)			Q'ty	Area (m2)	Mass (kg)	Pain are (m2)	Note
		Type of material	Material	Thk.	Length	Width					
1	P4	Plate steel	SS400	5	3786	1500	1	5.7	222.9	11.4	
TOTAL							1.0	5.7	222.9	11.4	

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

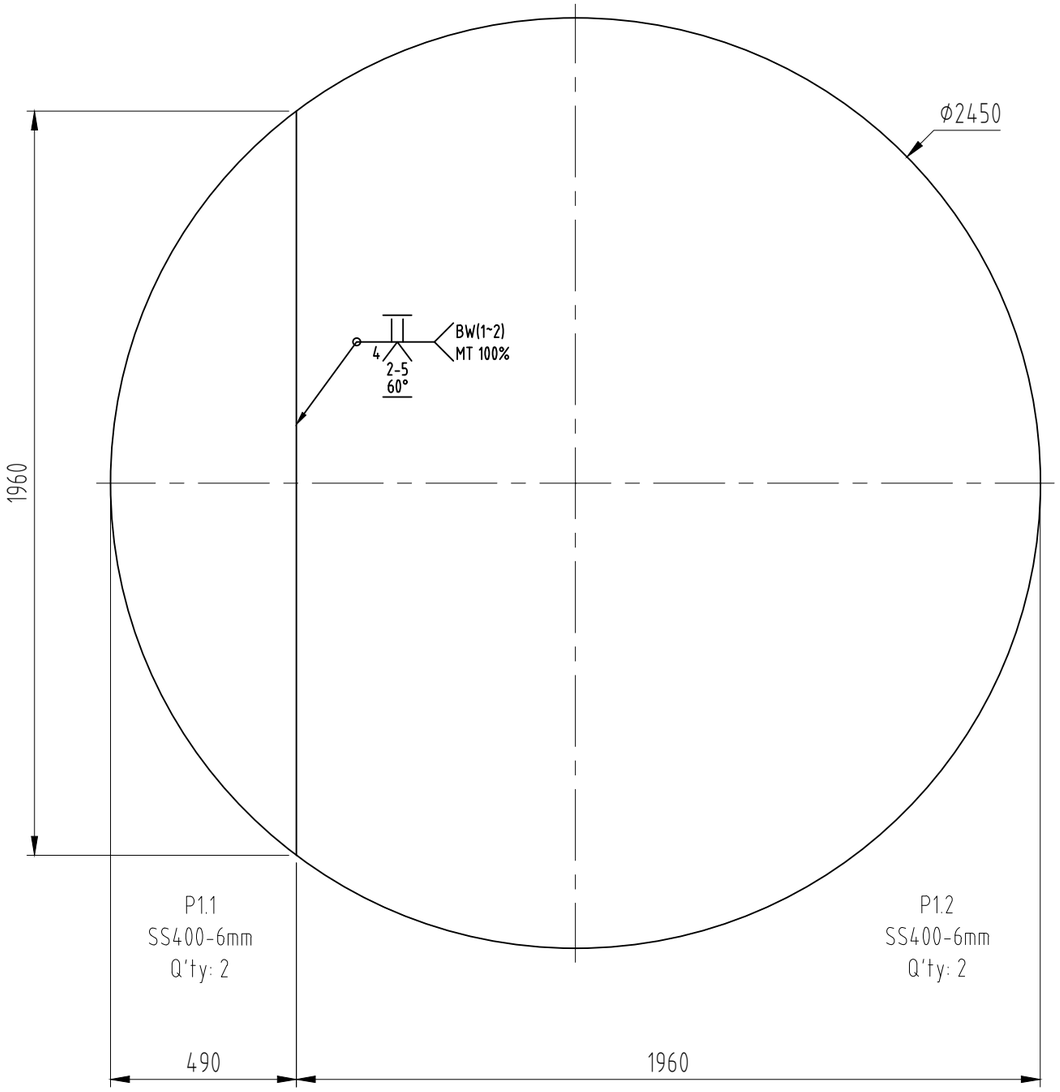
MANUFACTURER:
MARTECH BOILER
The energy Solutions
MARTECH JOINT STOCK COMPANY

PROJECT JOB CODE: TP2887

TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.02



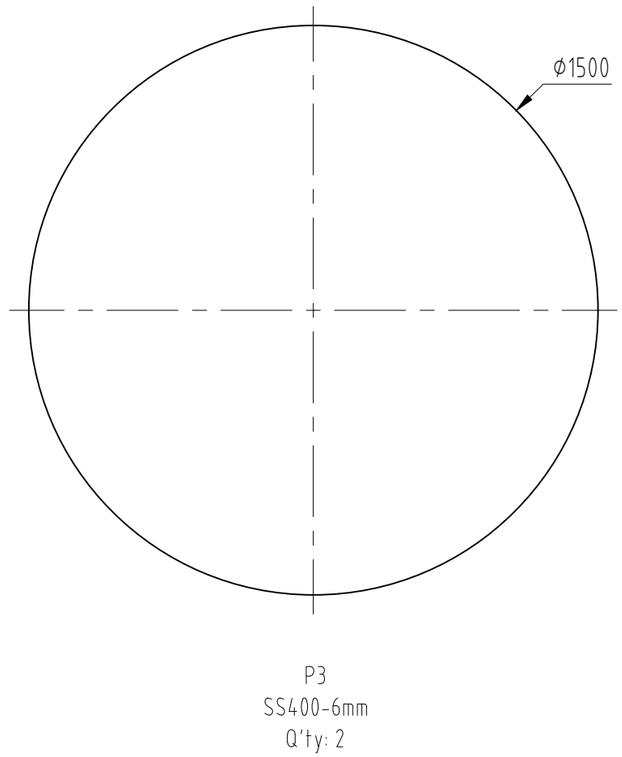


PLATE LIST

No.	Symbol	Material Decripsion		Specifications (mm)			Q'ty	Area (m2)	Mass (kg)	Pain are (m2)	Note
		Type of material	Material	Thk.	Length	Width					
1	P1.1	Plate steel	SS400	6	490	1960	2	0.7	63.1	2.7	
2	P1.2	Plate steel	SS400	6	2450	1960	2	4.0	376.8	16.0	
3	P3	Plate steel	SS400	6	1550	1550	2	1.8	169.6	7.2	
TOTAL							6.0	0.7	609.5	25.9	

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:
MARTECH BOILER
The energy Solutions
MARTECH JOINT STOCK COMPANY

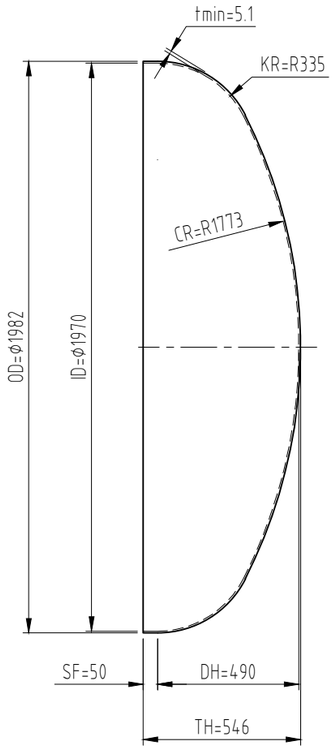
PROJECT JOB CODE: TP2887

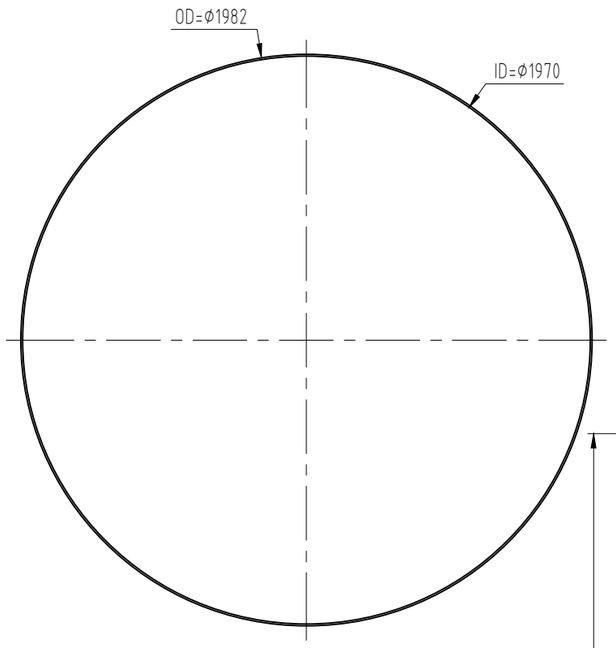
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

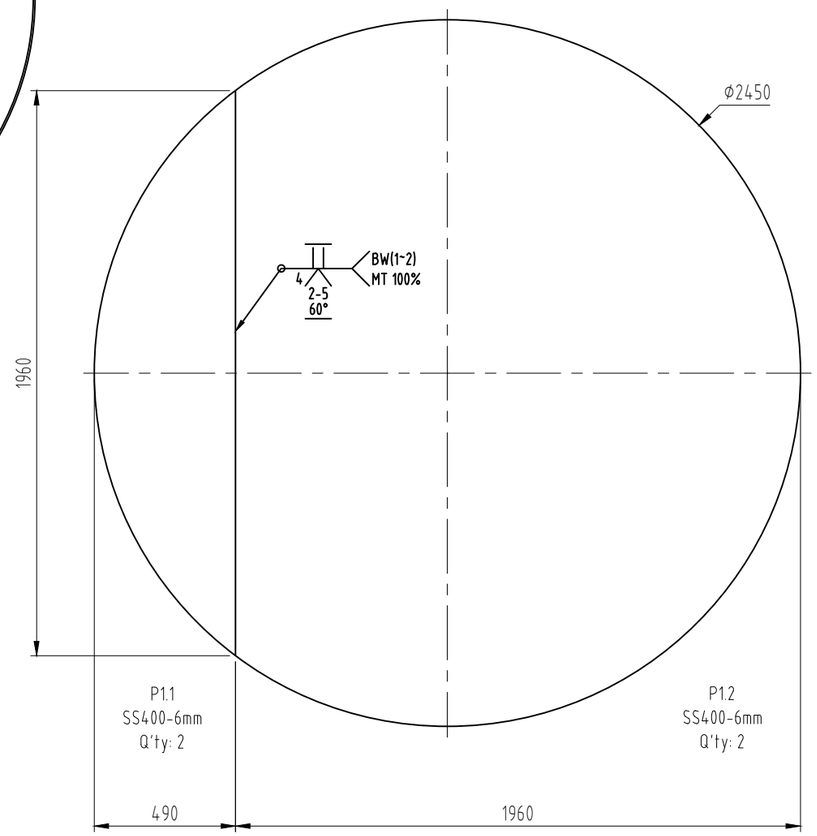
Scale	Size	Rev.
1:1	A3	00

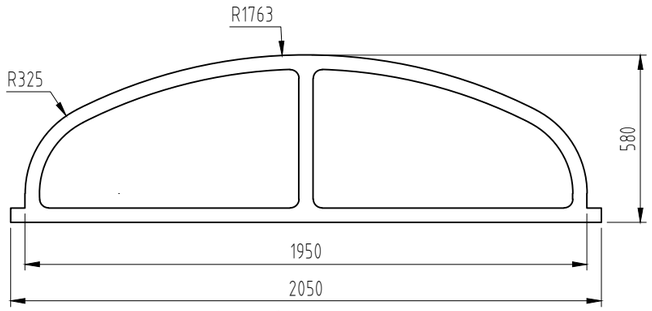
DRAWING NO.: 2887.0100.MDWG.03

BẢN VẼ GIA CÔNG NGOÀI









DUONG 1
SS400-5mm-Q'ty: 1
Marking: ELD-ID1970-CR1773-KR335
Lưu kho tận dụng dưỡng tồn kho

YÊU CẦU KĨ THUẬT:

a. Loại chỏm: Xấp xỉ Ellipsodal 2: 1

b. Yêu cầu vát mép: N/A

c. Yêu cầu bề mặt sau khi tạo hình không có khuyết tật bề mặt hình thành, VD: Như xước bề mặt, nứt, vết lõm, v.v.

d. Phạm vi dung sai:

- Độ tròn (I.D.max-I.D.min) của đoạn trụ thẳng không được vượt quá 1% I.D. danh nghĩa.
- Dung sai đoạn SF: ± 2 mm
- Tổng chiều sâu bên trong D.H + S.F: Xem mục III. của Bảng A. phía dưới
- Biên dạng bề mặt trong của chỏm phải được kiểm tra theo dưỡng đính kèm để đáp ứng yêu cầu dung sai của: UG-81 (Mục a), ASME VIII, Div.1. 2025 (Áp dụng cho đầu bo bình bồn).
- Dung sai độ lệch biên dạng trong: Xem mục III. của Bảng A. phía dưới.
- Trong trường hợp dung sai giữa các "Ghi chú chung" là xung đột, thì dung sai cấp cao hơn sẽ được áp dụng.
- Serial No: XXXXXX288701 - XXXXXX: 6 Số cuối mã dự án

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:
MARTECH BOILER The energy Solutions
MARTECH JOINT STOCK COMPANY

PROJECT JOB CODE: TP2887

TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

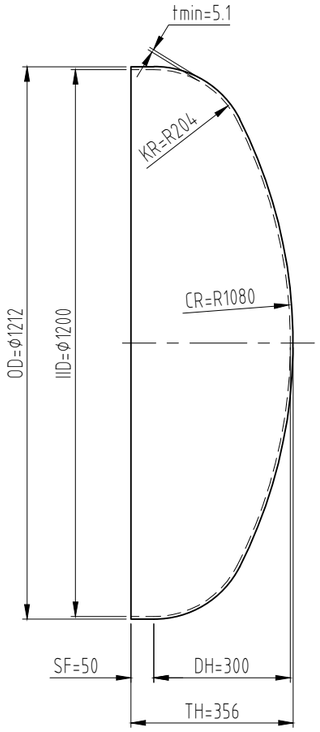
Scale	Size	Rev.
1:1	A3	00

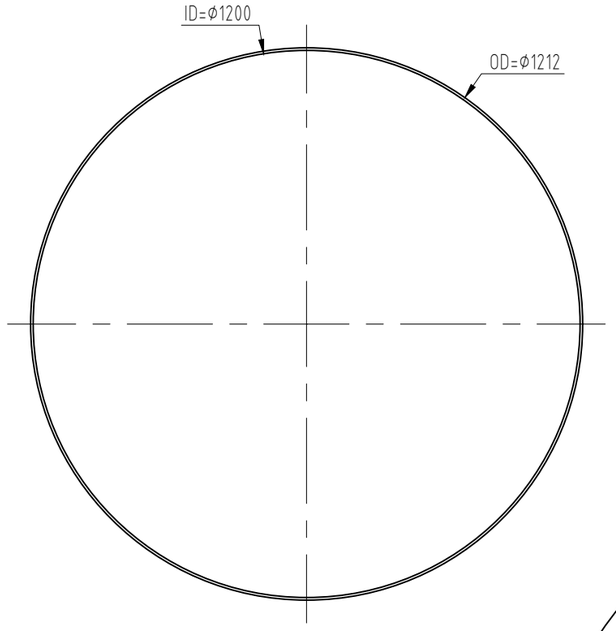
DRAWING NO.: 2887.0100.MDWG.04

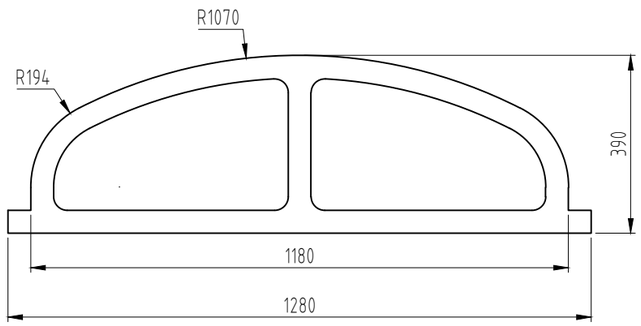
BẢNG A: ĐẦU BO XẤP XỈ ELLIPSOIDAL 2:1

STT	Mã DA	Tên TB	Số lượng	I. MÔ TẢ					II. Qui cách (mm)						III. DUNG SAI (mm)		KL/1 Cái (kg)	Ghi chú
				Vật liệu	t (mm)	OD (mm)	ID (mm)	t.min (mm)	C.R	K.R	D.H	S.F	T.H	D/K Phôi	DS Trên (Nhô ra)	DS dưới (Lõm vào)		
1	TP2887	BỒN NƯỚC 20m3	2	SS400	6	1982	1970	5.10	1773.0	335	490	50	546	2450	24.6	12.3	222.1	Code dung sai UG81

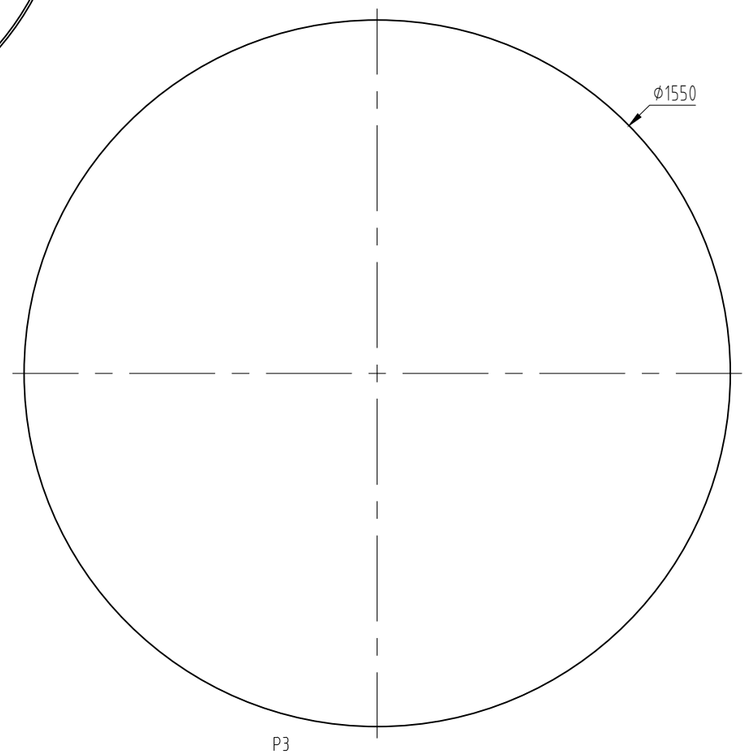
BẢN VẼ GIA CÔNG NGOÀI







DUONG 2
SS400-5mm-Q'ty: 1
Marking: ELD-ID1200-CR1080-KR204
Lưu kho tận dụng dưỡng tồn kho



P3
SS400-6mm
Q'ty: 2

YÊU CẦU KĨ THUẬT:

a. Loại chỏm: Xấp xỉ Ellipsodal 2: 1

b. Yêu cầu vát mép: N/A

c. Yêu cầu bề mặt sau khi tạo hình không có khuyết tật bề mặt hình thành, VD: Như xước bề mặt, nứt, vết lõm, v.v.

d. Phạm vi dung sai:

- Độ tròn (I.D.max-I.D.min) của đoạn trụ thẳng không được vượt quá 1% I.D. danh nghĩa.
- Dung sai đoạn SF: ± 2 mm
- Tổng chiều sâu bên trong D.H + S.F: Xem mục III. của Bảng A. phía dưới
- Biên dạng bề mặt trong của chỏm phải được kiểm tra theo dưỡng đính kèm để đáp ứng yêu cầu dung sai của: UG-81 (Mục a), ASME VIII, Div.1. 2025 (Áp dụng cho đầu bo bình bồn).
- Dung sai độ lệch biên dạng trong: Xem mục III. của Bảng A. phía dưới.
- Trong trường hợp dung sai giữa các "Ghi chú chung" là xung đột, thì dung sai cấp cao hơn sẽ được áp dụng.
- Serial No: XXXXXX288701 - XXXXXX: 6 Số cuối mã dự án

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:
MARTECH BOILER The Energy Solutions
MARTECH JOINT STOCK COMPANY

PROJECT JOB CODE: TP2887

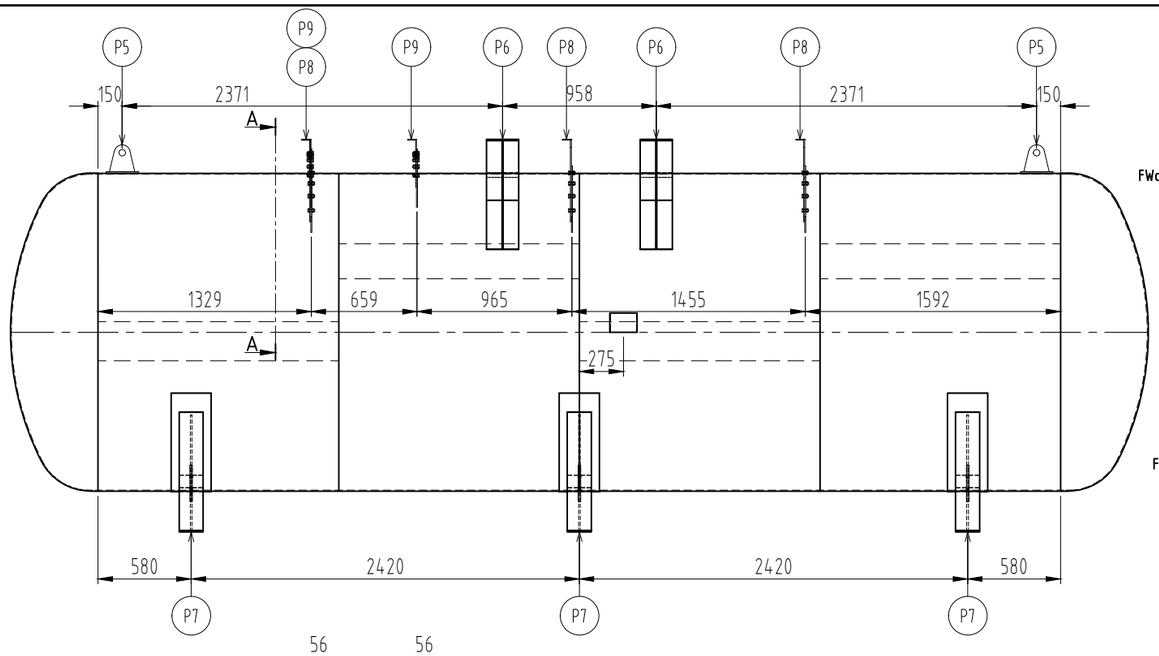
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

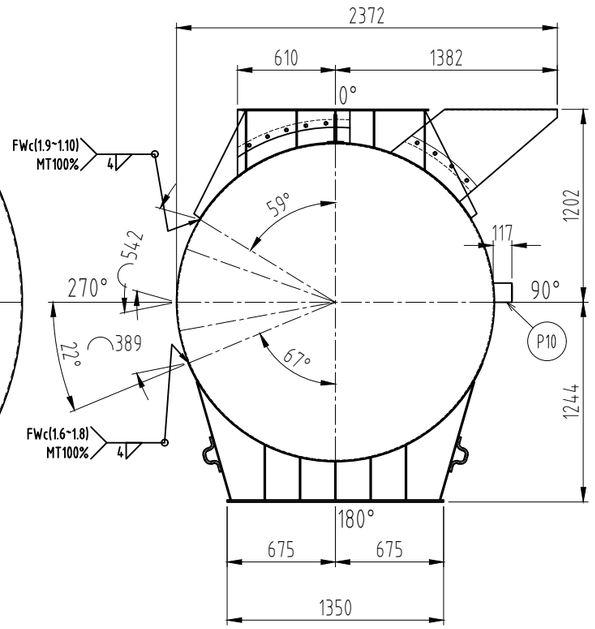
Scale	Size	Rev.
1:1	A3	00

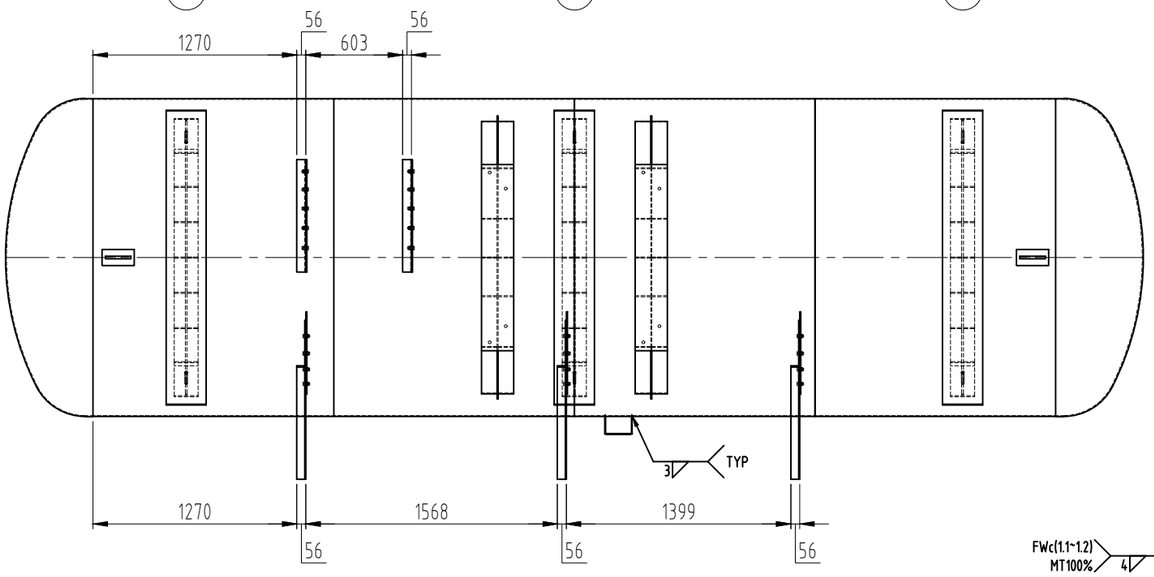
DRAWING NO.: 2887.0100.MDWG.05

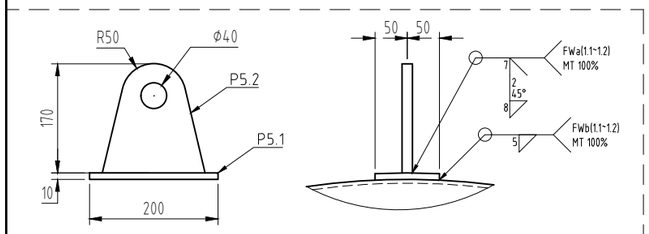
BẢNG A: ĐẦU BO XẤP XỈ ELLIPSOIDAL 2:1

STT	Mã DA	Tên TB	Số lượng	I. MÔ TẢ					II. Qui cách (mm)						III. DUNG SAI (mm)		KL/1 Cái (kg)	Ghi chú
				Vật liệu	t (mm)	OD (mm)	ID (mm)	t.min (mm)	C.R	K.R	D.H	S.F	T.H	D/K Phôi	DS Trên (Nhô ra)	DS dưới (Lõm vào)		
1	TP2887	BỘ KHỬ KHÍ 2.15m3	2	SS400	6	1212	1200	5.10	1080.0	204.0	300	50	356	1550	15.0	7.5	88.9	Code dung sai UG81

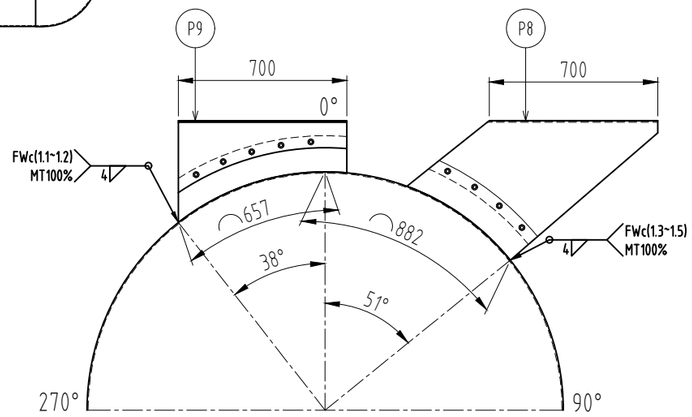








P5
Q'ty:2-SC: 4/1



SECTION A-A

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

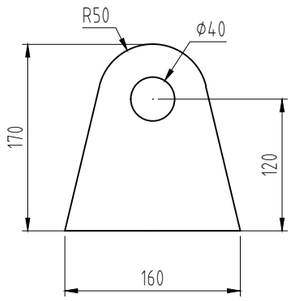
MANUFACTURER: MARTECH BOILER The energy Solutions — MARTECH JOINT STOCK COMPANY

PROJECT JOB CODE: TP2887

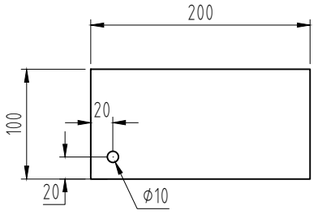
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

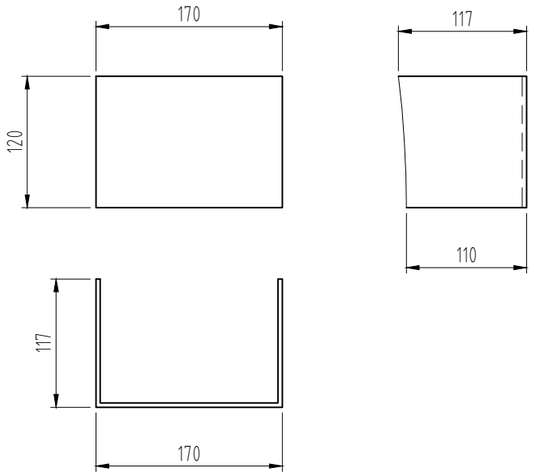
DRAWING NO.: 2887.0100.MDWG.06

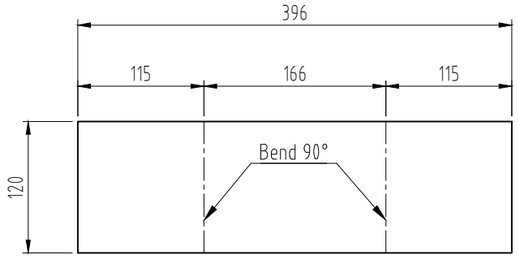


P5.2
A36-16mm
Q'ty:2-SC:3/1

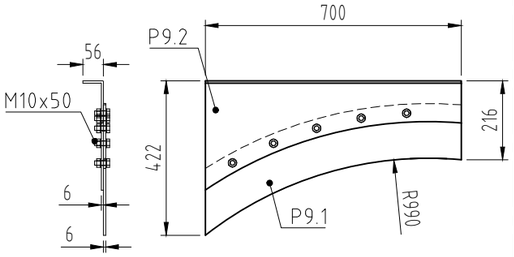


P5.1
SS400-10mm
Q'ty:2-SC:3/1

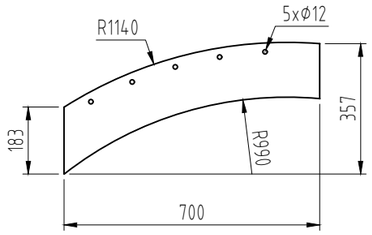




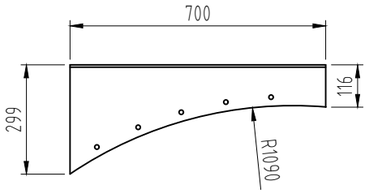
P10
SS400-4mm
Q'ty:1-SC:3/1



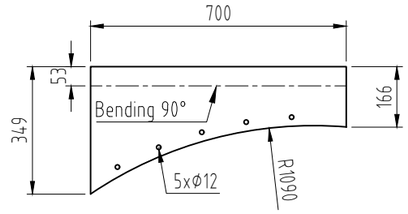
P9
Q'ty:2



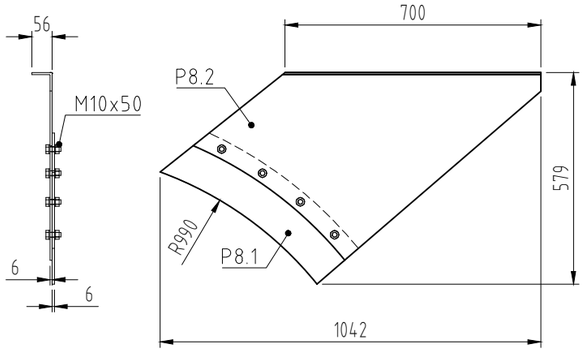
P9.1
SS400-6mm
Q'ty:2



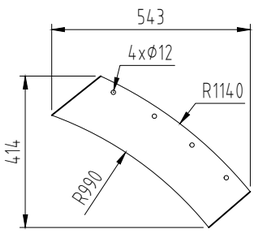
P9.2
SS400-6mm
Q'ty:2



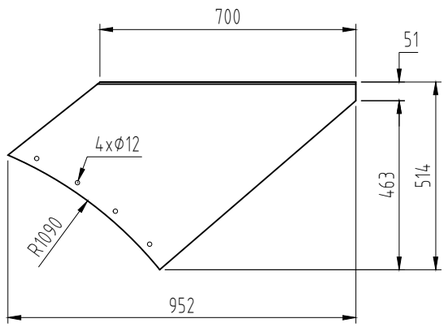
Expansion P9.2
SS400-6mm
Q'ty:2



P8
Q'ty:3



P8.1
SS400-6mm
Q'ty:3



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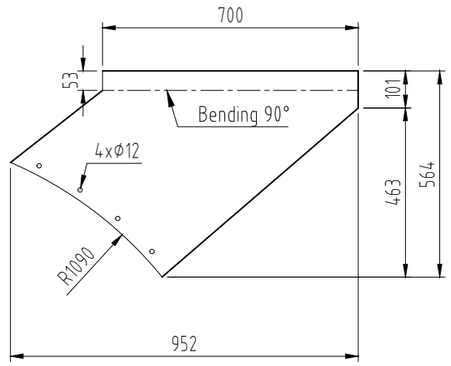
SS400-6mm
Q'ty:3



Expansion P8.2
SS400-6mm
Q'ty:3

Rev.	Date	Description	Drawn	Checked	Approved
3					
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0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

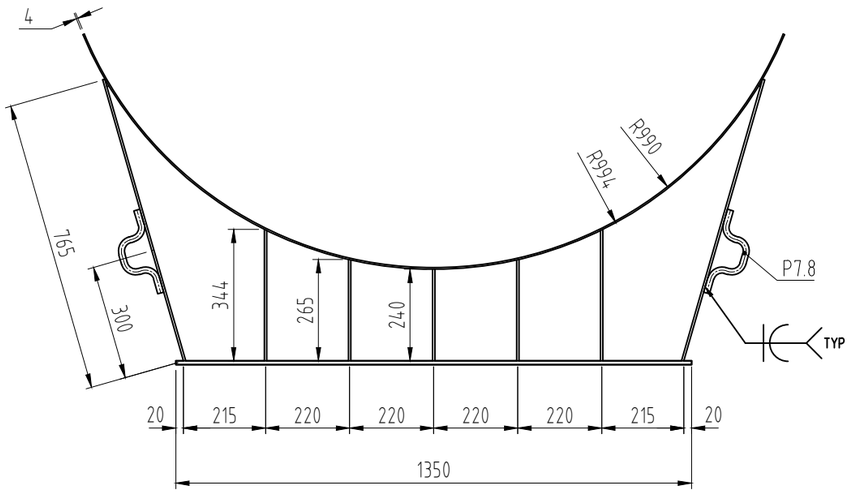
MANUFACTURER:
MARTECH BOILER
The energy Solutions
MARTECH JOINT STOCK COMPANY

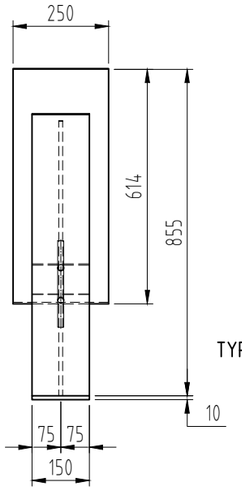
PROJECT JOB CODE: TP2887

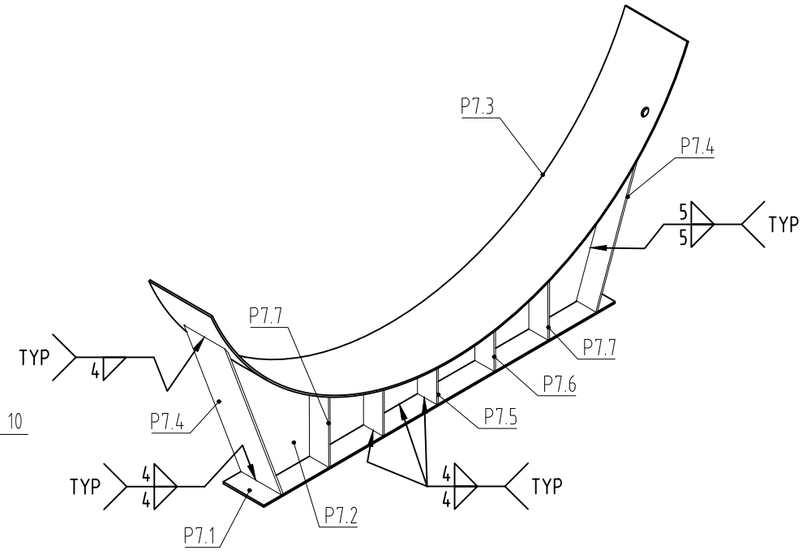
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

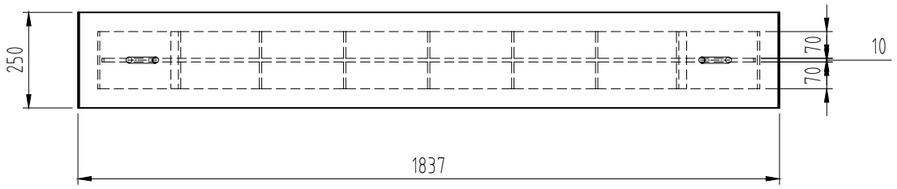
DRAWING NO.: 2887.0100.MDWG.07

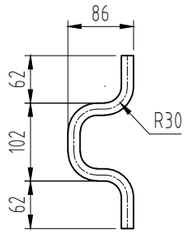




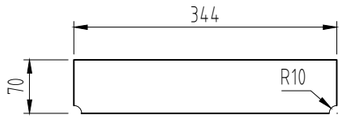


P7
Q'ty:03

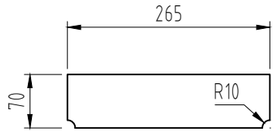




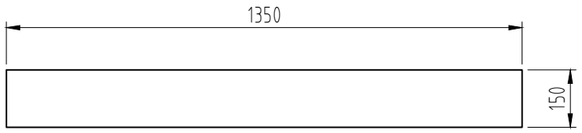
P7.8
Shape steel Φ16-315mm
SC:2/1-Q'ty:6



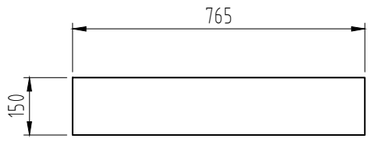
P7.7
SS400-6mm
SC:2/1-Q'ty:12



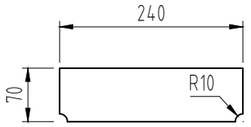
P7.6
SS400-6mm
SC:2/1-Q'ty:12



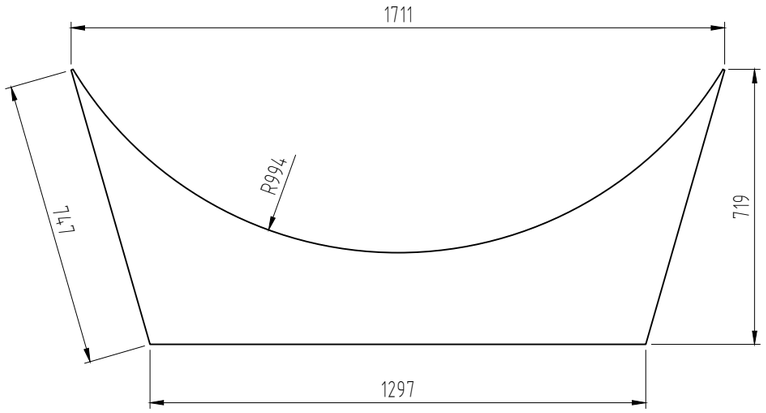
P7.1
SS400-10mm
Q'ty:3



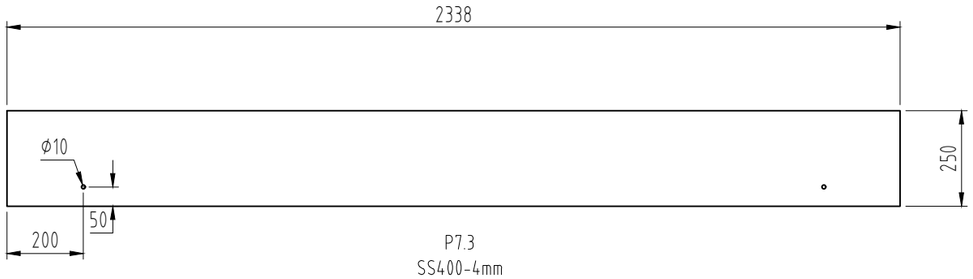
P7.4
SS400-6mm
Q'ty:6



P7.5
SS400-6mm
SC:2/1-Q'ty:6



P7.2
SS400-10mm
Q'ty:3



P7.3
SS400-4mm
Q'ty:3

Rev.	Date	Description	Drawn	Checked	Approved
3					
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1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

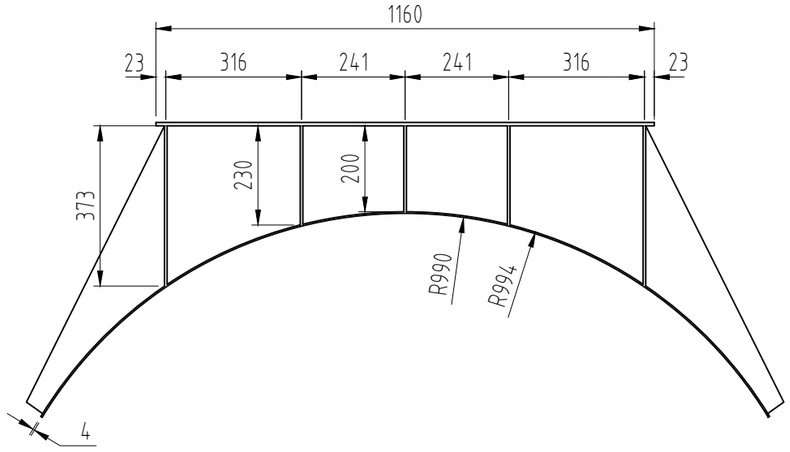
MANUFACTURER:
MARTECH BOILER The Energy Solutions
MARTECH JOINT STOCK COMPANY

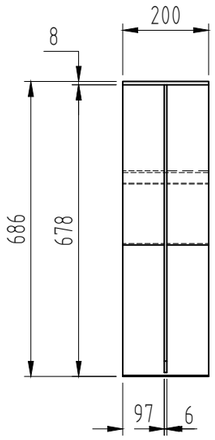
PROJECT JOB CODE:
TP2887

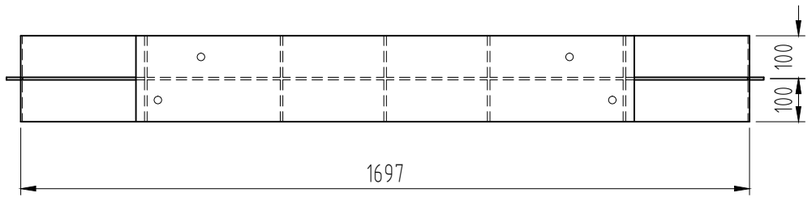
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

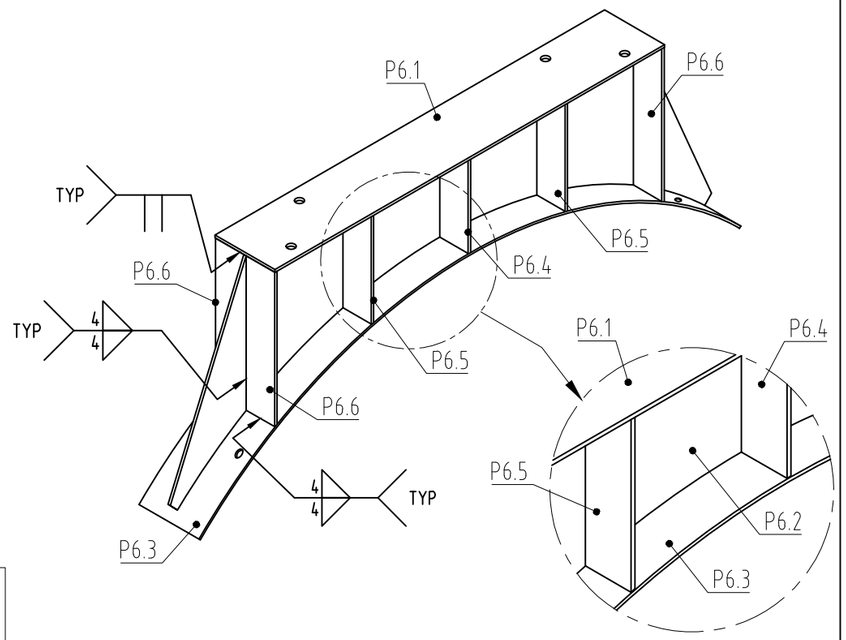
Scale	Size	Rev.
1:1	A3	00

DRAWING NO.:
2887.0100.MDWG.08

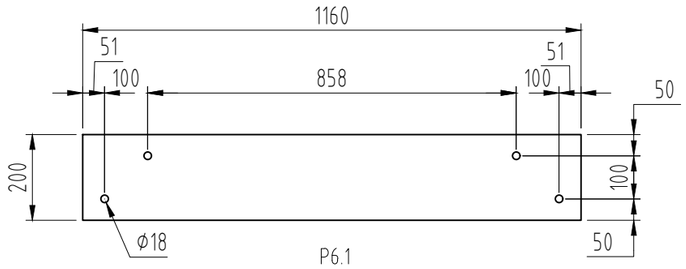




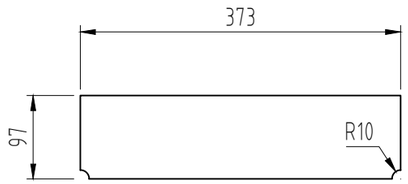




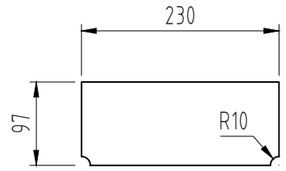
P6
Q'ty:02



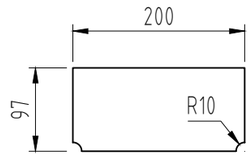
P6.1
SS400-8mm
Q'ty:2



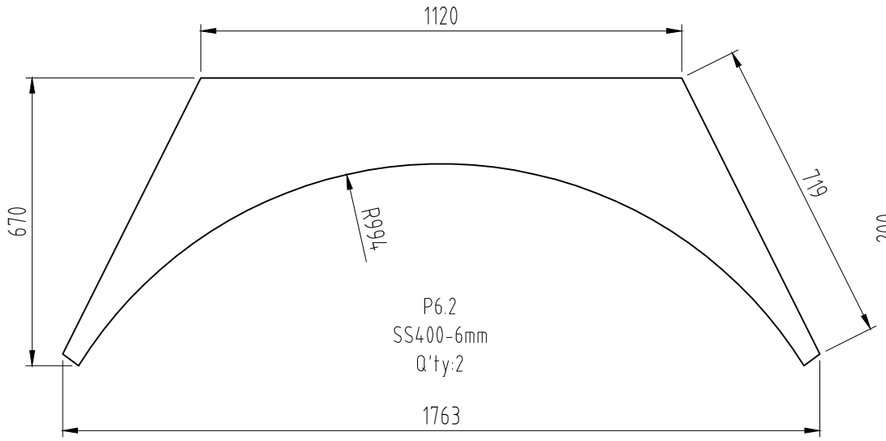
P6.6
SS400-6mm
SC:2/1-Q'ty:8



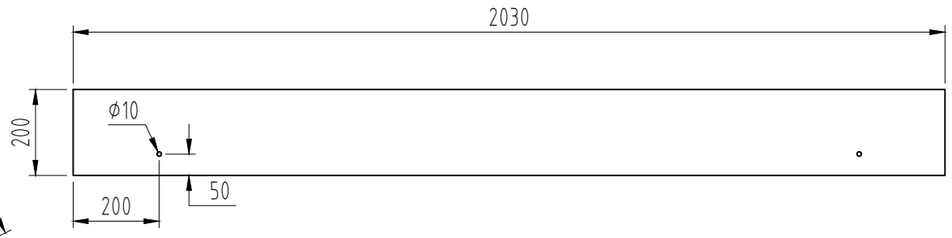
P6.5
SS400-6mm
SC:2/1-Q'ty:8



P6.4
SS400-6mm
SC:2/1-Q'ty:4



P6.2
SS400-6mm
Q'ty:2



P6.3
SS400-4mm
Q'ty:2

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

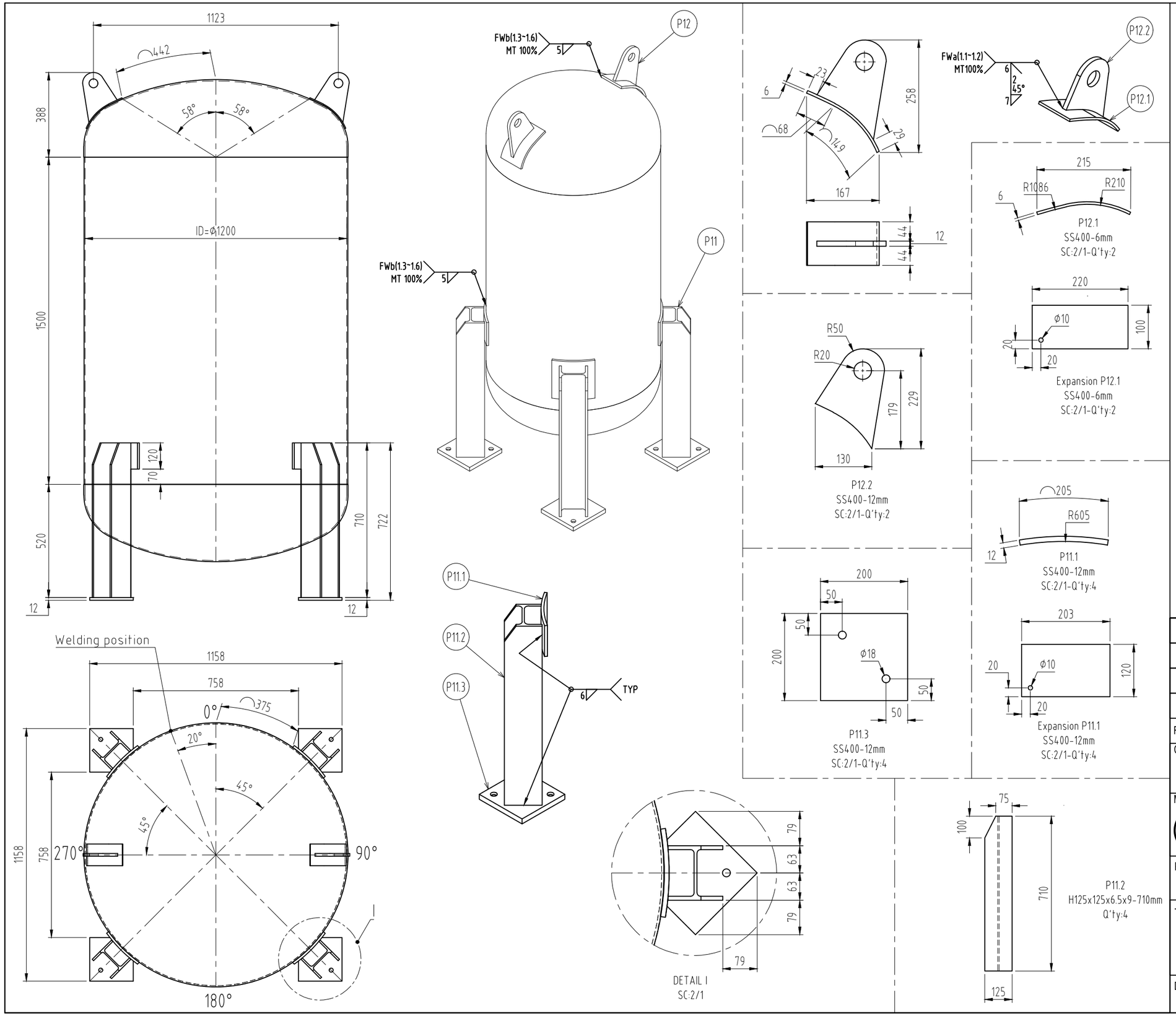
MANUFACTURER:
MARTECH BOILER
The Energy Solutions
MARTECH JOINT STOCK COMPANY

PROJECT JOB CODE: TP2887

TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.09



Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

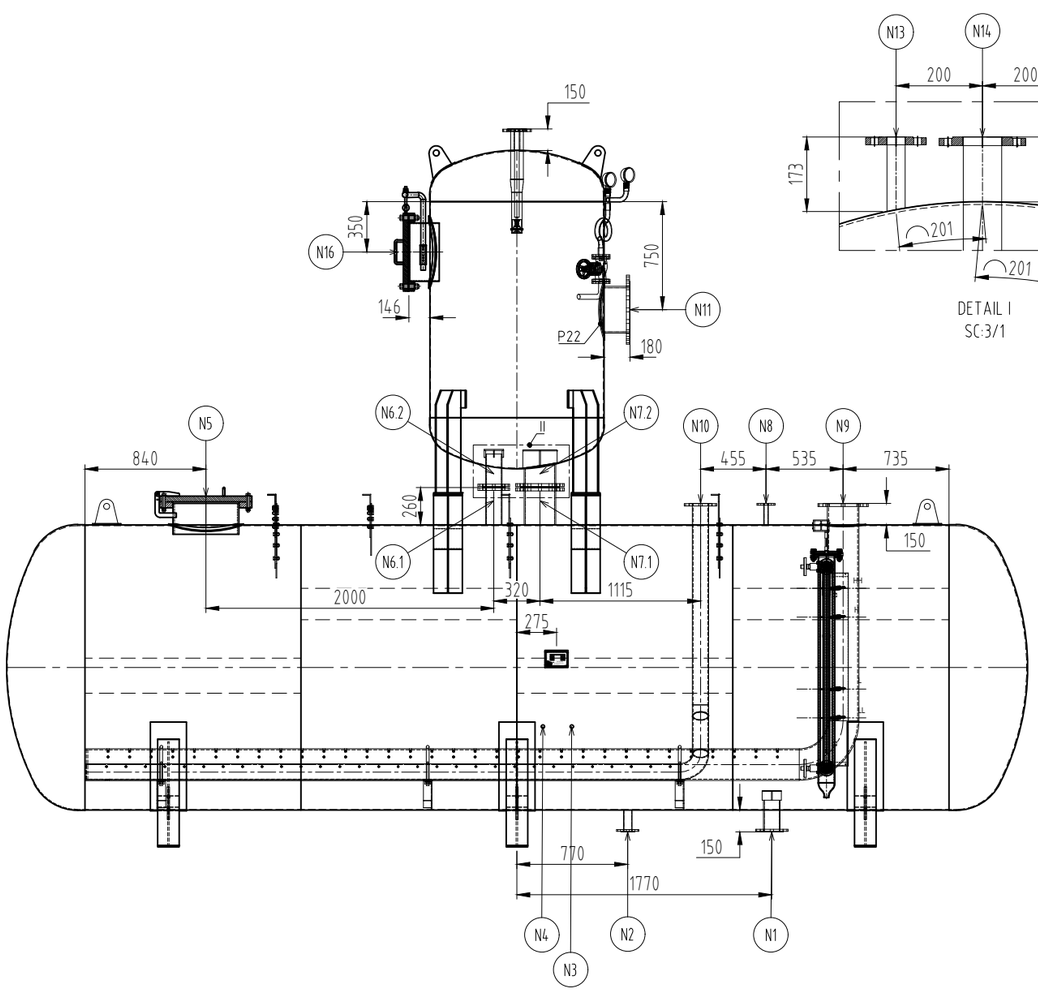
MANUFACTURER: MARTECH BOILER The energy Solutions — MARTECH JOINT STOCK COMPANY

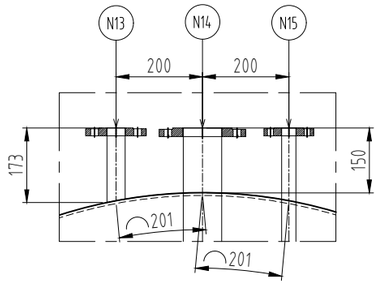
PROJECT JOB CODE: TP2887

TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

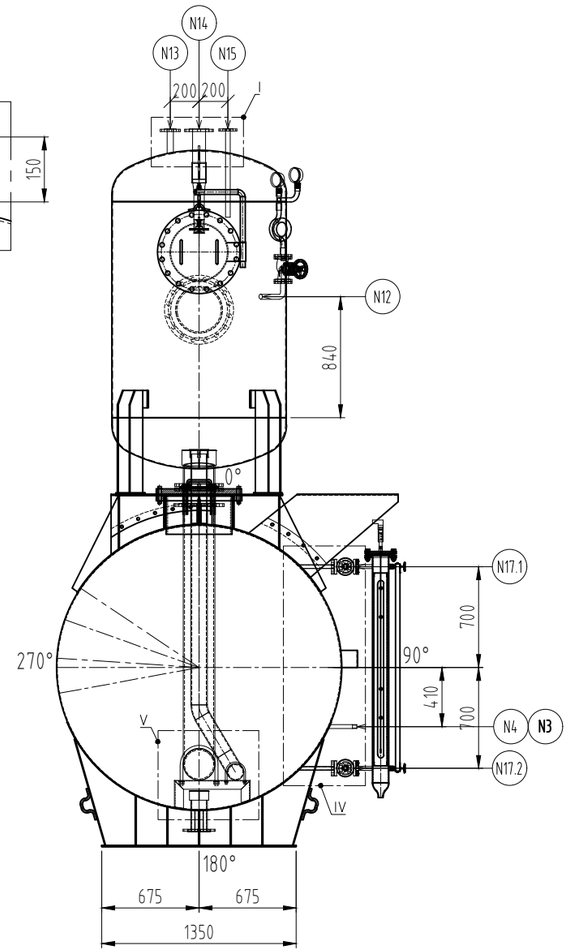
Scale	Size	Rev.
1:1	A3	00

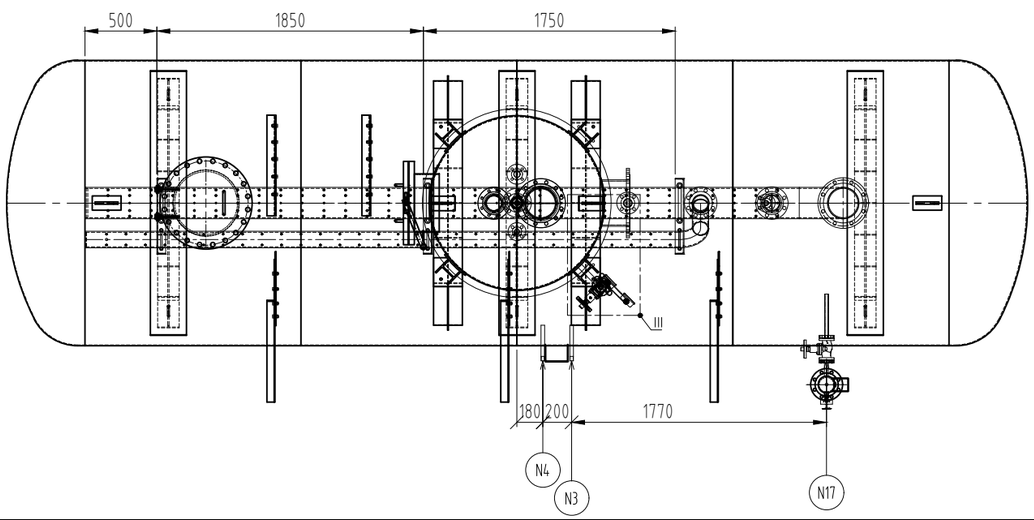
DRAWING NO.: 2887.0100.MDWG.10

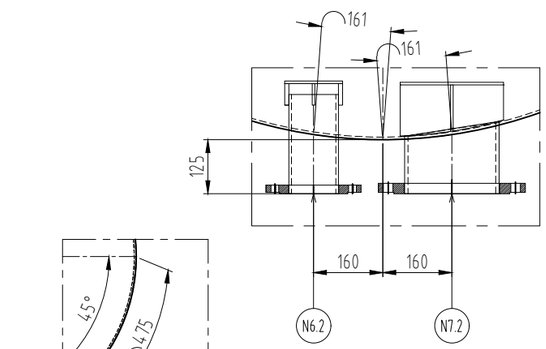




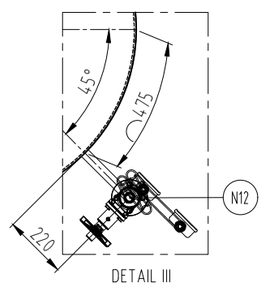
DETAIL I
SC:3/1







DETAIL II
SC:3/1



DETAIL III
SC:2/1

CONNECTION SIZE

No	Service	Q'ty	Nozzle neck			Flange			Remark
			Size	Sch.	Material	Type	Rating	Material	
N1	Ống nước đến cụm bơm	1	DN100	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N2	Ống xả đáy	1	DN50	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N3	Ống gắn nhiệt kế	1	DN20	40	SA/A106	Măng xông M21			-
N4	Ống gắn cảm biến nhiệt độ	1	DN20	40	SA/A106	Măng xông M21			-
N5	Cửa người chui	1	DN450	6T	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N6	Ống cân bằng hơi và nước	1	DN100	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N7	Ống xả xuống bồn nước	1	DN200	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N8	Ống gắn van xả áp chân không	1	DN25	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N9	Ống hơi cấp vào bồn nước	1	DN200	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N10	Ống nước hồi cấp vào bồn	1	DN100	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N11	Ống cấp hơi gia nhiệt bộ khử khí	1	DN300	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N12	Ống gắn TT, TG	1	DN25	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N13	Ống gắn van an toàn	1	DN32	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N14	Ống cấp nước bổ sung	1	DN80	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N15	Ống xả khí	1	DN25	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N16	Cửa vệ sinh bộ khử khí	1	DN400	6T	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-
N17	Ống gắn kính thủy tối	2	DN20	40	SA/A106	EN 1092-1,SORF, Type 01	PN16	SS400/ A516	-

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

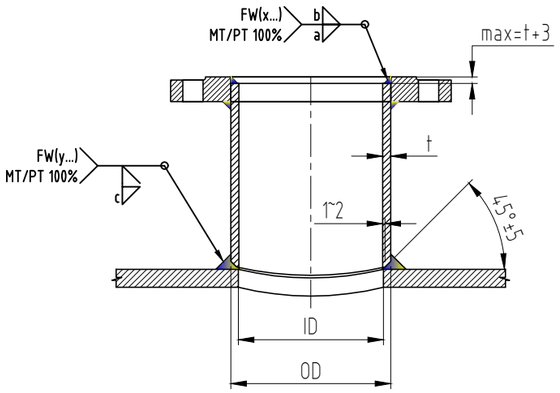
MANUFACTURER:
MARTECH BOILER
The energy Solutions
MARTECH JOINT STOCK COMPANY

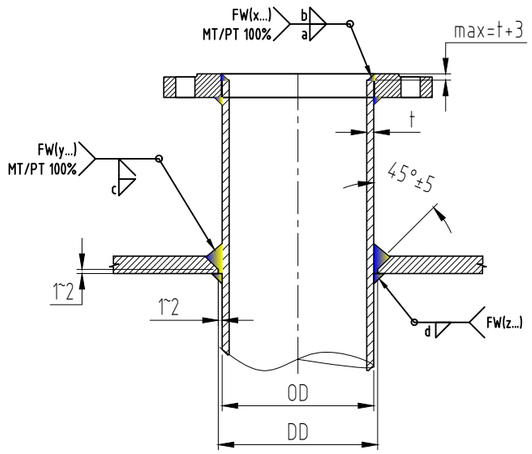
PROJECT JOB CODE:
TP2887

TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

DRAWING NO.:
2887.0100.MDWG.11



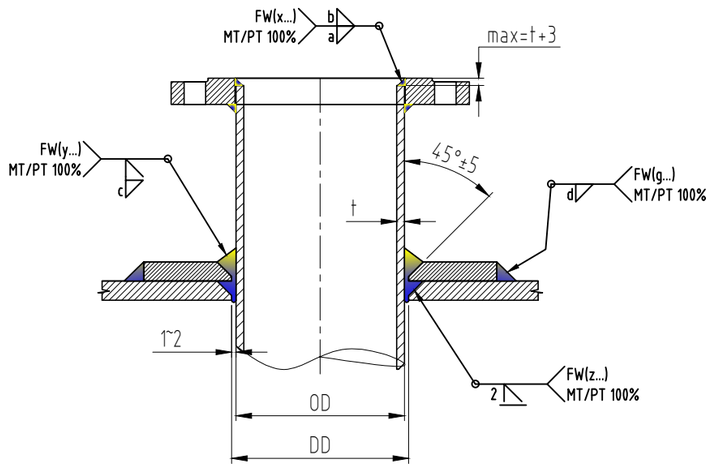


Welding detail list drum nozzle

No.	Symbol	Description	Detail Nozzle(mm)			No. Weld		Size weld		
			OD	t	ID	FW(x)	FW(y)	a	b	c
1	N2	Ống xả đáy	60.3	3.91	52.5	2.1	2.2	3	4	3
2	N3	Ống gắn nhiệt kế	26.7	2.87	21.0		3.1	-	-	2
3	N4	Ống gắn cảm biến nhiệt độ	26.7	2.87	21.0		4.1	-	-	2
4	N6.1	Ống cân bằng hơi và nước	114.3	6.02	102.3	6.1	6.2	5	7	5
5	N7.1 N7.2	Ống xả xuống bồn nước	219.1	8.18	202.7	7.1 7.3	7.2 7.4	7	9	6
6	N8	Ống gắn van xả áp chân không	33.4	3.38	26.6	8.1	8.2	3	4	3
7	N12	Ống gắn TT, TG	33.4	3.38	26.6	12.1	12.2	3	4	3
8	N13	Ống gắn van an toàn	42.2	3.56	35.1	13.1	13.2	3	4	3
9	N17.1 N17.2	Ống gắn kính thủy tối	26.7	2.87	21.0	17.1 17.3	17.2 17.4	2	3	2

Welding detail list drum nozzle

No.	Symbol	Description	Detail Nozzle (mm)			No. Weld				Size weld			
			OD	t	DD	FW(x)	FW(y)	FW(z)	FW(g)	a	b	c	d
1	N1	Ống nước đến cụm bơm	114.3	6.02	115.3	1.1	1.2	1.3	-	5	7	5	5
2	N6.2	Ống cân bằng hơi và nước	114.3	6.02	115.3	6.3	6.4	6.5	-	5	7	5	5
3	N10	Ống nước hồi cấp vào bồn	114.3	6.02	115.3	10.1	10.2	10.3	-	5	7	5	5
4	N14	Ống cấp nước bổ sung	88.9	5.49	89.9	14.1	14.2	14.3	-	5	6	4	4
5	N15	Ống xả khí	33.4	3.38	34.4	15.1	15.2	15.3	-	3	4	3	3



Welding detail list drum nozzle

No.	Symbol	Description	Detail Nozzle (mm)			No. Weld				Size weld			
			OD	t	DD	FW(x)	FW(y)	FW(z)	FW(g)	a	b	c	d
1	N5	Cửa người chui	457.0	6.00	459.0	5.1	5.2	5.3	5.4	5	7	5	6
2	N9	Ống hơi cấp vào bồn nước	219.1	8.18	221.1	9.1	9.2	9.3	9.4	7	9	6	6
3	N11	Ống cấp hơi gia nhiệt bộ khử khí	323.9	10.31	325.9	11.1	11.2	11.3	11.4	9	11	8	6
4	N16	Cửa người chui	406.4	6.00	408.4	16.1	16.2	16.3	16.4	5	7	5	6

Rev.	Date	Description	Drawn	Checked	Approved
3					
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1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:

MARTECH BOILER The Energy Solutions

MARTECH JOINT STOCK COMPANY

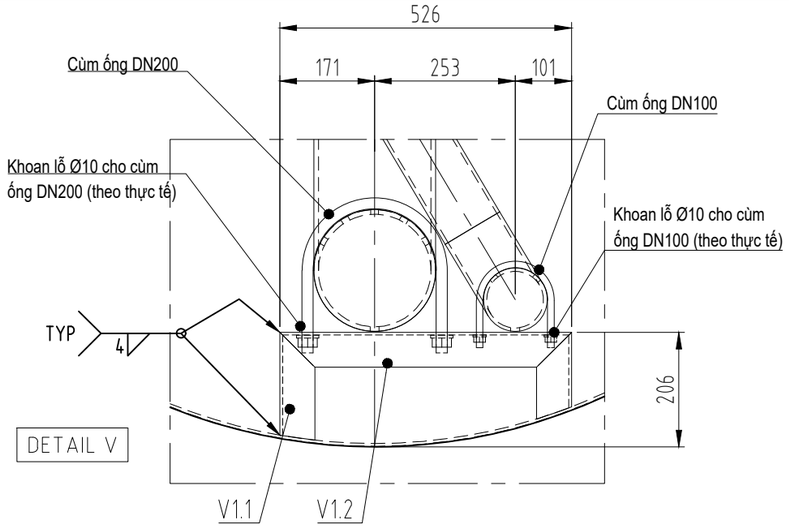
PROJECT JOB CODE: TP2887

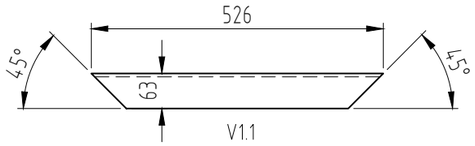
TITLE: DETAIL DRAWING

DEAERATOR AND WATER TANK

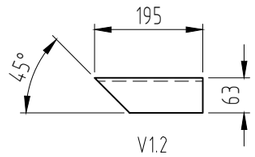
Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.12

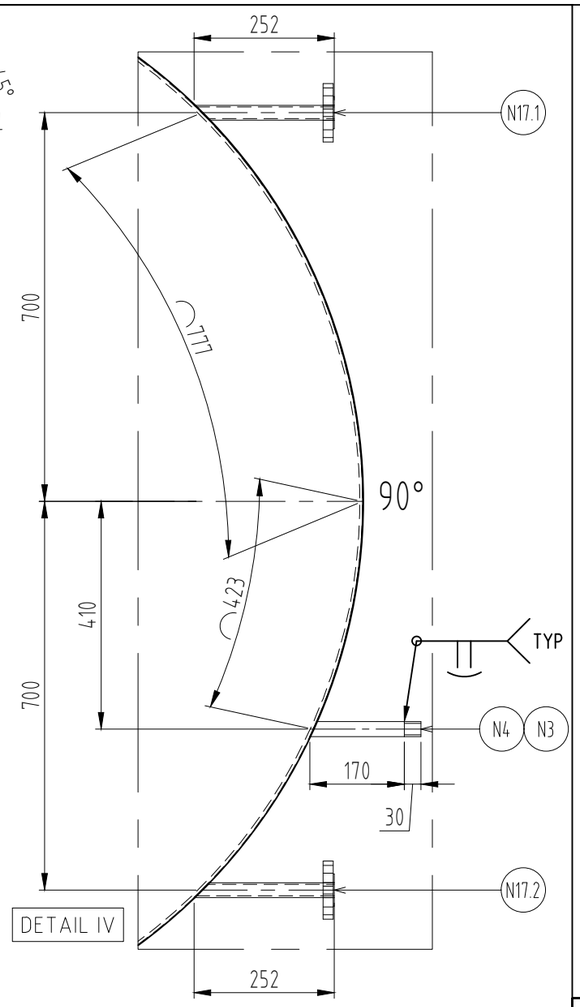


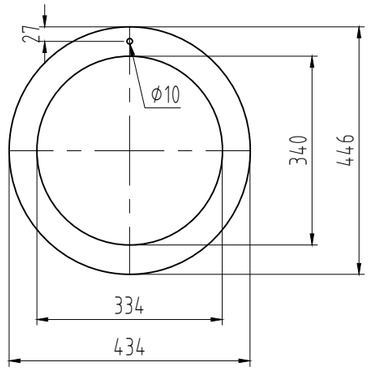


V1.1
V63x63x6-526mm
Q'ty:3

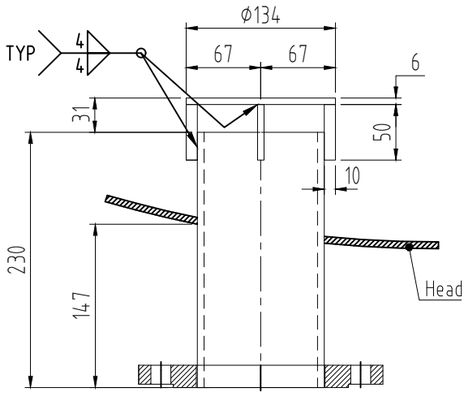


V1.2
V63x63x6-195mm
Q'ty:6

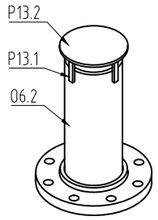


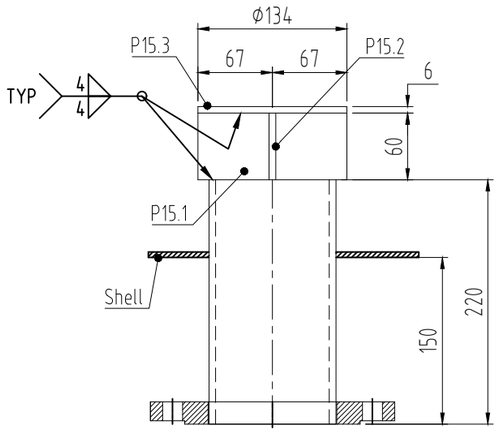


P22(GIA CUONG N11)
SS400-6mm
Q'ty:1

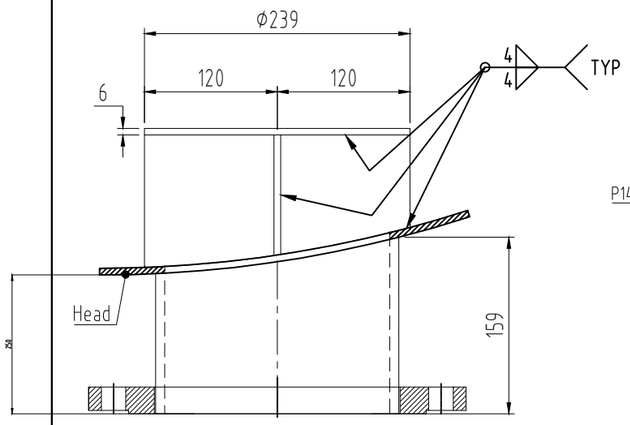


N6.2 (DN100-PN16)
SC:2/1

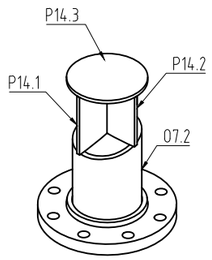


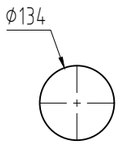


N1 (DN100-PN16)
SC:2/1

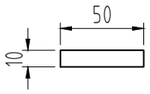


N7.2 (DN200-PN16)
SC:2/1





P13.2
SS400-6mm
Q'ty:1



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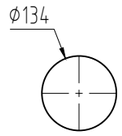
SS400-6mm
SC:3/1-Q'ty:4



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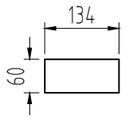
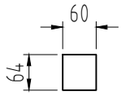
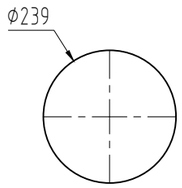
SS400-6mm
Q'ty:1



P15.2
SS400-6mm
Q'ty:1



P15.3
SS400-6mm
Q'ty:2



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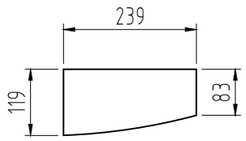
SS400-6mm
Q'ty:1



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SS400-6mm
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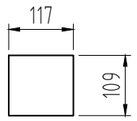


P14.2
SS400-6mm
Q'ty:2

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:
MARTECH BOILER
The Energy Solutions
MARTECH JOINT STOCK COMPANY

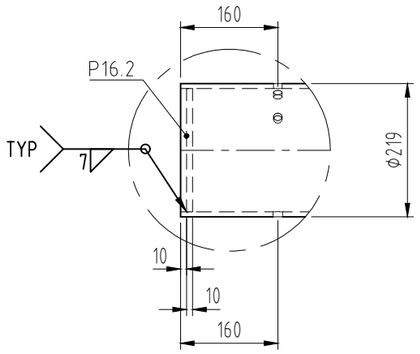
PROJECT JOB CODE:
TP2887

TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

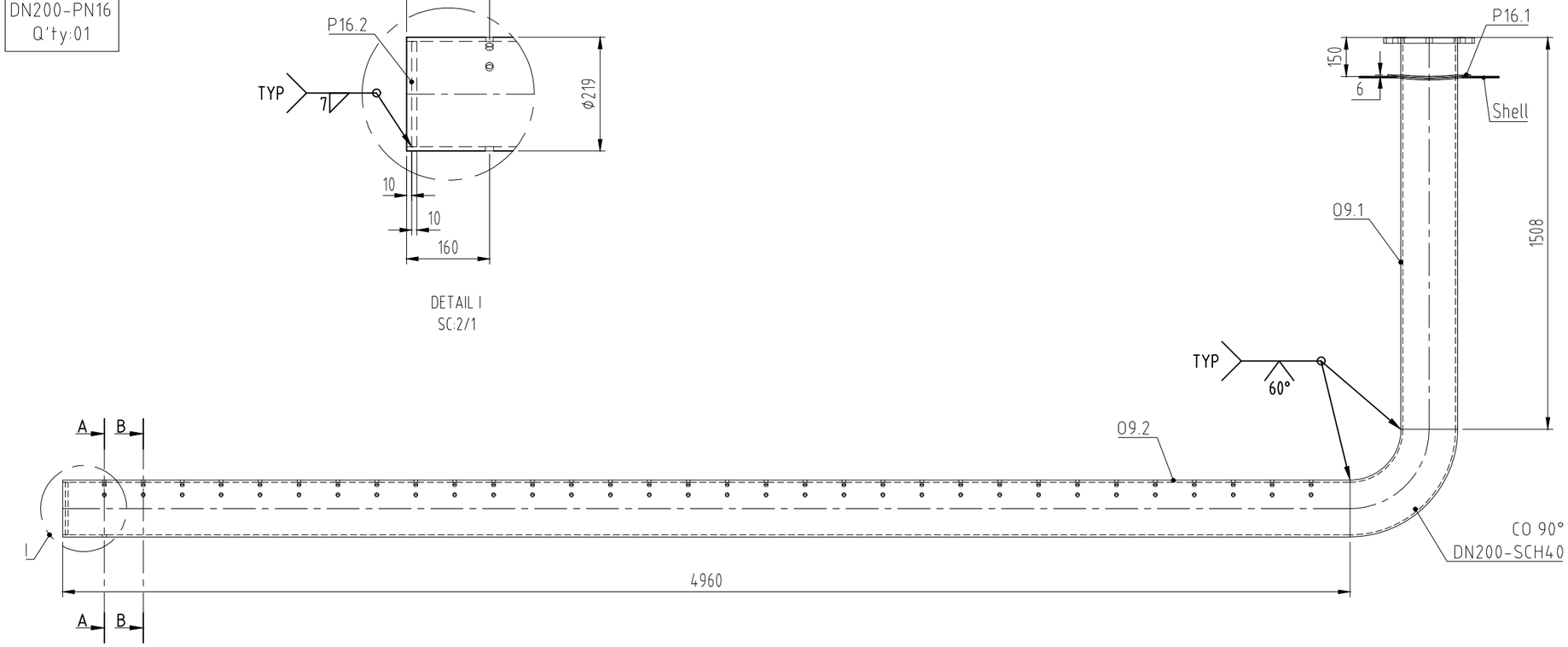
Scale	Size	Rev.
1:1	A3	00

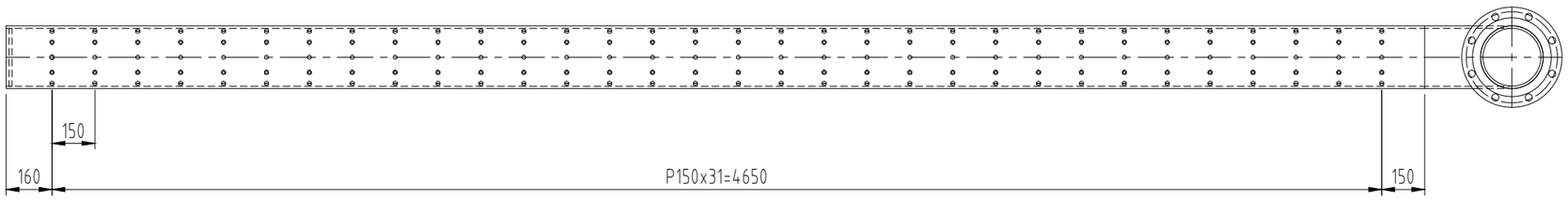
DRAWING NO.:
2887.0100.MDWG.13

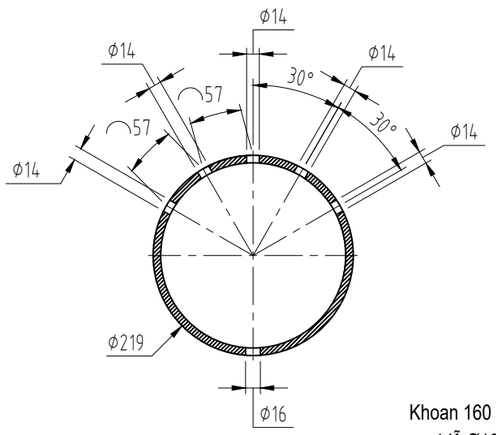
N9
DN200-PN16
Q'ty:01



DETAIL I
SC:2/1

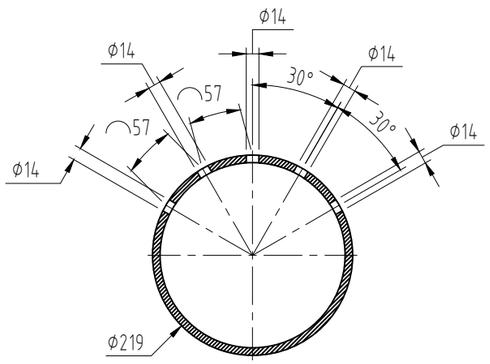




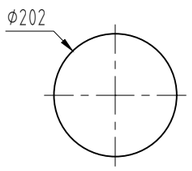


SECTION A-A
SC:3/1

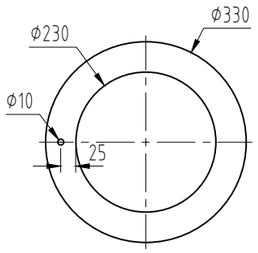
Khoan 160 lỗ Ø14mm,
1 lỗ Ø16 xả cặn



SECTION B-B
SC:3/1



P16.2
SS400-10mm
SC:2/1-Q'ty:01



P16.1
SS400-6mm
SC:2/1-Q'ty:01

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:
MARTECH BOILER
The Energy Solutions
MARTECH JOINT STOCK COMPANY

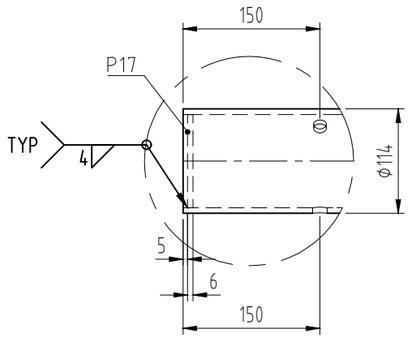
PROJECT JOB CODE:
TP2887

TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

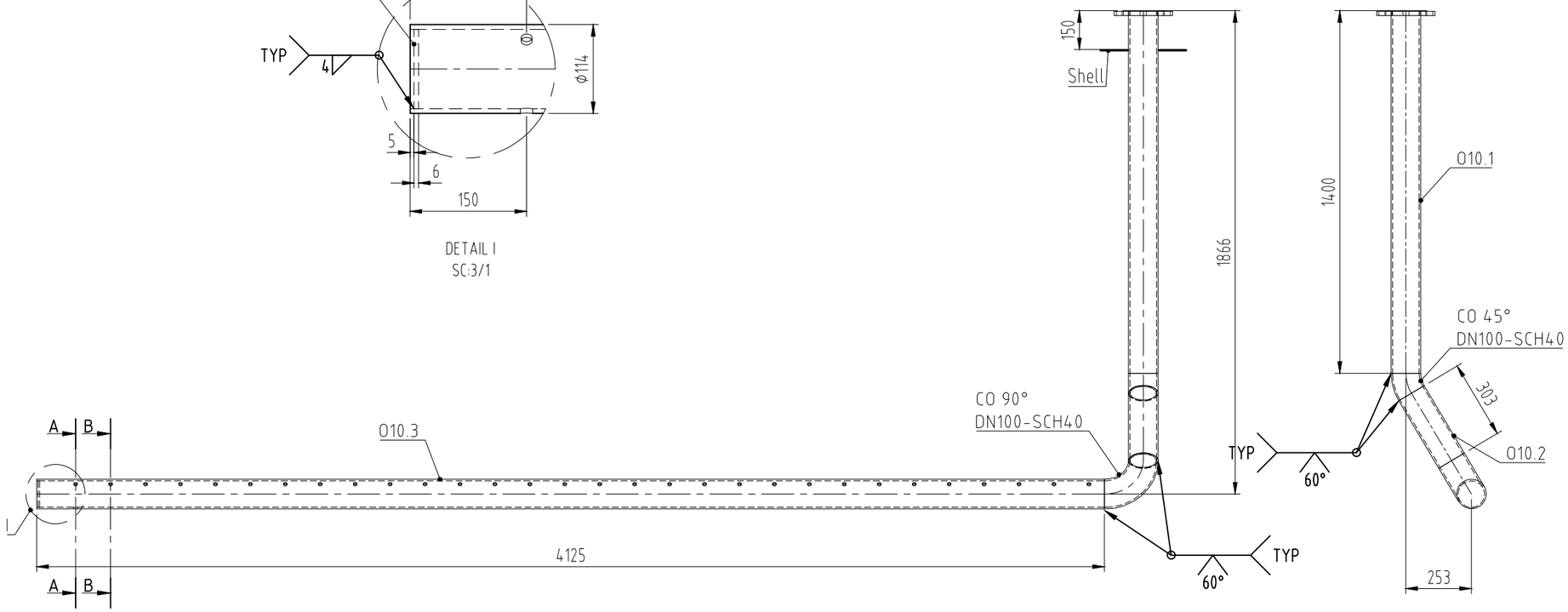
Scale	Size	Rev.
1:1	A3	00

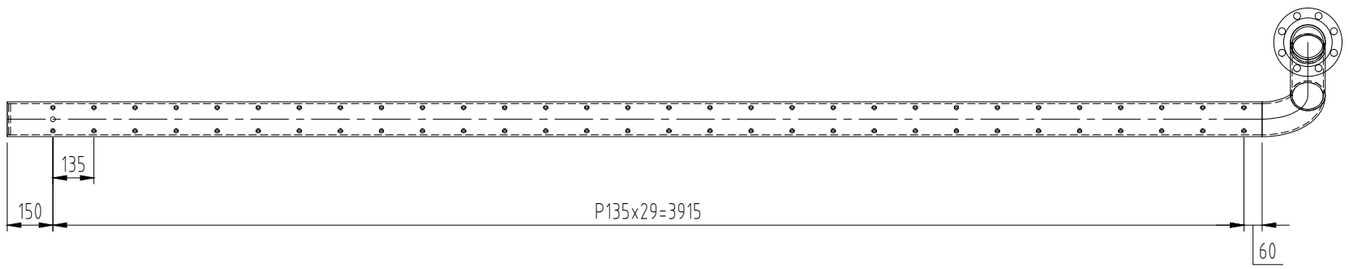
DRAWING NO.:
2887.0100.MDWG.14

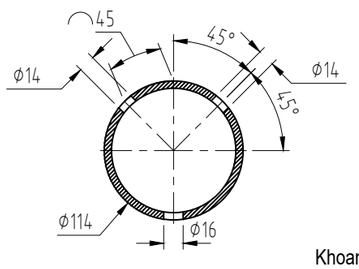
N10
DN100-PN16
Q'ty:01



DETAIL I
SC:3/1

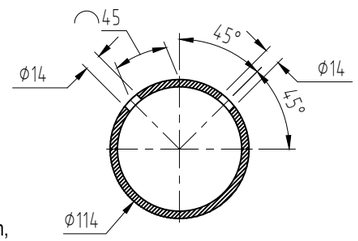




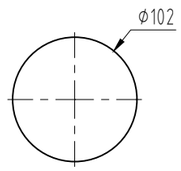


SECTION A-A
SC:4/1

Khoan 60 lỗ Ø14mm,
1 lỗ Ø16 xả cặn



SECTION B-B
SC:4/1



P17
SS400-6mm
SC:4/1-Q'ty:01

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:
MARTECH BOILER
The energy Solutions
MARTECH JOINT STOCK COMPANY

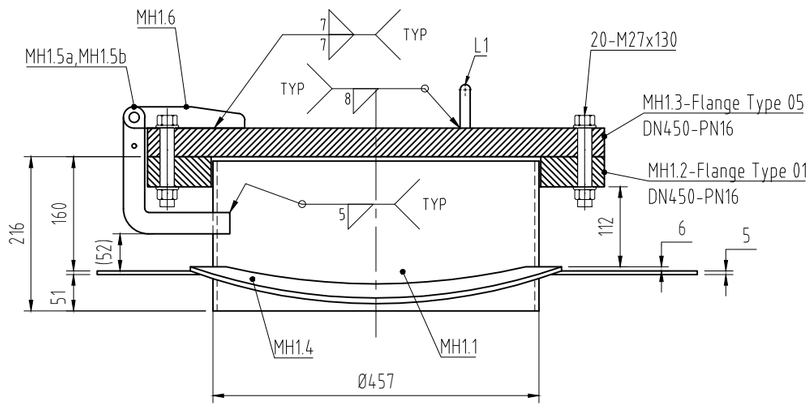
PROJECT JOB CODE:
TP2887

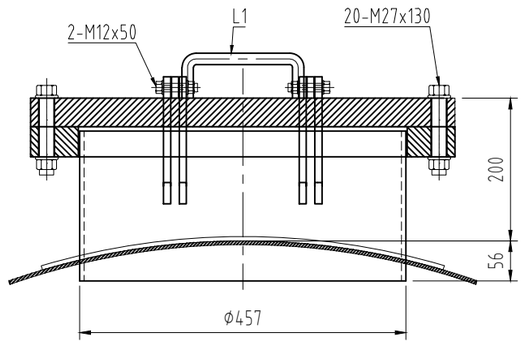
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

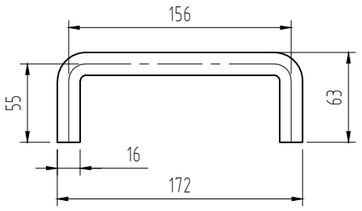
Scale	Size	Rev.
1:1	A3	00

DRAWING NO.:
2887.0100.MDWG.15

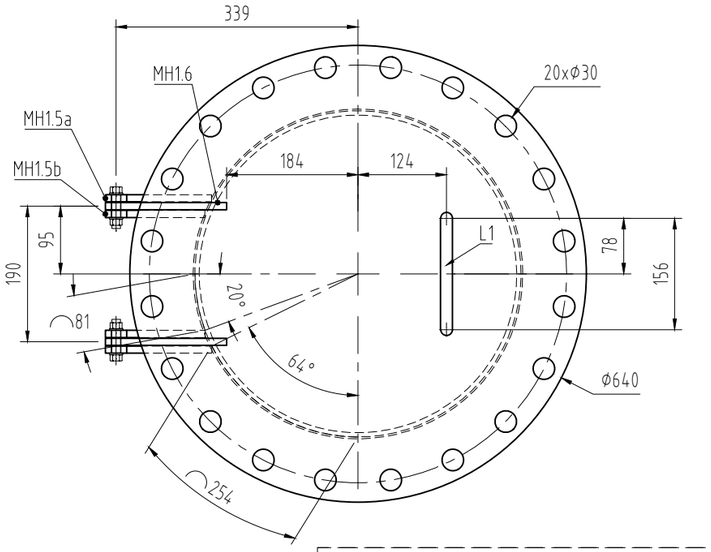
N5-Manhole 1
SL: 01

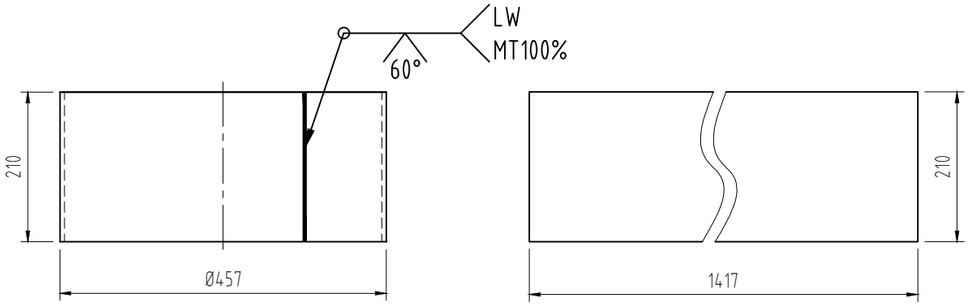






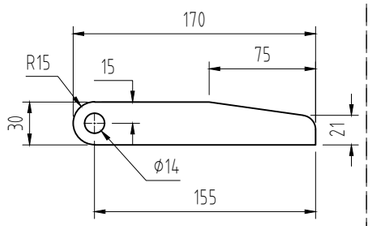
L1
Shape steel Φ16x260mm
Q'ty:01-SC: 2/1



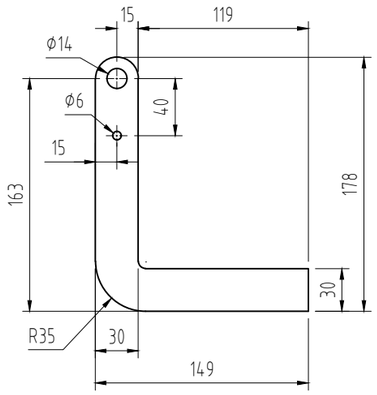


M1.1
SS400-6mm
Q'ty:01

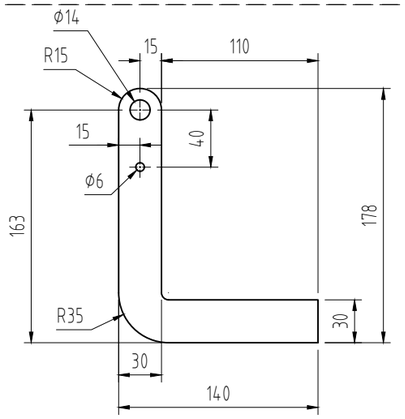
Expansion M1.1
SS400-6mm
Q'ty:01



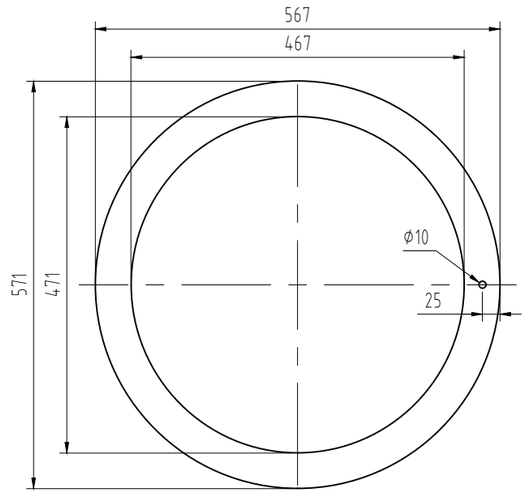
MH1.6
SS400-10mm
Q'ty:02-SC: 2/1



MH1.5a
SS400-10mm
Q'ty:02-SC: 2/1



MH1.5b
SS400-10mm
Q'ty:02-SC: 2/1



MH1.4
SS400-6mm
Q'ty:01

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
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0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:
MARTECH BOILER
The energy Solutions
MARTECH JOINT STOCK COMPANY

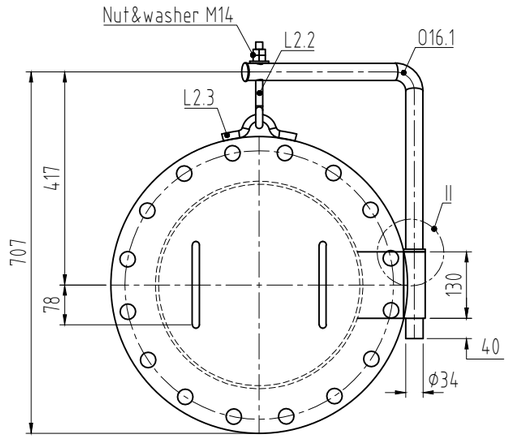
PROJECT JOB CODE: TP2887

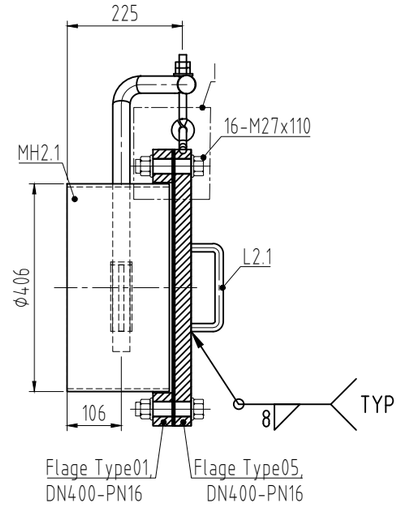
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

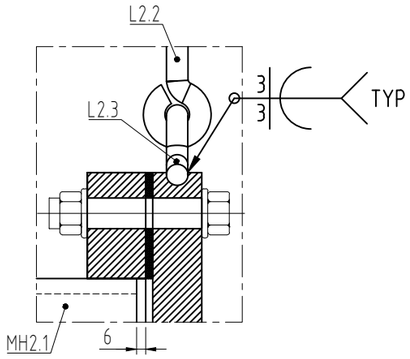
Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.16

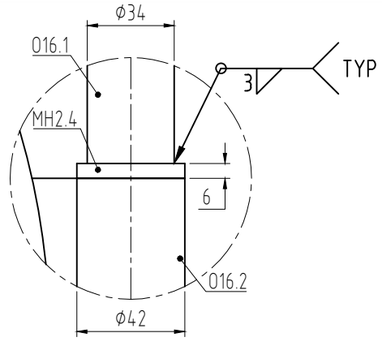
N16 (Manhole)
Q'ty:01



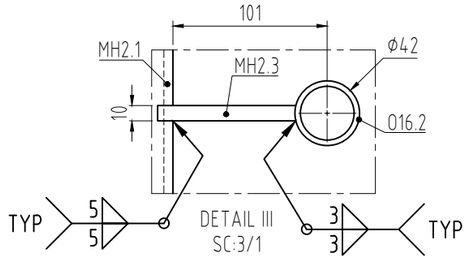




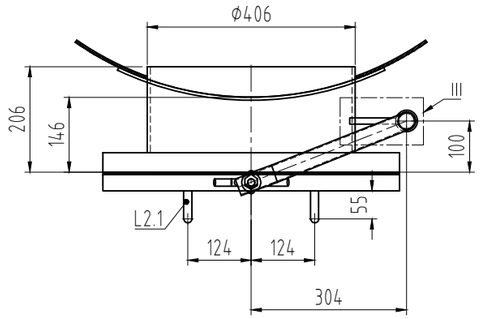
DETAIL I
SC:3/1

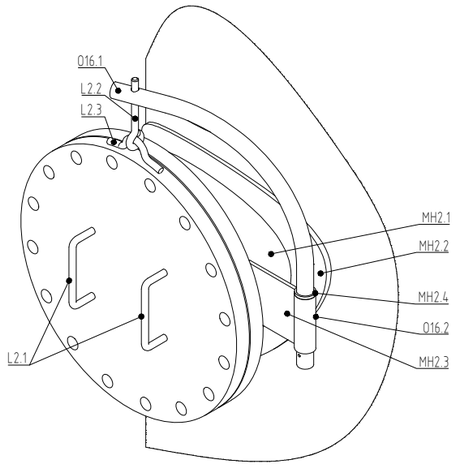


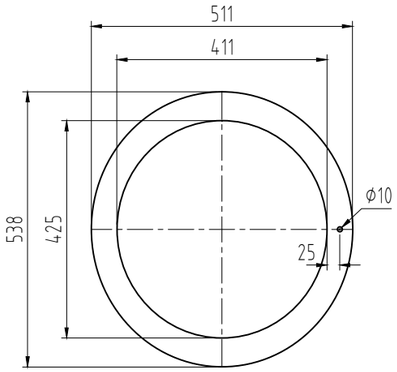
DETAIL II
SC:5/1



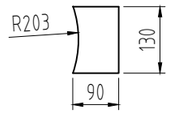
DETAIL III
SC:3/1



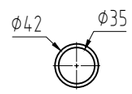




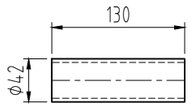
MH2.2
SS400-6mm
Q'ty:1



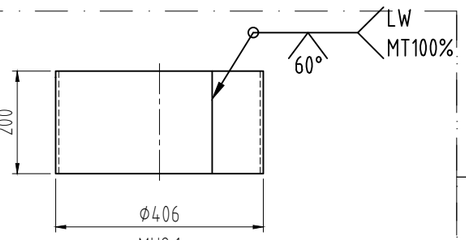
MH2.3
SS400-10mm
Q'ty:1



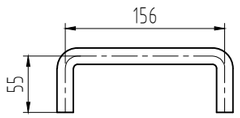
MH2.4
SS400-6mm
SC:2/1-Q'ty:1



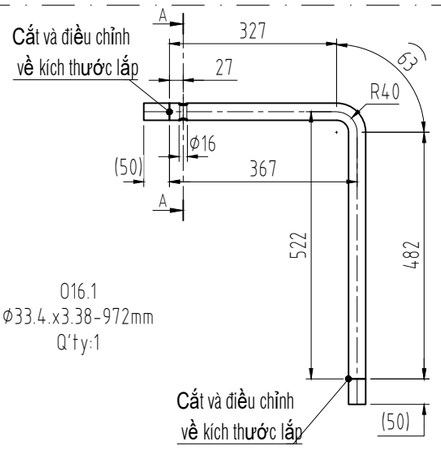
O16.2
Φ42.2x3.56-130mm
SC:2/1-Q'ty:1



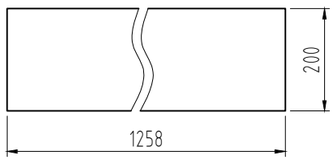
MH2.1
SS400-6mm
Q'ty:1



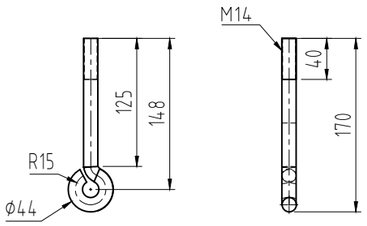
L2.1
Shape steel Φ16-260mm
SC:2/1-Q'ty:2



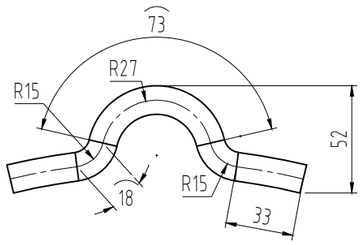
O16.1
Φ33.4.x3.38-972mm
Q'ty:1



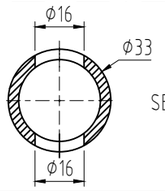
Expansion MH2.1
SS400-6mm
Q'ty:1



L2.2
Shape steel Φ14-250mm
SC:2/1-Q'ty:1



L2.3
Shape steel Φ14-350mm
SC:4/1-Q'ty:1



SECTION A-A

MATERIAL LIST

No.	Symbol	Material Decripsion		Specifications (mm)			Q'ty	Area (m2)	Mass (kg)	Pain are (m2)	Note
		Type of material	Material	Thk.	Length	Width					
1	MH2.1	Plate steel	SS400/ A36	6	1258	200	1	0.3	11.9	0.3	
2	MH2.2	Plate steel	SS400/ A36	6	538	511	1	0.3	12.9	0.3	
3	MH2.3	Plate steel	SS400/ A36	10	130	90	1	0.0	0.9	0.0	
4	MH2.4	Plate steel	SS400/ A36	6	42	42	1	0.0	0.1	0.0	
5	L2.1	Shape steel Φ16	S45C		260	16	2	-	0.8	0.0	
6	L2.2	Shape steel Φ14	S45C		250	14	1	-	0.4	0.0	
7	L2.3	Shape steel Φ14	S45C		350	14	1	-	0.6	0.0	
8	O16.1	Tube Φ33.4x3.38	SA106 Gr.B	3.38	972	33.40	1	-	2.4	0.1	
9	O16.2	Tube Φ42.2x3.56	SA106 Gr.B	3.56	130	42.20	1	-	0.4	0.0	
TOTAL							7.0	0.5	27.0	0.6	

Rev.	Date	Description	Drawn	Checked	Approved
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0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

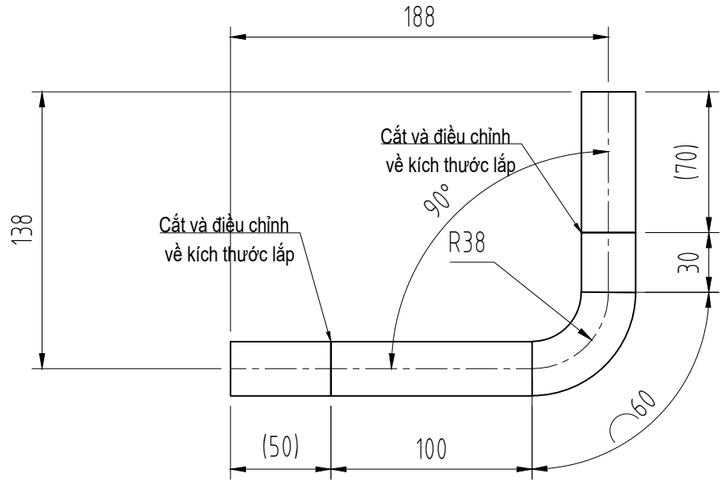
MANUFACTURER:
MARTECH BOILER The Energy Solutions
MARTECH JOINT STOCK COMPANY

PROJECT JOB CODE:
TP2887

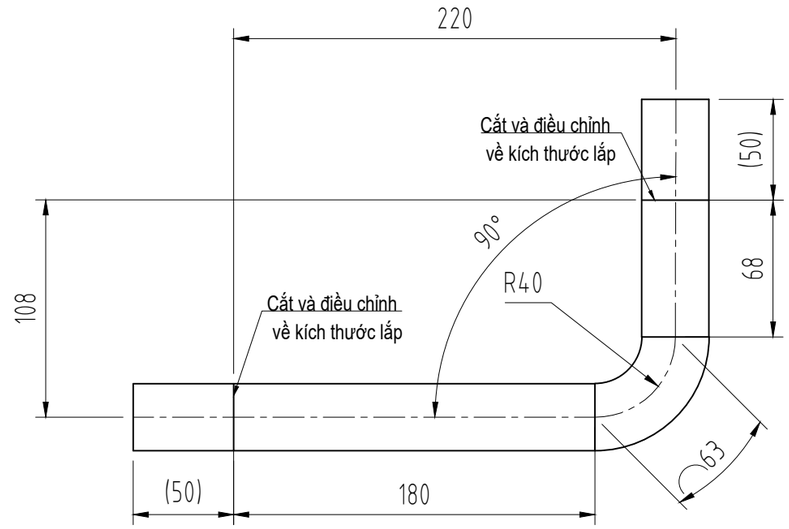
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

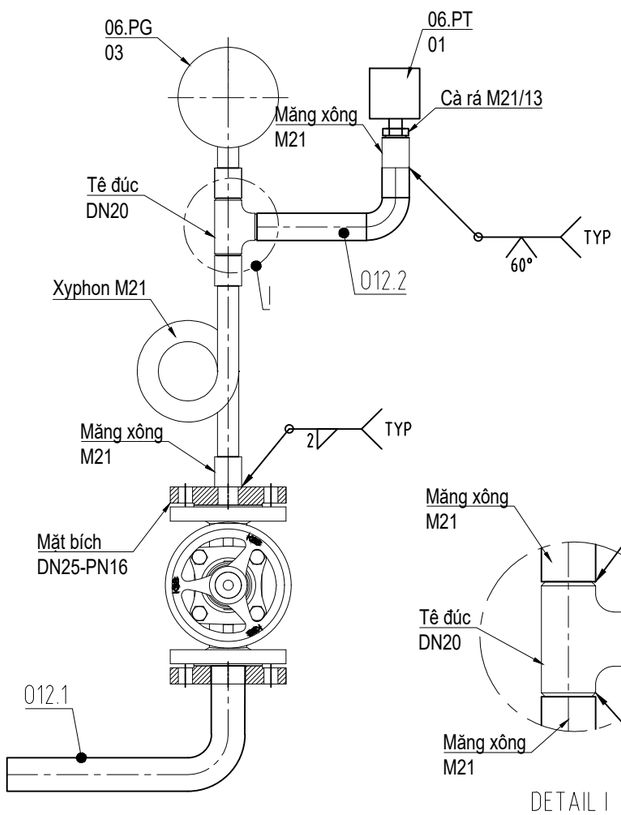
DRAWING NO.:
2887.0100.MDWG.17



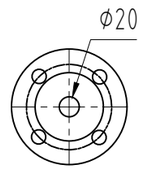
012.2
Φ26.7x2.87-310mm
SC:2/1-Q'ty:01



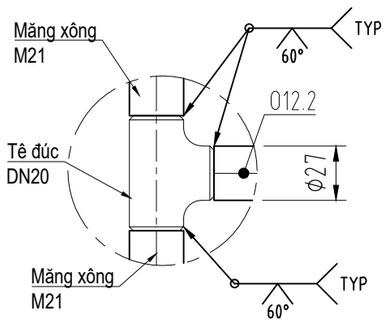
012.1
Φ33.4x3.38-415mm
SC:2/1-Q'ty:01



N12
DN25-PN16
Q'ty:01



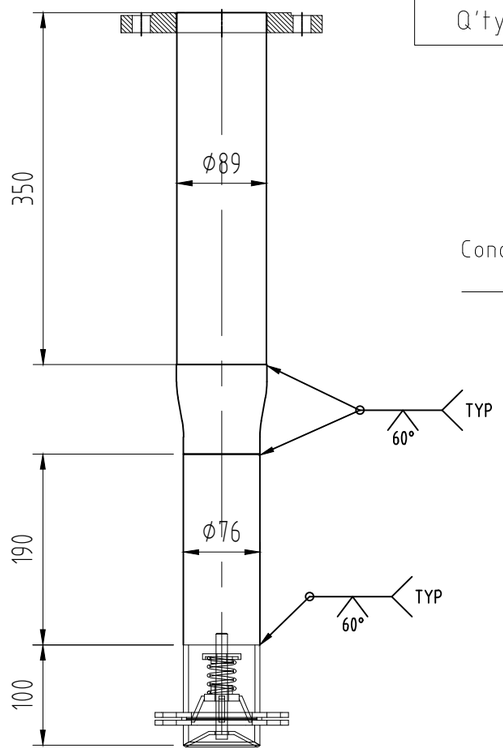
Type05 DN25-PN16
Q'ty:01

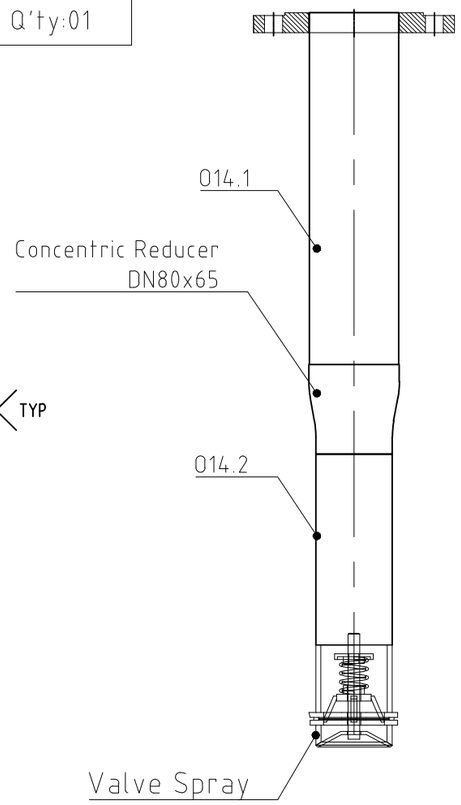


DETAIL I
SC:2/1

Tagname:XXXXXX-2887-03

N14
DN80-PN16
Q'ty:01





Rev.	Date	Description	Drawn	Checked	Approved
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0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

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MANUFACTURER:
MARTECH BOILER
The Energy Solutions
MARTECH JOINT STOCK COMPANY

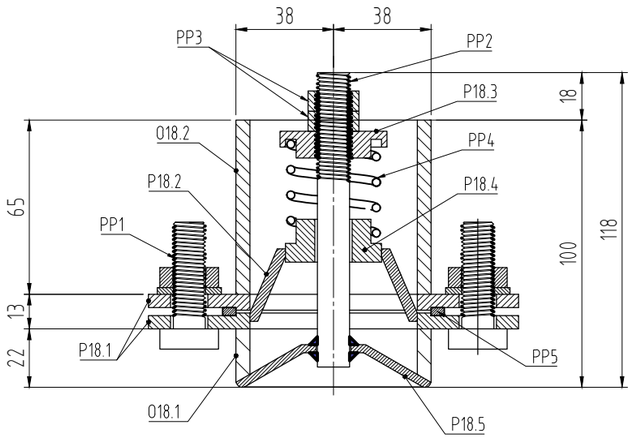
PROJECT JOB CODE:
TP2887

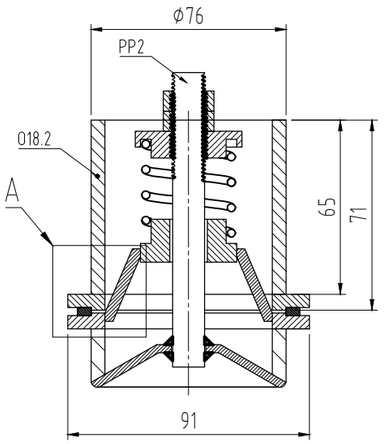
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

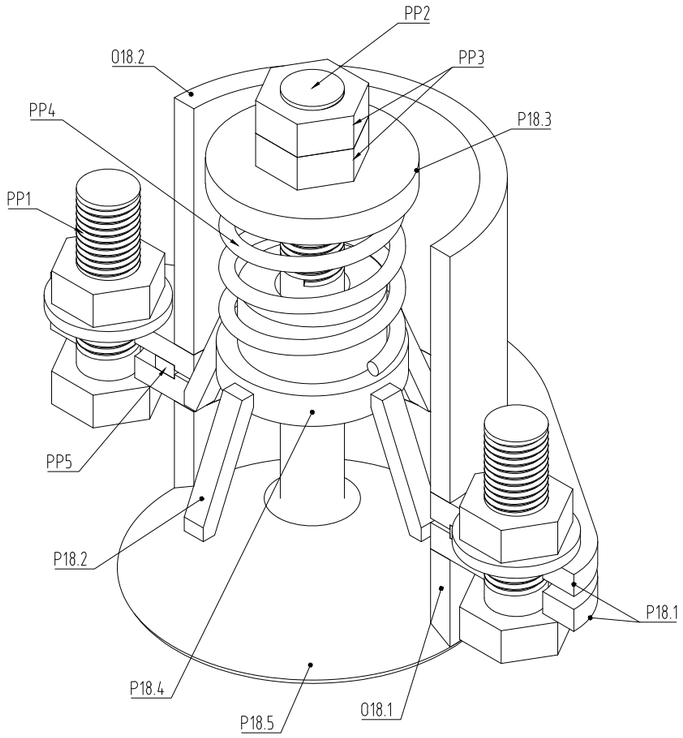
Scale	Size	Rev.
1:1	A3	00

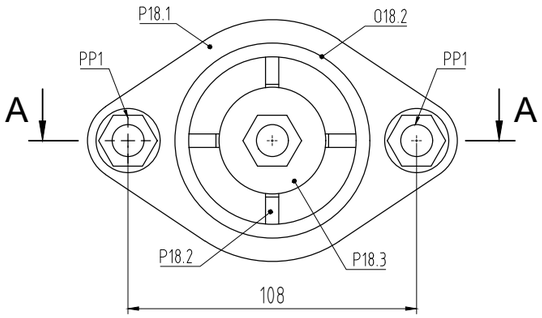
DRAWING NO.:
2887.0100.MDWG.18

Valve Spray
Q'ty: 01

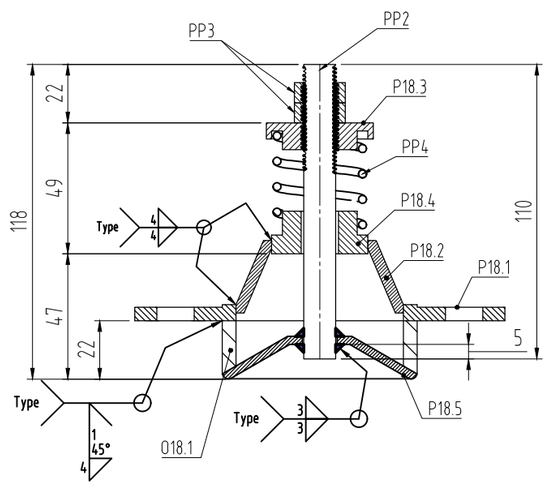




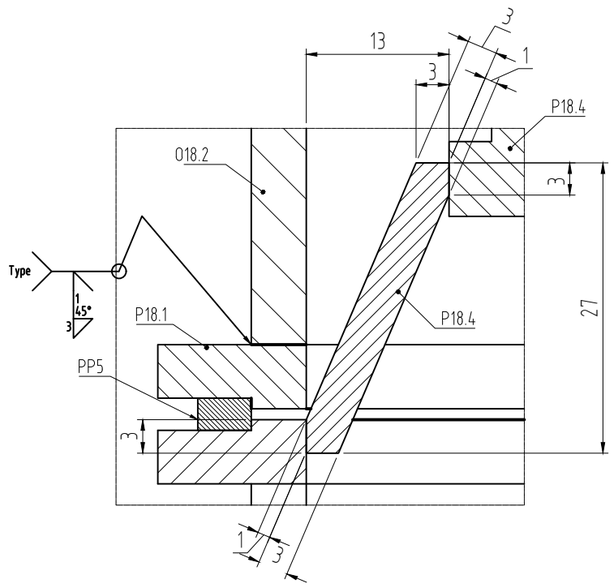




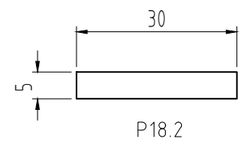
LIST VẬT TƯ									
No.	Mô tả	Vật liệu	Kích thước (mm)			Số lượng/bộ	Số bộ	Số lượng	Đơn vị
			Dày	Dài	Rộng				
O18.1	A312 TP304, Ø76x5.16mm	Inox 304	5,16	20		1	1	1	ống
P18.1	Tấm đỡ	Inox 304	6	139	91	2	1	2	tấm
PP1	Bulong M12x40mm (Tán+long đền)	Inox				2	1	2	bộ
P18.2	Gân gia cường	Inox 304	6	30	5	4	1	4	tấm
O18.2	A312 TP304, Ø76x5.16mm	Inox 304	5,16	65		1	1	1	ống
PP2	Ty ren M12	Inox		110		1	1	1	đoạn
PP3	Tán M12	Inox				2	1	2	con
P18.3	Ốp đỡ 1	Inox 304	16	40	40	1	1	1	tấm
PP4	Lò xo Ø30 dày 3mm, dài 40mm	Inox				1	1	1	cái
P18.4	Ốp đỡ 2	Inox 304	16	36	36	1	1	1	tấm
PP5	Vòng đệm	Cao su tấm	3	83	83	1	1	1	tấm
P18.5	Vành côn	Inox 304	16	73	73	1	1	1	tấm



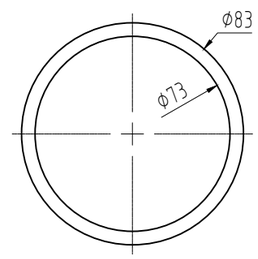
Section A-A



Detail A



P18.2
SUS304-6mm
SC:2/1-Q'ty:04



PP5
Cao su-3mm
Q'ty:01

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

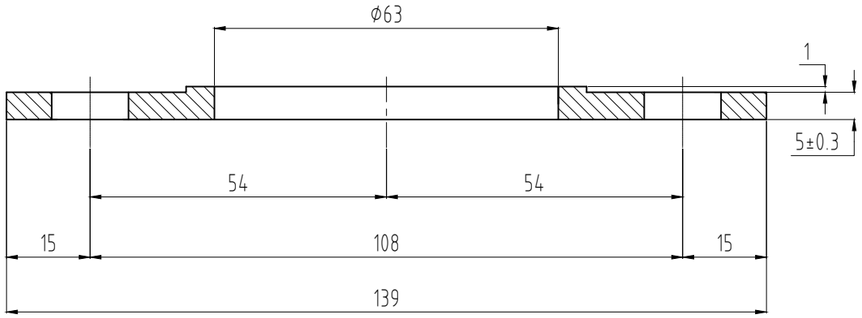
MANUFACTURER: MARTECH BOILER The Energy Solutions — MARTECH JOINT STOCK COMPANY

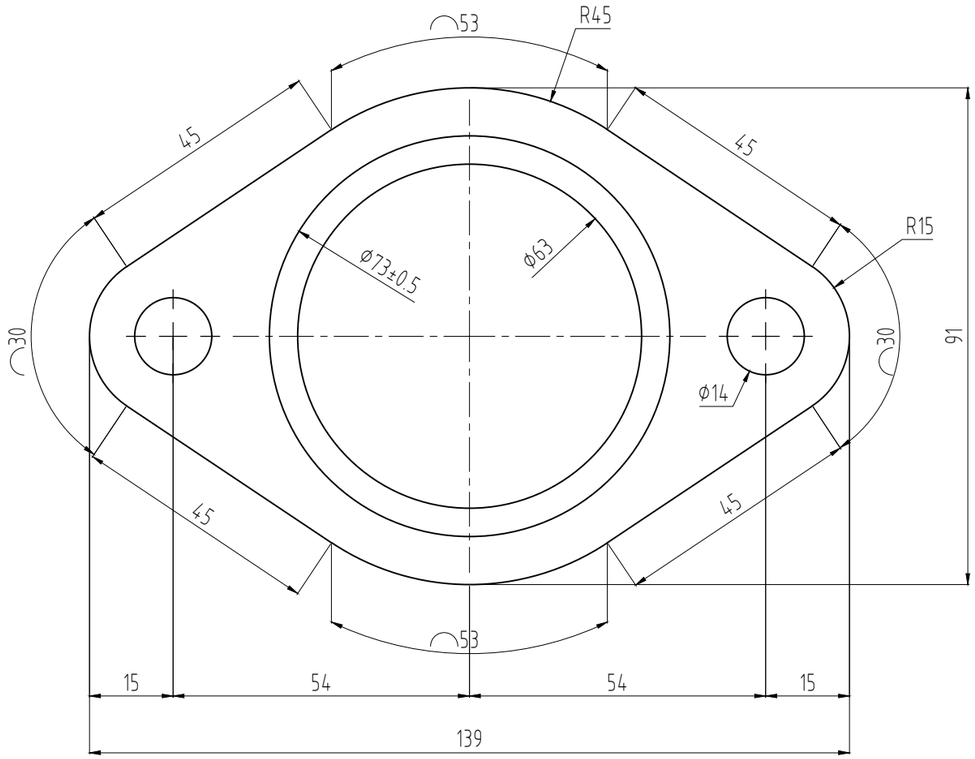
PROJECT JOB CODE: TP2887

TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

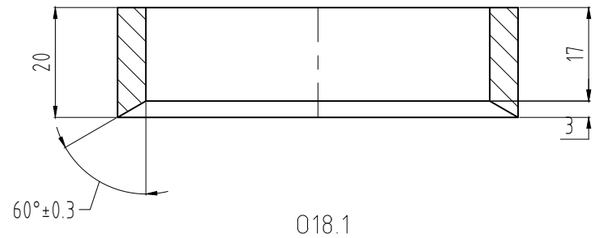
Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.19





P18.1
SUS304-6mm
Q'ty:2



O18.1
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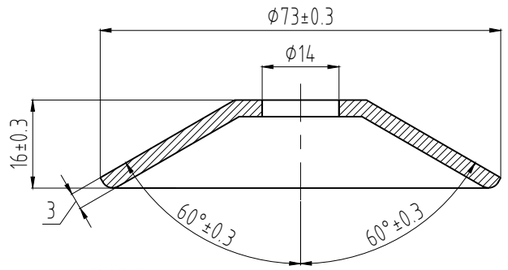
SL:01



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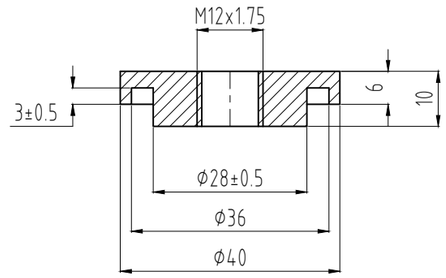
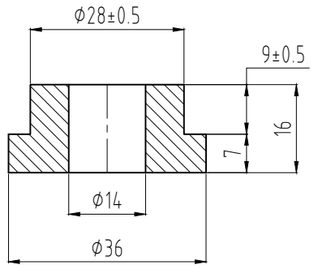
SUS304-16mm
Q'ty:01



P18.3
SUS304-16mm
Q'ty:01



P18.4
SUS304-16mm
Q'ty:01

Rev.	Date	Description	Drawn	Checked	Approved
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2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

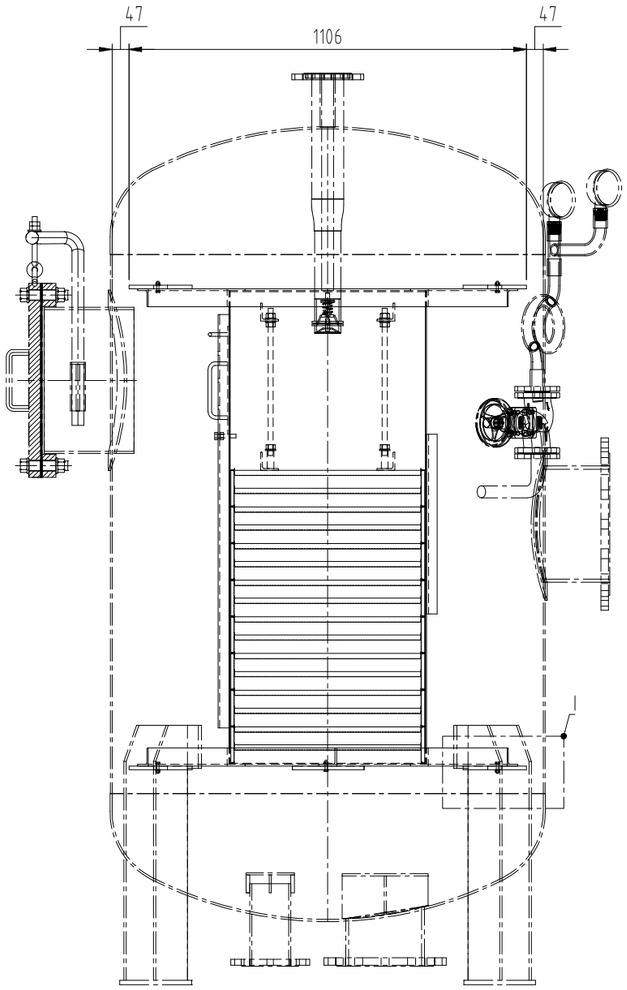
MANUFACTURER:
MARTECH BOILER The energy Solutions
MARTECH JOINT STOCK COMPANY

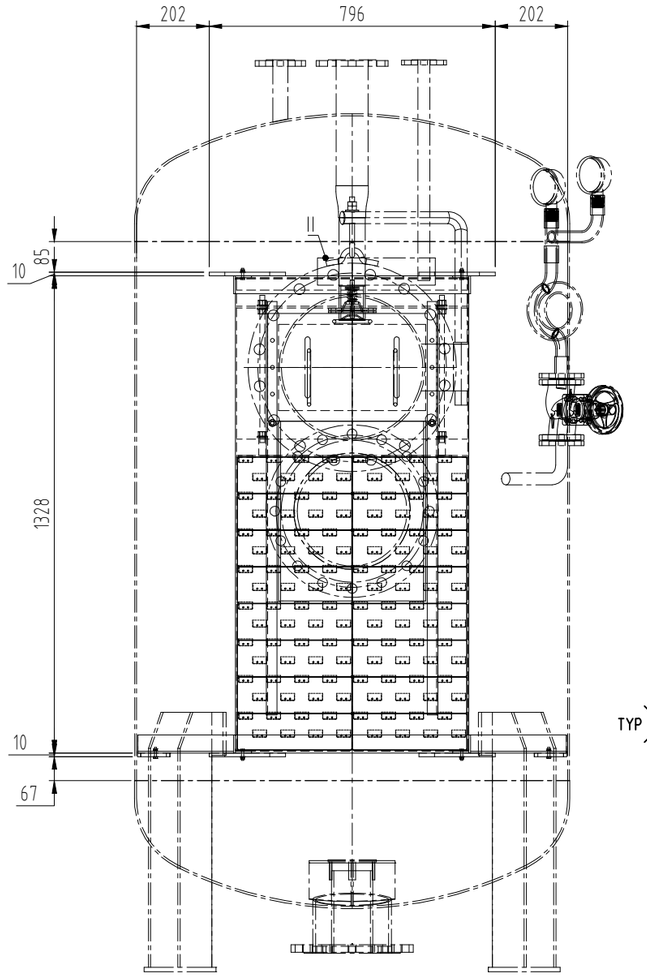
PROJECT JOB CODE: TP2887

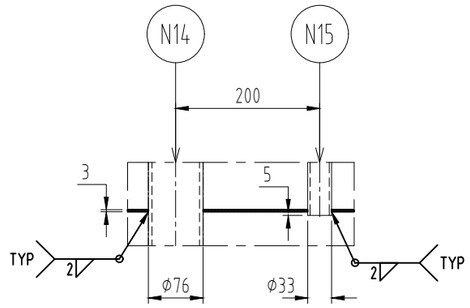
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

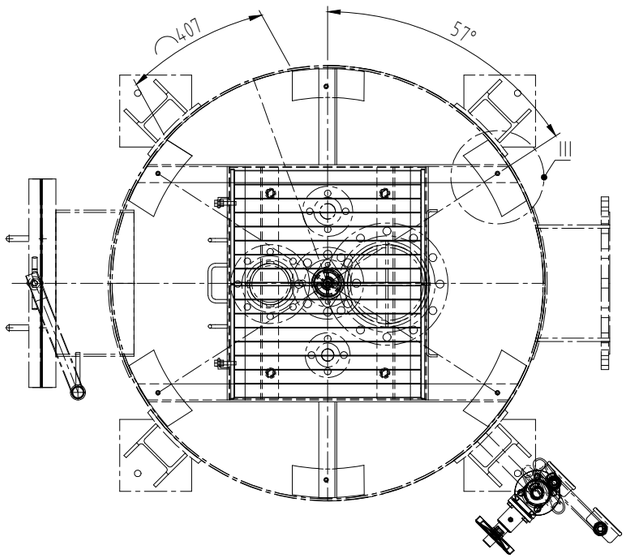
DRAWING NO.: 2887.0100.MDWG.20

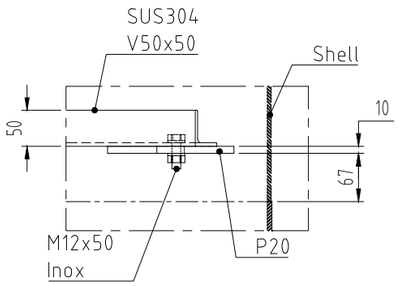






DETAIL II
SC:2/1





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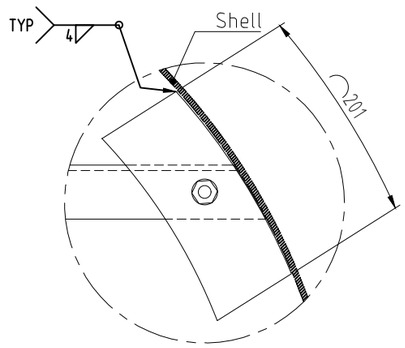
SC:2/1



DETAIL III
SC:3/1

Rev.	Date	Description	Drawn	Checked	Approved
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2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

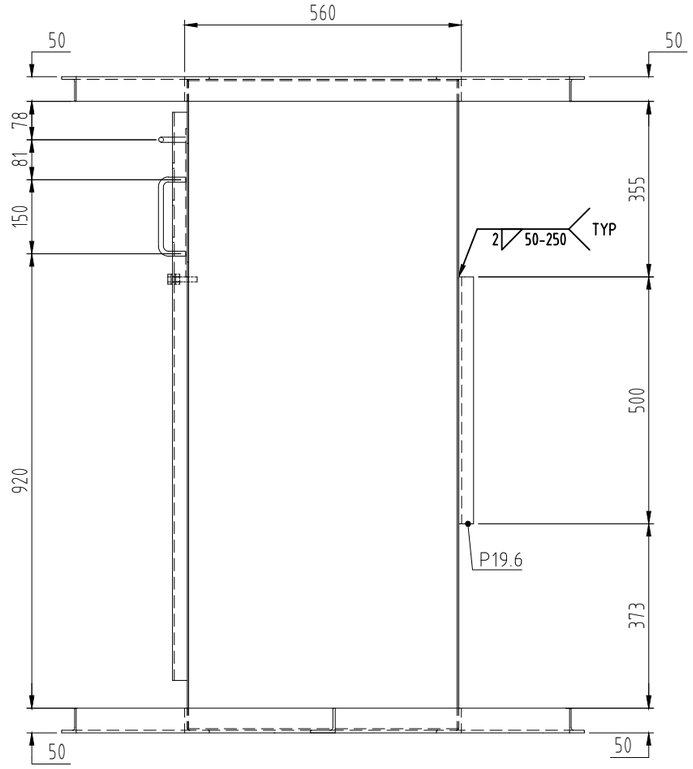
MANUFACTURER:
MARTECH BOILER
The Energy Solutions
MARTECH JOINT STOCK COMPANY

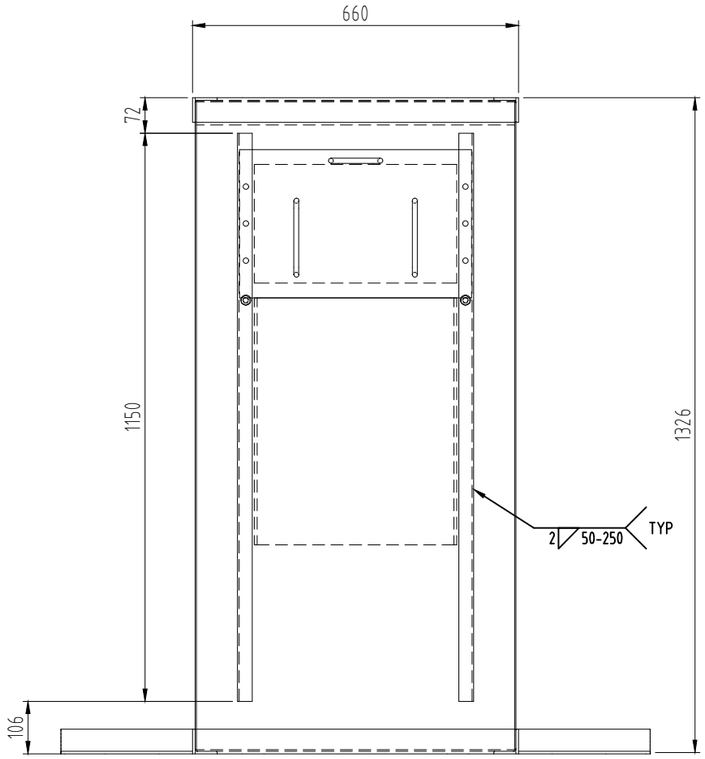
PROJECT JOB CODE: TP2887

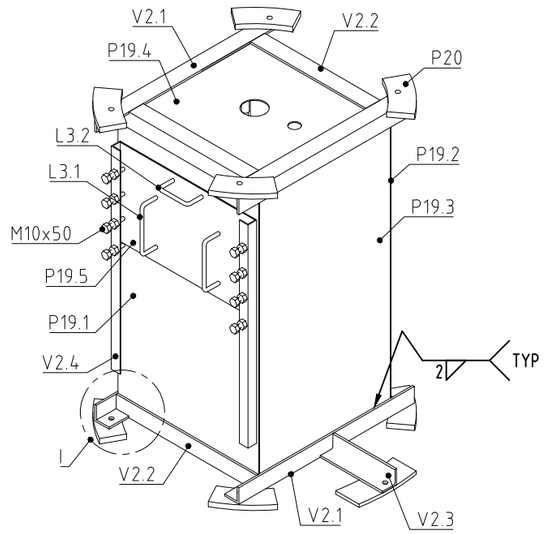
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

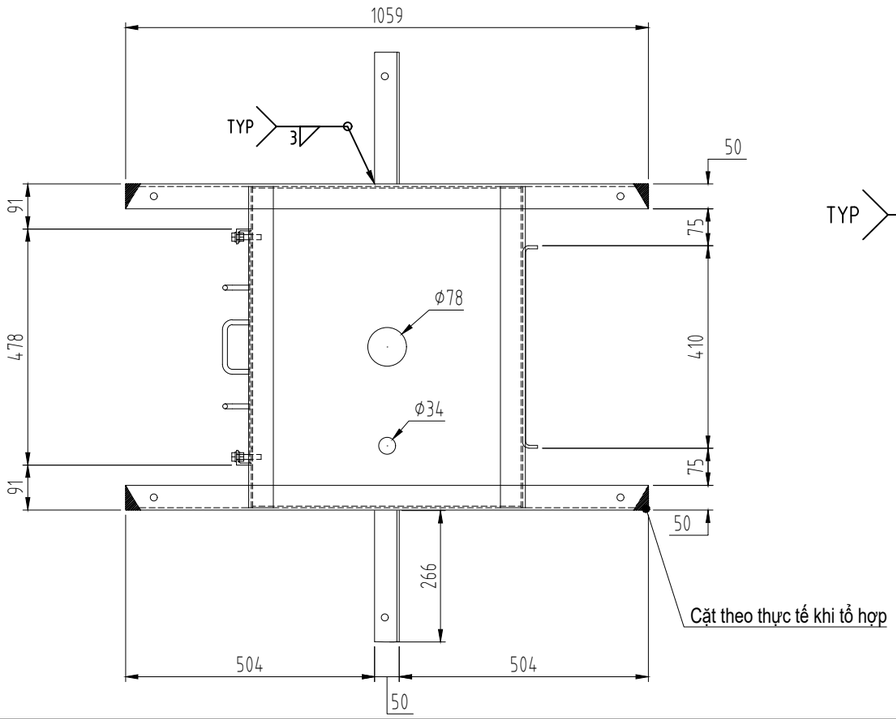
Scale	Size	Rev.
1:1	A3	00

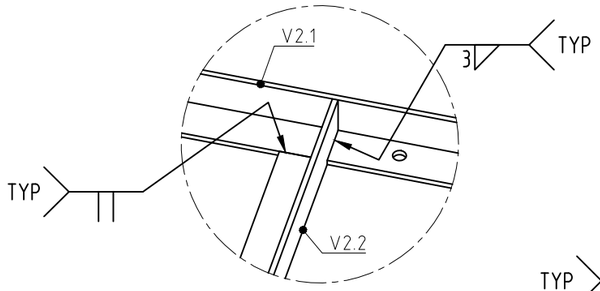
DRAWING NO.: 2887.0100.MDWG.21



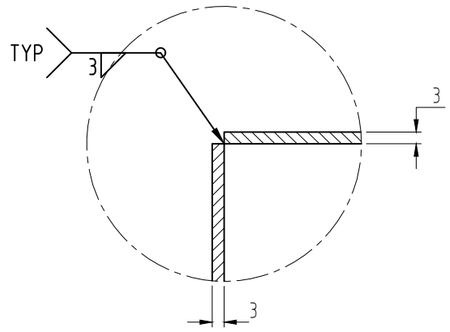








DETAIL I



Detail hàn tole vỏ bộ khử khí

Rev.	Date	Description	Drawn	Checked	Approved
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2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

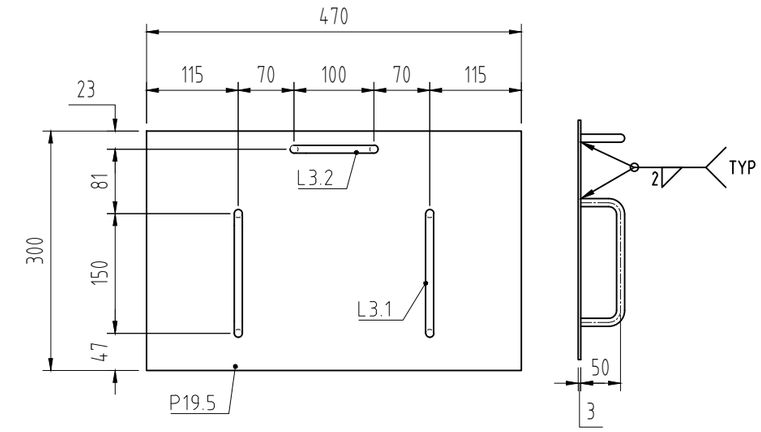
MANUFACTURER: MARTECH BOILER — MARTECH JOINT STOCK COMPANY

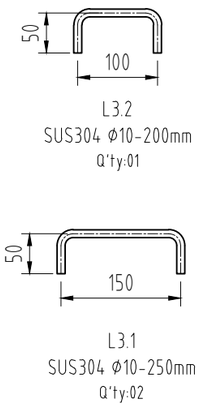
PROJECT JOB CODE: TP2887

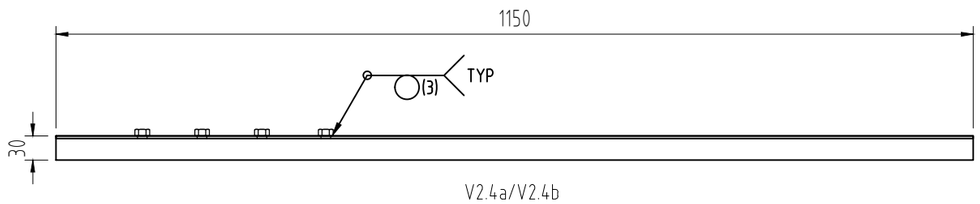
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

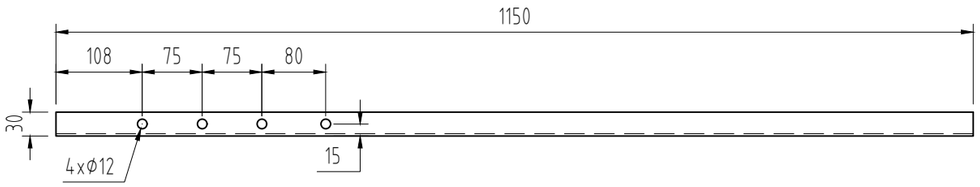
DRAWING NO.: 2887.0100.MDWG.22

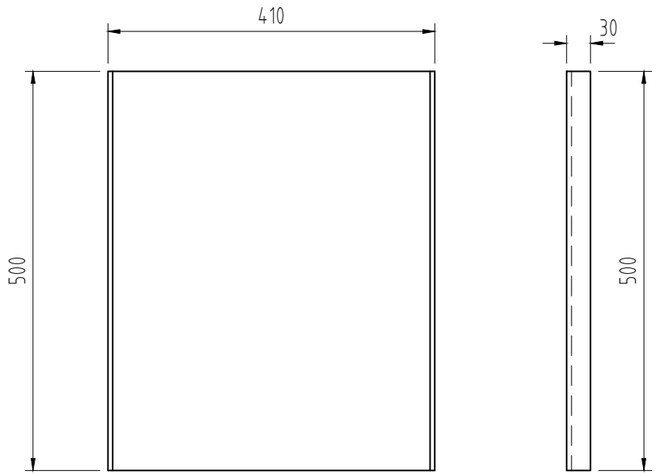




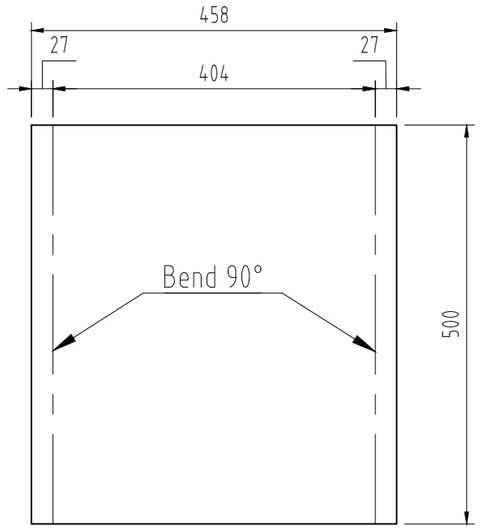


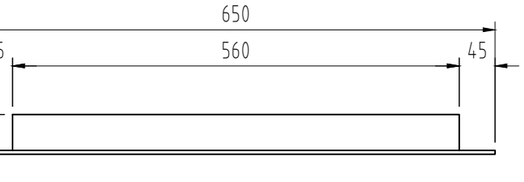
V2.4a/V2.4b
SUS304 V30x30x3-1150mm
Q'ty:02-1 cặp đối xứng



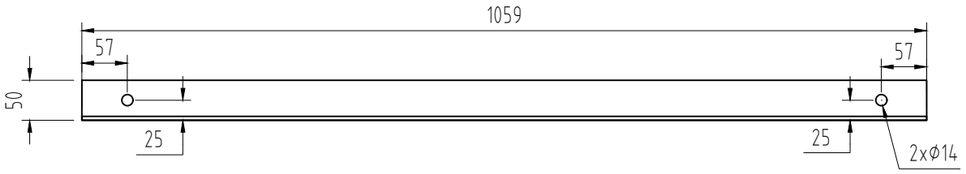


Expansion P19.6
A240 304-6mm
Q'ty:01





V2.2
SUS304 V50x50x5-650mm
Q'ty:04



V2.1
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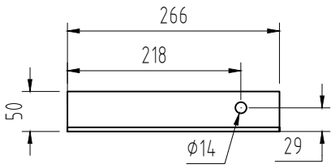
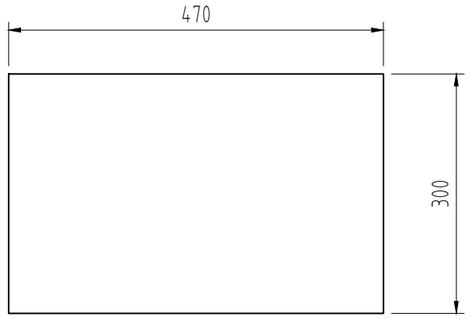


V2.3a/V2.3b
SUS304 V50x50x5-266mm
Q'ty:02-1 cặp đối xứng



P19.5
A240 304-3mm
Q'ty:01

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

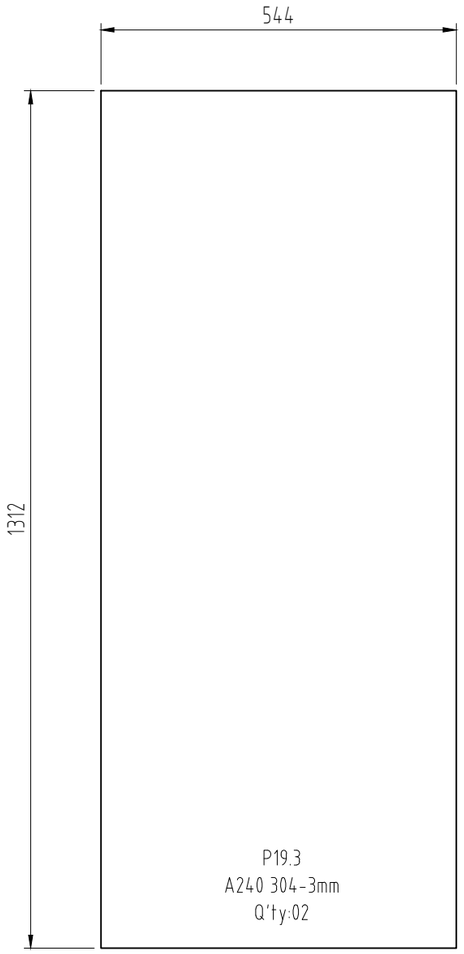
MANUFACTURER:
MARTECH BOILER
The Energy Solutions
MARTECH JOINT STOCK COMPANY

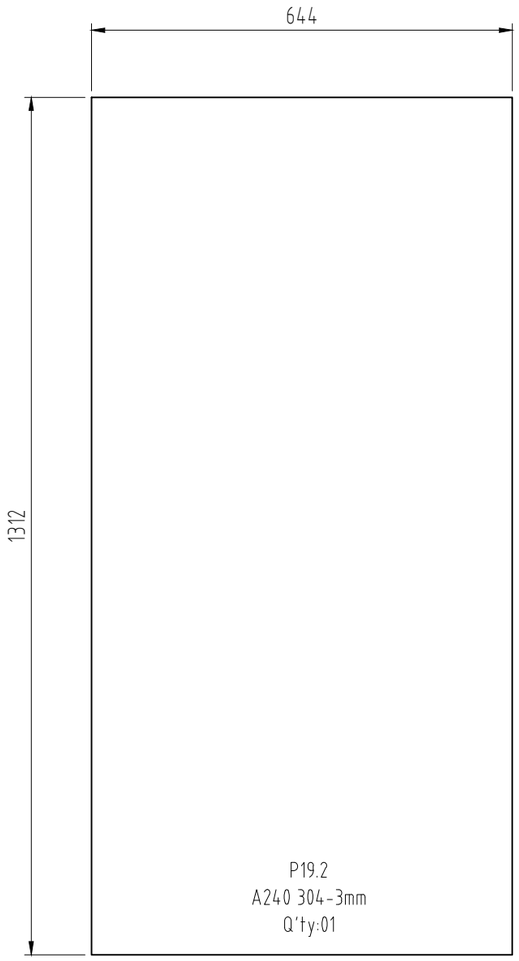
PROJECT JOB CODE: TP2887

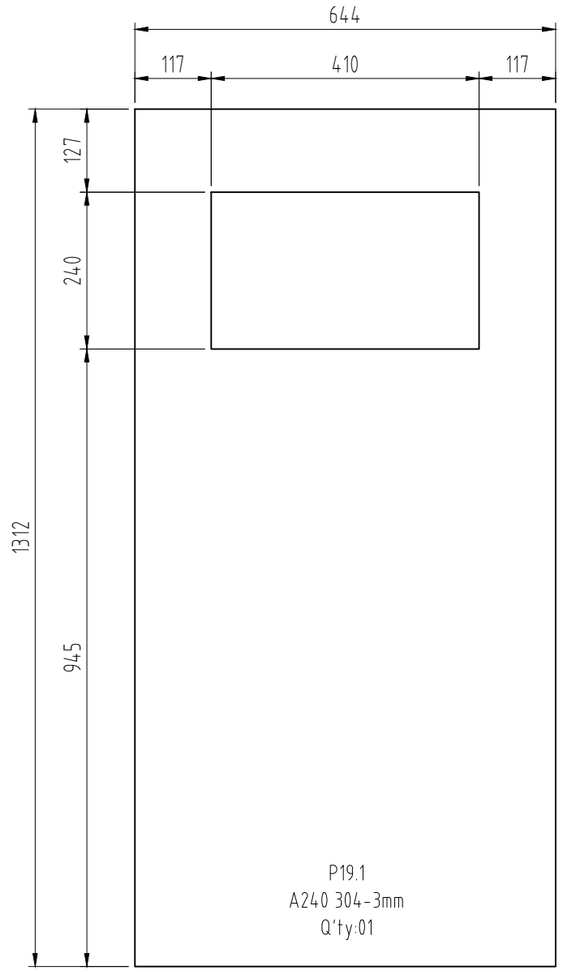
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

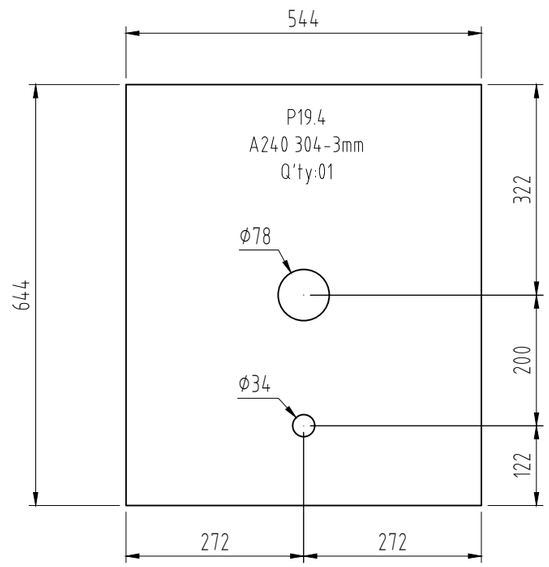
Scale	Size	Rev.
1:1	A3	00

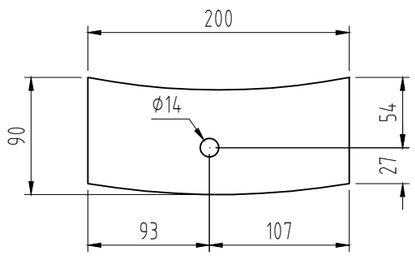
DRAWING NO.: 2887.0100.MDWG.23











P20
A240 304-10mm
SC:2/1-Q'ty:10

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:
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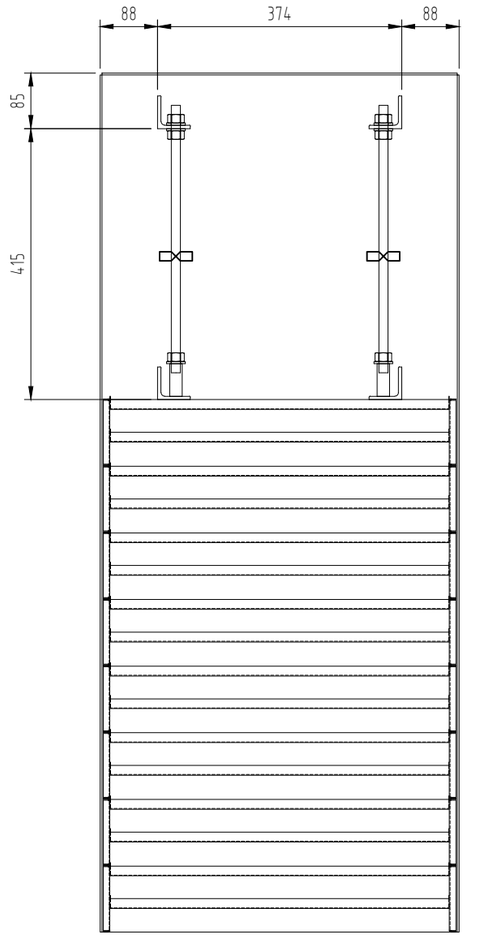
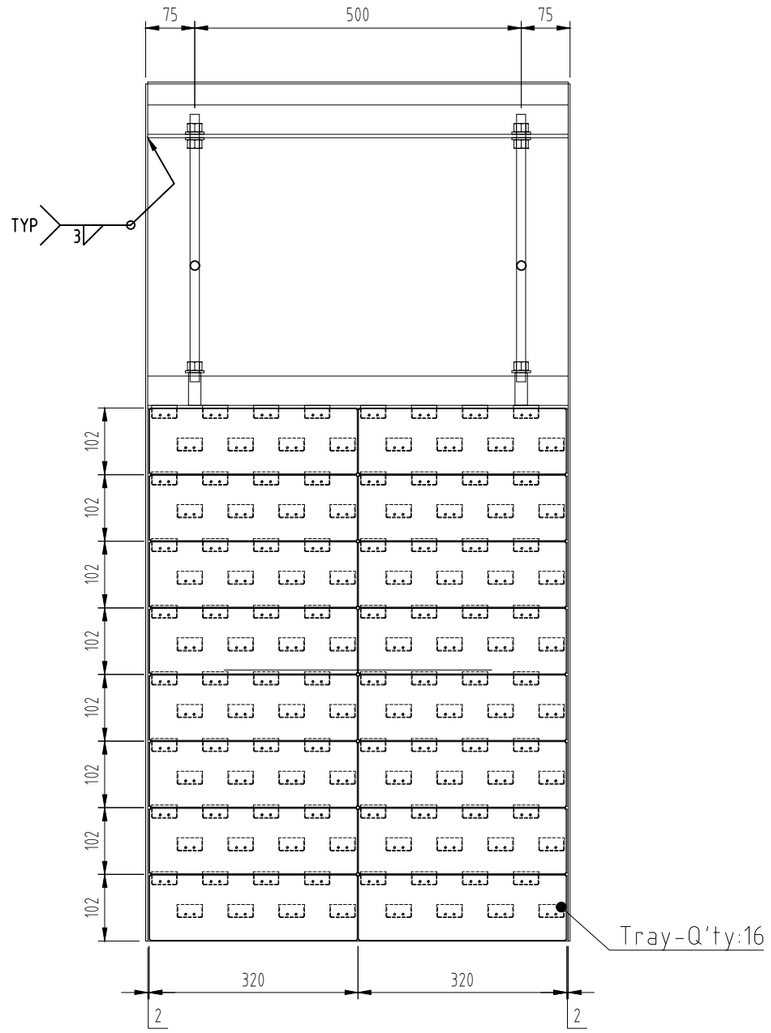
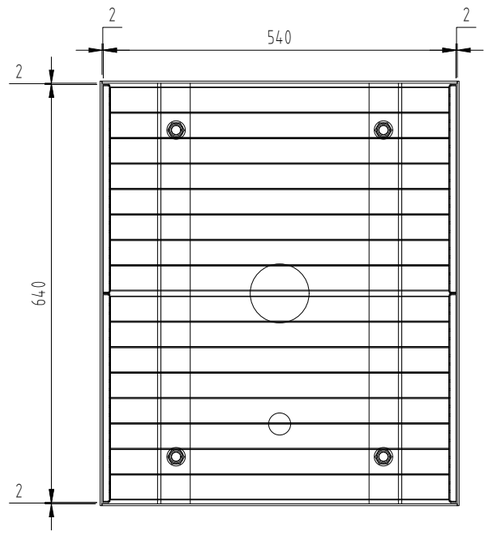
DEAERATOR AND WATER TANK

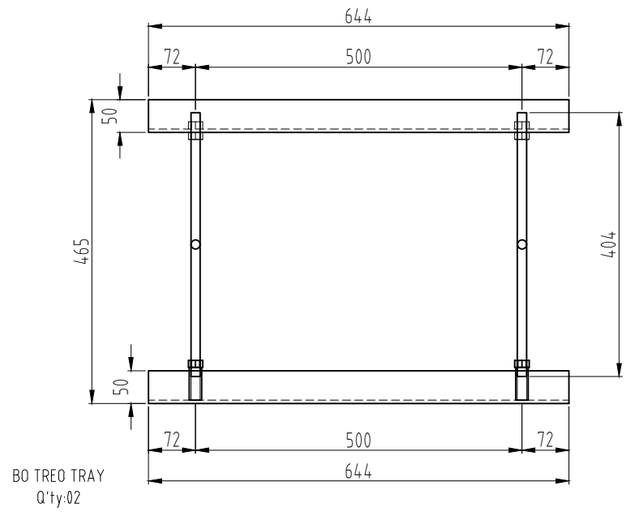
Scale	Size	Rev.
1:1	A3	00

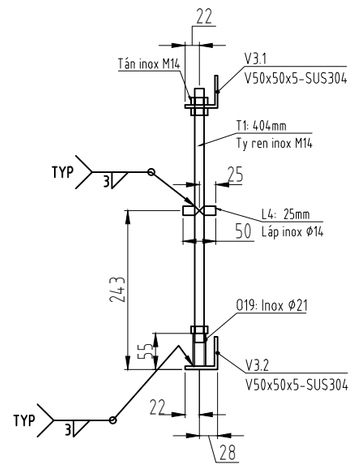
DRAWING NO.: 2887.0100.MDWG.24











3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong
Rev.	Date	Description	Drawn	Checked	Approved

CLIENT:

MANUFACTURER:

MARTECH BOILER The Energy Solutions

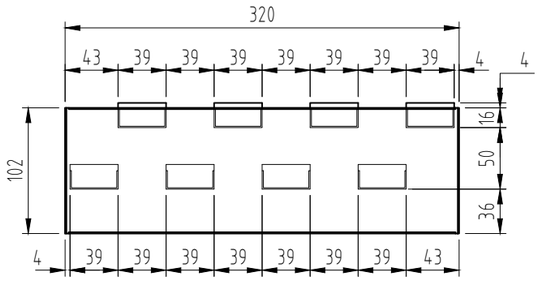
MARTECH JOINT STOCK COMPANY

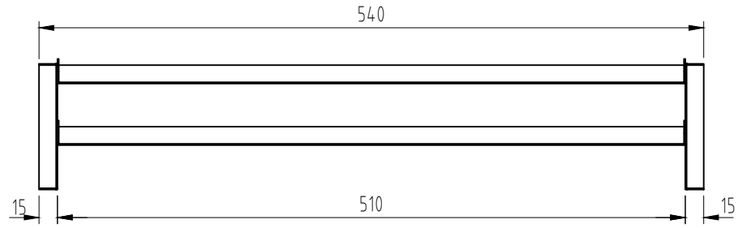
PROJECT JOB CODE: TP2887

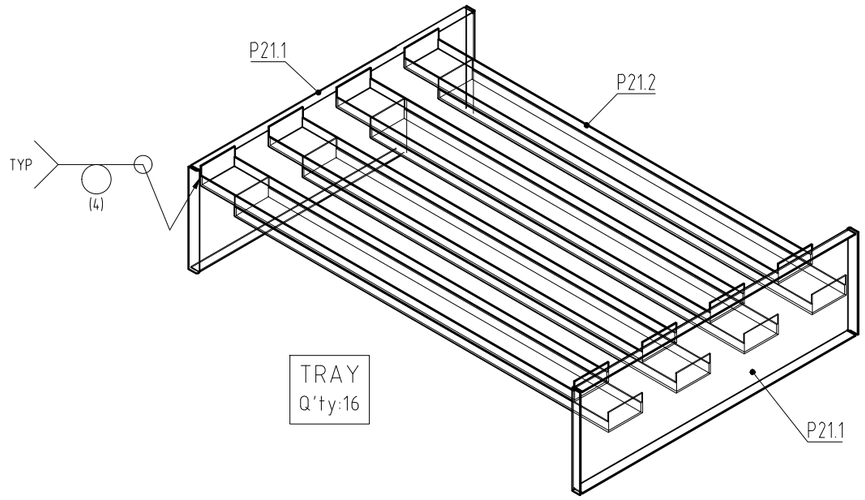
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

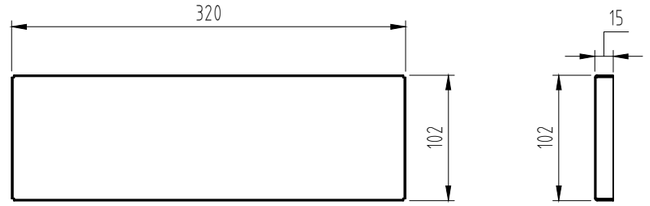
Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.25

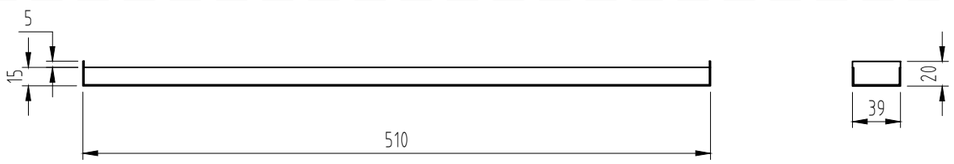




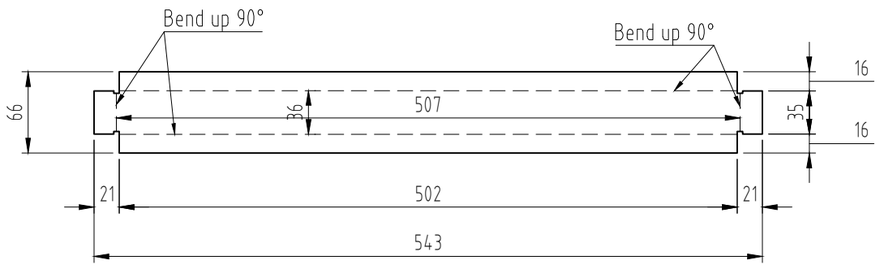




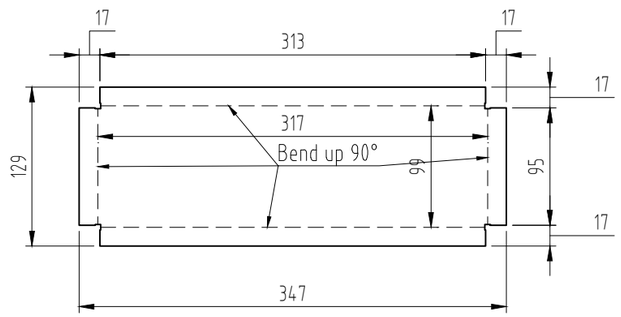
P21.1
A240 304-2mm
Q'ty:32



P21.2
A240 304-2mm
Q'ty:128



Expansion P21.2
A240 304-2mm
Q'ty:128



Expansion P21.1
A240 304-2mm
Q'ty:32

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

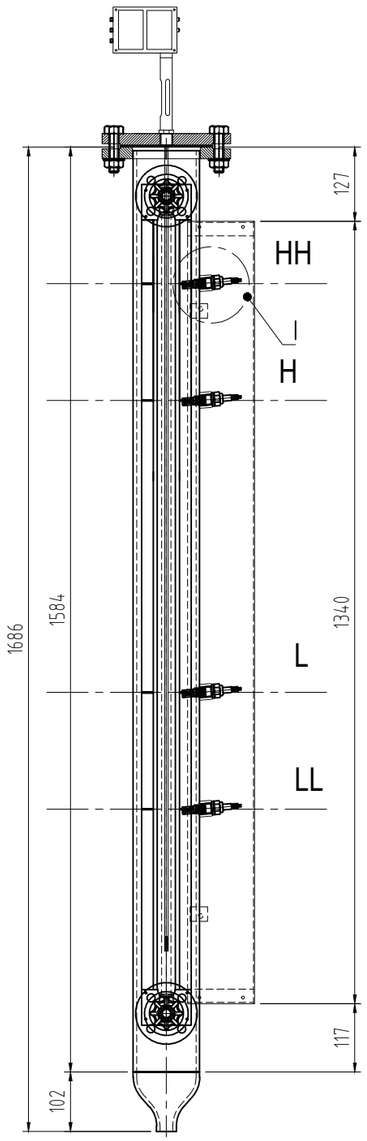
MANUFACTURER:
MARTECH BOILER The energy Solutions
MARTECH JOINT STOCK COMPANY

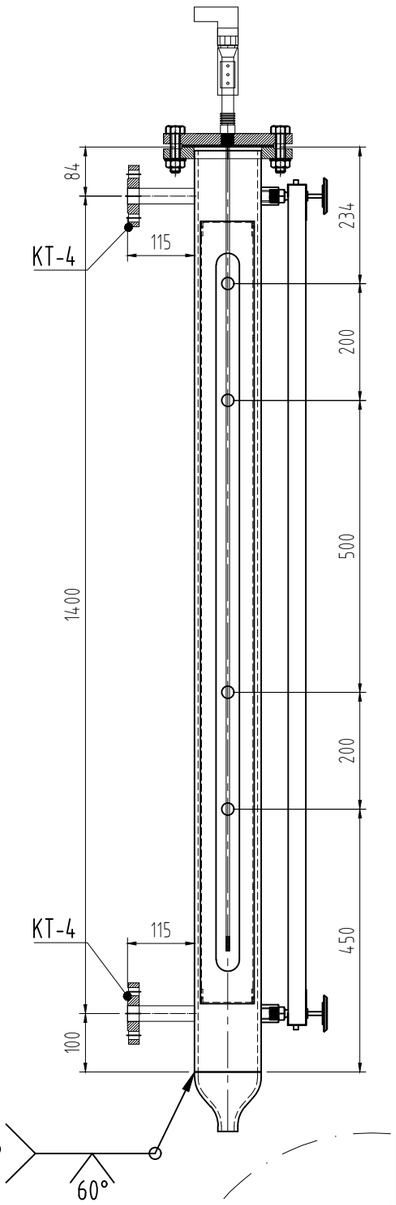
PROJECT JOB CODE: TP2887

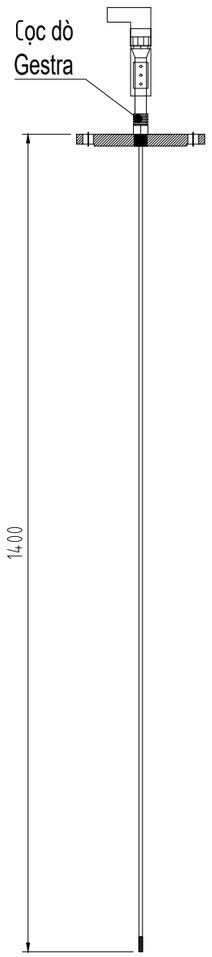
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

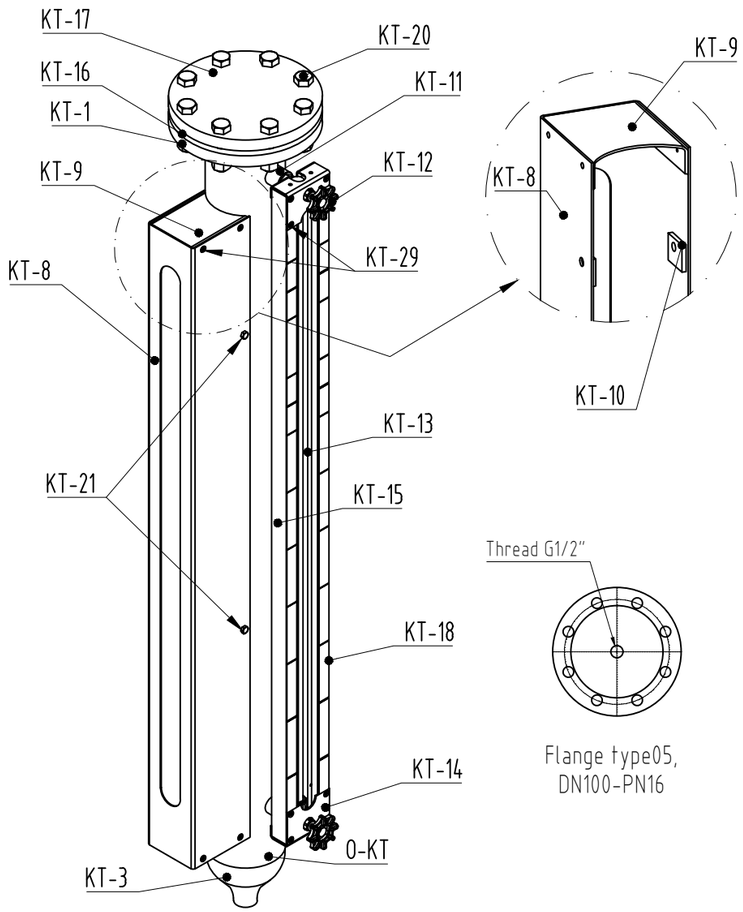
Scale	Size	Rev.
1:1	A3	00

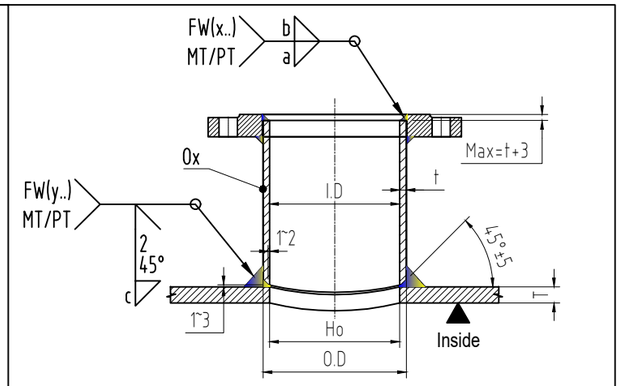
DRAWING NO.: 2887.0100.MDWG.26



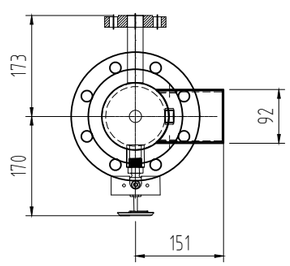


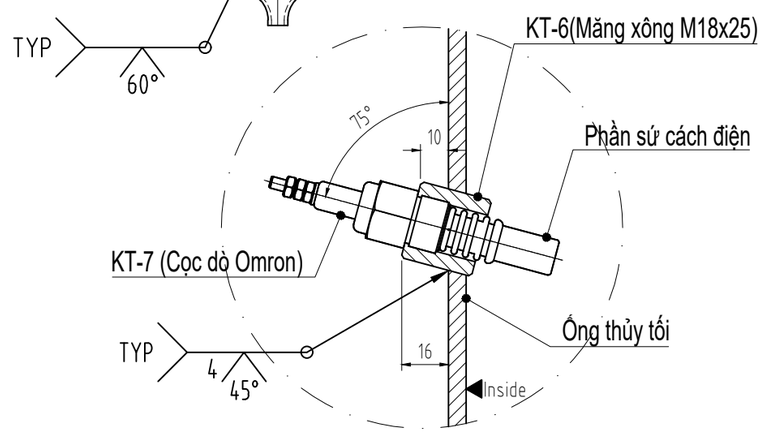






No	Symbol	Description	Detail Nozzle (mm)					No. Weld		Size weld		
			OD	t	ID	Ho	Ox	FW(x)	FW(y)	a	b	c
1	OKT2.1	Đường nối ống thủy tối	26.7	2.87	21.0	21	OKT2.1	1.1	1.2	2	4	4
2	OKT2.2	Đường nối ống thủy tối	26.7	2.87	21.0	21	OKT2.2	1.3	1.4	2	4	4
4	OKT1	Ống thủy tối	114.3	6.02	102.3	-	OKT1	1.5	-	5	6	-





Detail I
5:1

Cụm kính thủy-SL:1
Tagname: XXXXXX-2887-04

BẢNG KÊ CHI TIẾT CỤM KÍNH THỦY

STT	CHI TIẾT	MÔ TẢ	SỐ LƯỢNG	ĐƠN VỊ TÍNH	KHỐI LƯỢNG (Kg)
1	KT-1	Mặt bích Type 01, DN100 PN16	1	Cái	
2	OKT1	Ống đúc Ø114.3x6.02	1	Cái	
3	KT-3	Bầu giảm đúc đồng tâm, DN100x25 Sch40	1	Cái	
4	KT-4	Mặt bích Type 01, DN20 PN16	2	Cái	
5	OKT2.1	Ống đúc Ø26.7x2.87	1	Cái	
6	OKT2.2	Ống đúc Ø26.7x2.87	1	Cái	
7	KT-6	Măng xông ren suốt A-105, M18x25	4	Cái	
8	KT-7	Cọc dò Omron	4	Cái	
9	KT-8	Thép tấm A240 304, dày 2mm	1	Cái	
10	KT-9	Thép tấm A240 304, dày 2mm	2	Cái	
11	KT-10	Thép tấm SS400, dày 4mm	4	Cái	
12	KT-11	Măng xông ren suốt, G 1/2 x 30mm, Sch40	2	Cái	
13	KT-12	Van kính thủy sáng lắp ren Ø16	2	Cái	
14	KT-13	Kính thủy bồn nước Ø16	1	Cái	
15	KT-14	Thép tấm A240 304, dày 2mm	2	Cái	
16	KT-15	Thép tấm A240 304, dày 2mm	2	Cái	
17	KT-16	Roong mặt bích SW DN100 PN10/16, 316L FG, 316L IR, CS OR 4.5mmT	1	Cái	
18	KT-17	Mặt bích Type 05, DN100 PN16	1	Cái	
19	KT-18	Thước chia mức nước	2	Cái	
21	KT-19	Vít đầu dù đuôi cá M4.2x12mm	20	Con	
22	KT-20	Bulong M16x70mm	8	Bộ	
24	KT-21	Bulong M6x20mm	4	Bộ	

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

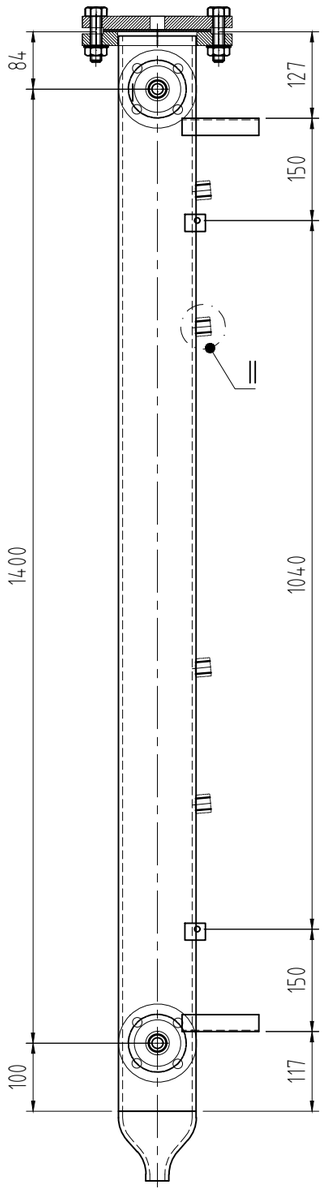
MANUFACTURER:
MARTECH BOILER
The Energy Solutions
MARTECH JOINT STOCK COMPANY

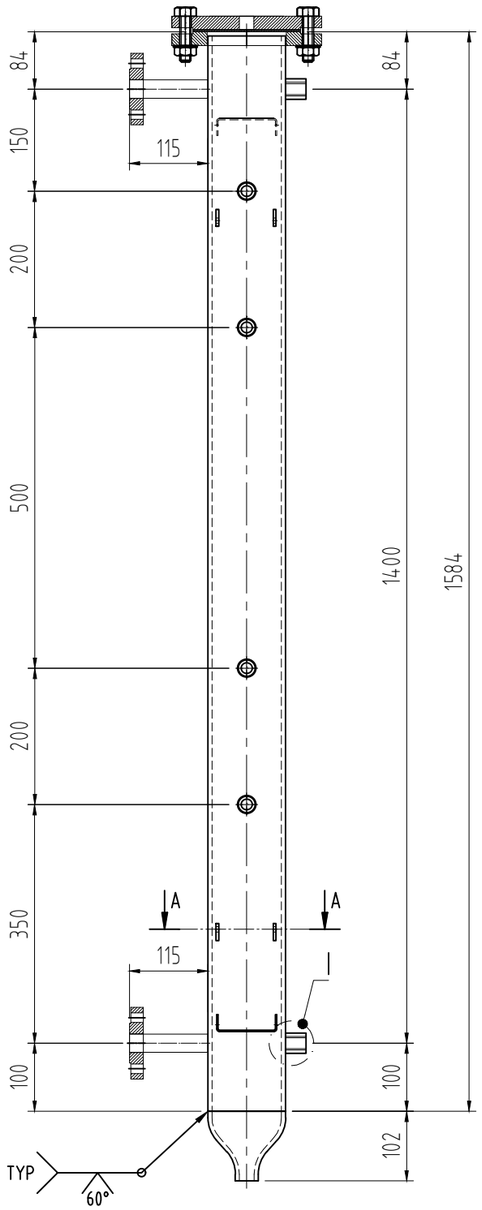
PROJECT JOB CODE:
TP2887

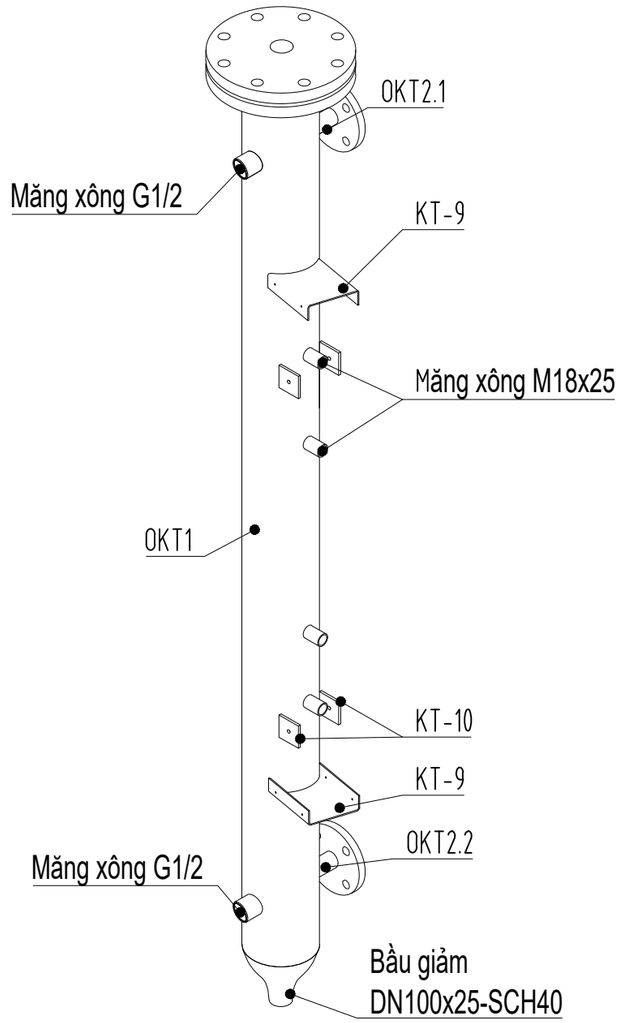
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

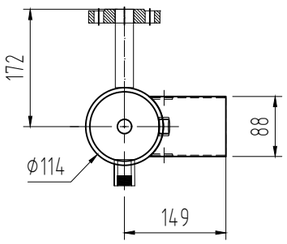
Scale	Size	Rev.
1:1	A3	00

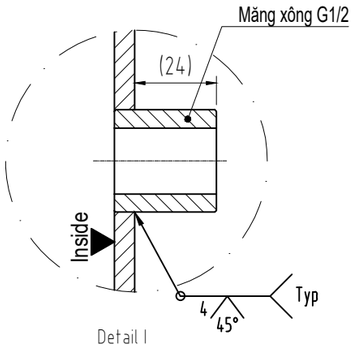
DRAWING NO.:
2887.0100.MDWG.27



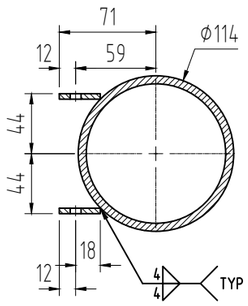




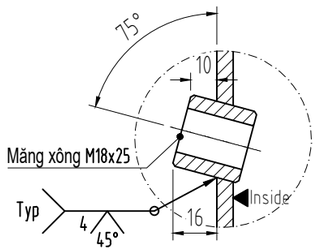




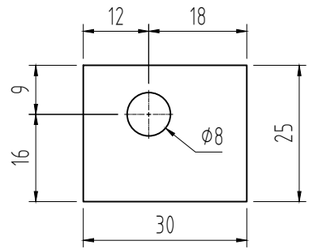
Detail I



Section A-A
Scale 2:1



Detail II



KT-10
VL: SS400/S(A)36-4mm
Q'ty:4-SC:8/1

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

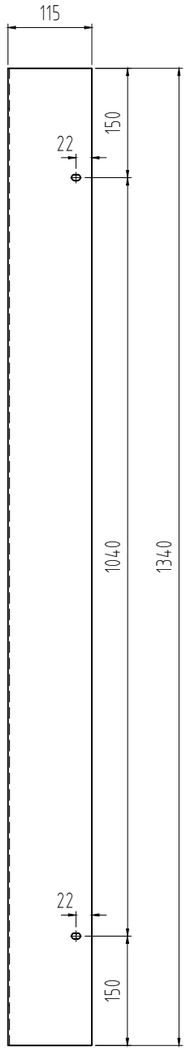
MANUFACTURER: MARTECH BOILER The energy Solutions — MARTECH JOINT STOCK COMPANY

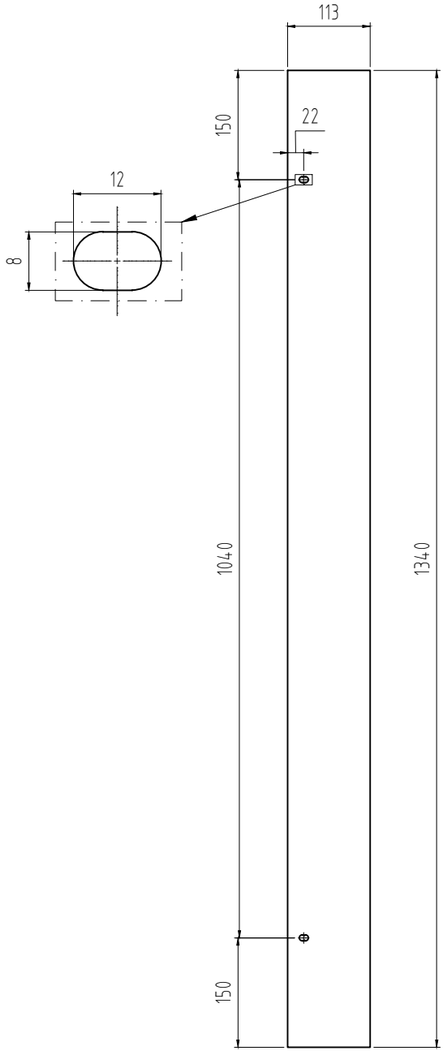
PROJECT JOB CODE: TP2887

TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

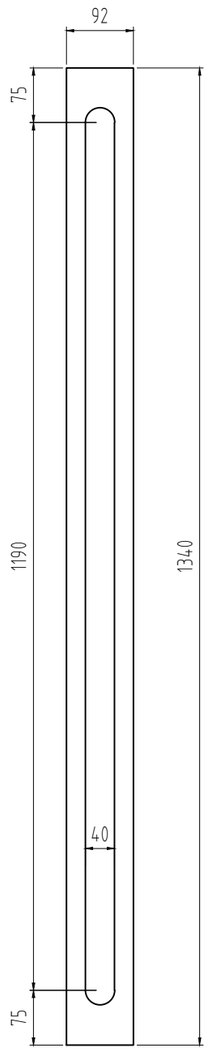
Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.28

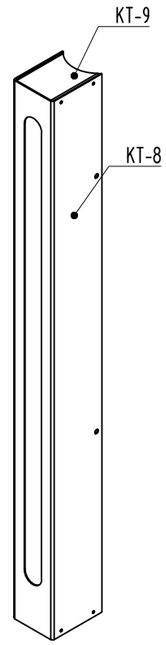


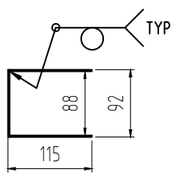


KT-8.1
VL: Thép tấm A240 304-2mm
SL: 2

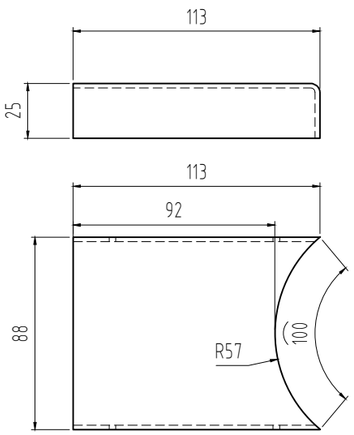


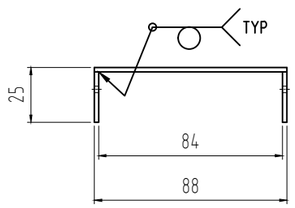
KT-8.2
VL: Thép tấm A240 304-2mm
SL: 1



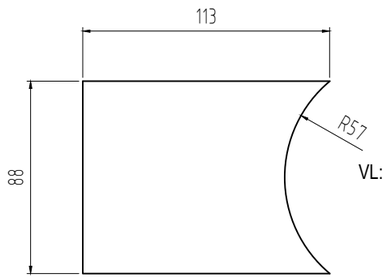


KT-8
VL: Thép tấm A240 304-2mm
SL: 1

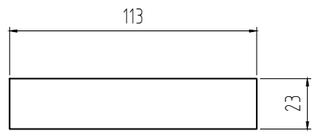




KT-9
VL: Thép tấm A240 304-2mm
SL: 2-Scale 3:1



KT-9.1
VL: Thép tấm A240 304-2mm
SL: 2-Scale 3:1



KT-9.2
VL: Thép tấm A240 304-2mm
SL: 4-Scale 3:1

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

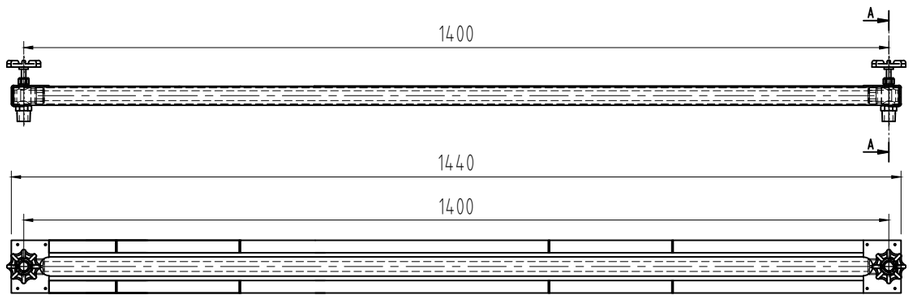
MANUFACTURER: MARTECH BOILER — MARTECH JOINT STOCK COMPANY

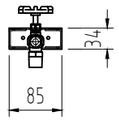
PROJECT JOB CODE: TP2887

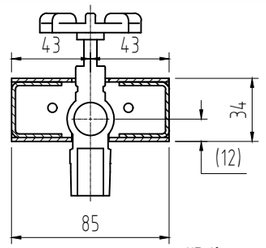
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

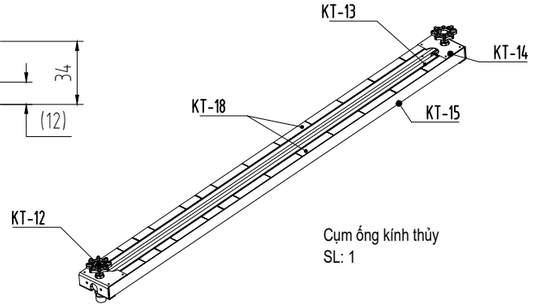
DRAWING NO.: 2887.0100.MDWG.29



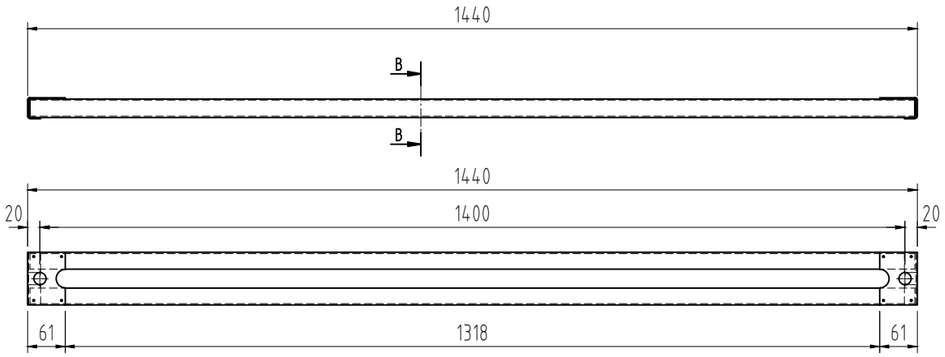


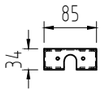


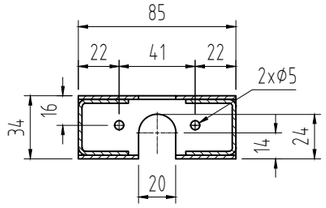
Section A-A
Scale 3:1



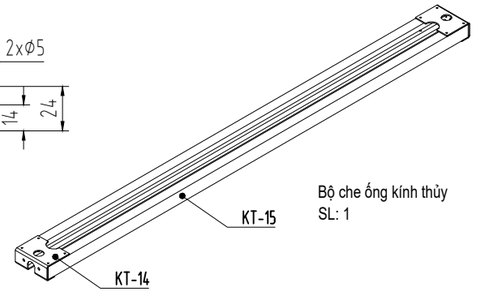
Cụm ống kính thủy
SL: 1



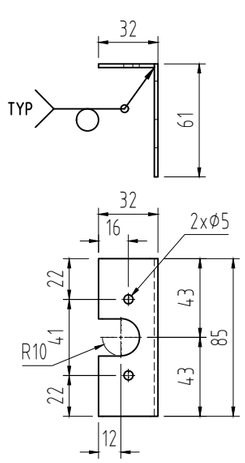


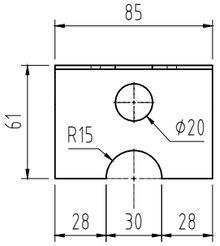


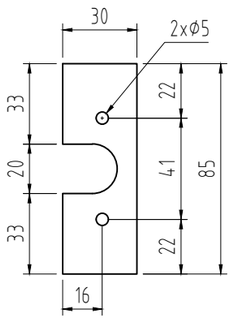
Section B-B
Scale 3:1



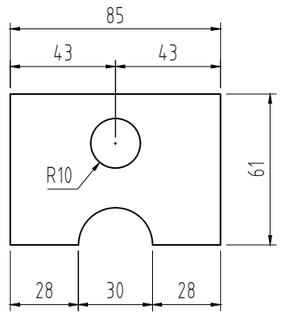
Bộ che ống kính thủy
SL: 1



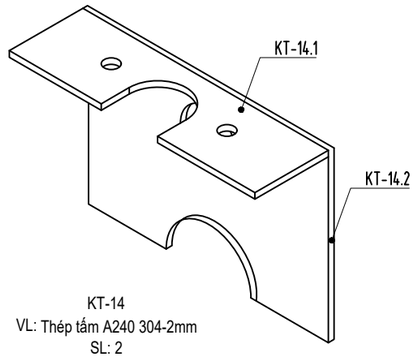




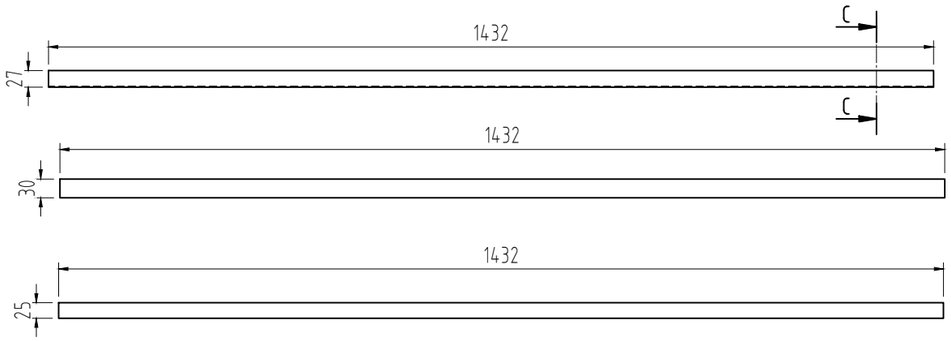
KT-14.1
A240 304-2mm
SL: 2-SC 4:1



KT-14.2
A240 304-2mm
SL: 2-SC 4:1

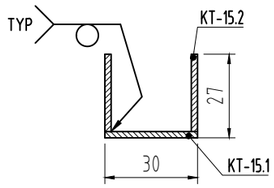


KT-14
VL: Thép tấm A240 304-2mm
SL: 2

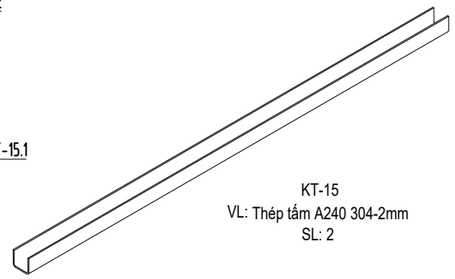


KT-15.1
A240 304-2mm
SL: 2

KT-15.2
A240 304-2mm
SL: 4



Section C-C
Scale 5:1



KT-15
VL: Thép tấm A240 304-2mm
SL: 2

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

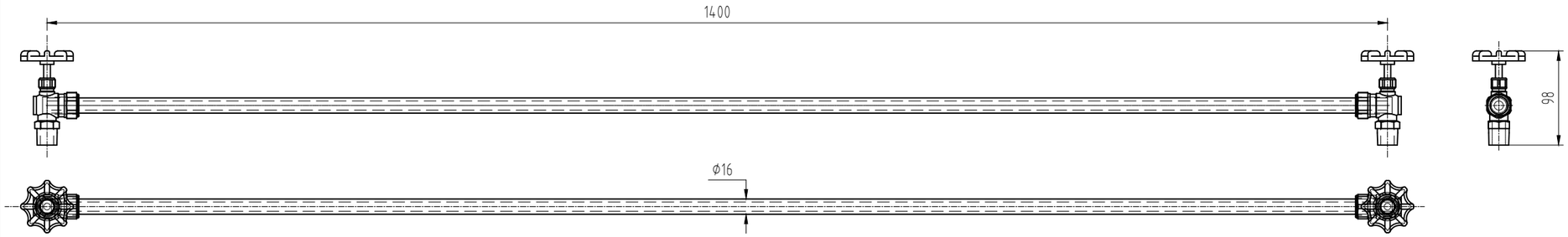
MANUFACTURER:
MARTECH BOILER
The Energy Solutions
MARTECH JOINT STOCK COMPANY

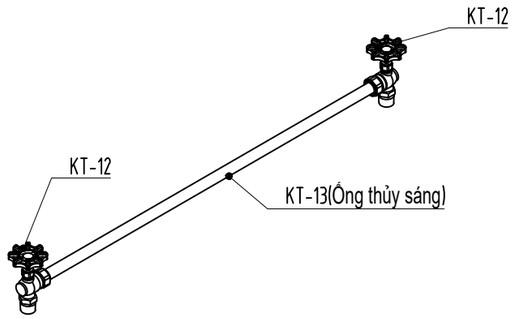
PROJECT JOB CODE:
TP2887

TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

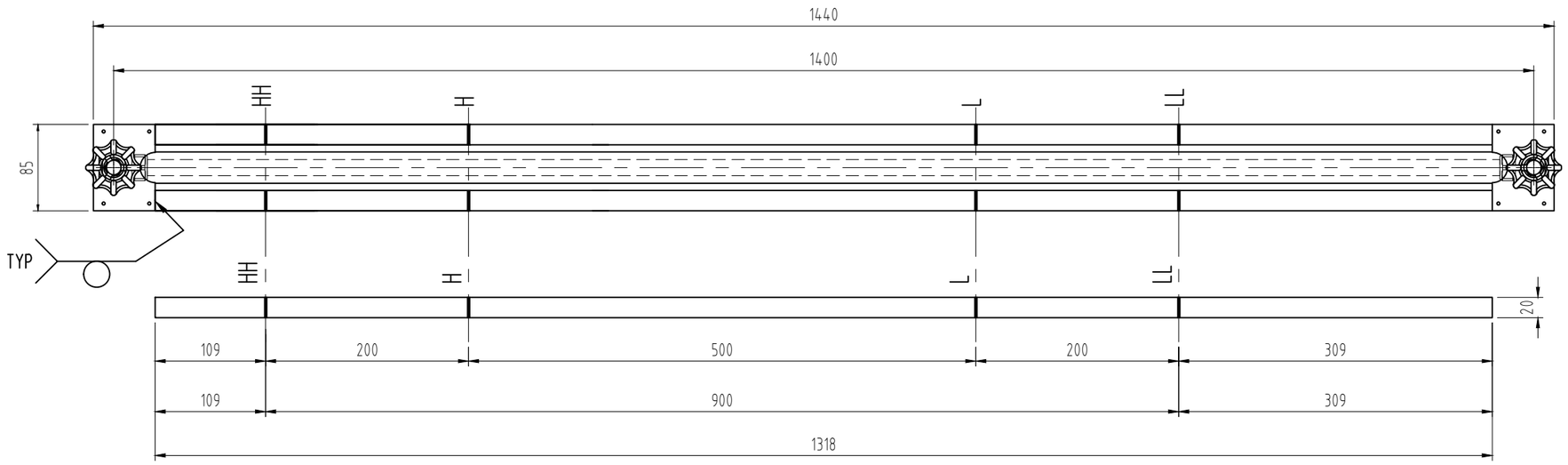
DRAWING NO.:
2887.0100.MDWG.30





Ghi chú:
Mức điều khiển:
- HH: Dừng cấp nước
- H : Cấp nước vào bồn
- L : Cảnh báo cạn
- LL : Dừng bơm nước cấp lò hơi

- Thước chia mức điều khiển 1318mm, 01 bộ gồm 2 thước đối xứng gắn 2 bên ống thủy sáng.
- Chế tạo hoàn thiện bộ ống thủy tại xưởng trước khi đưa đi công trình.



KT-18(Thước chia mức nước)
SL: 2

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:
MARTECH BOILER The energy Solutions
MARTECH JOINT STOCK COMPANY

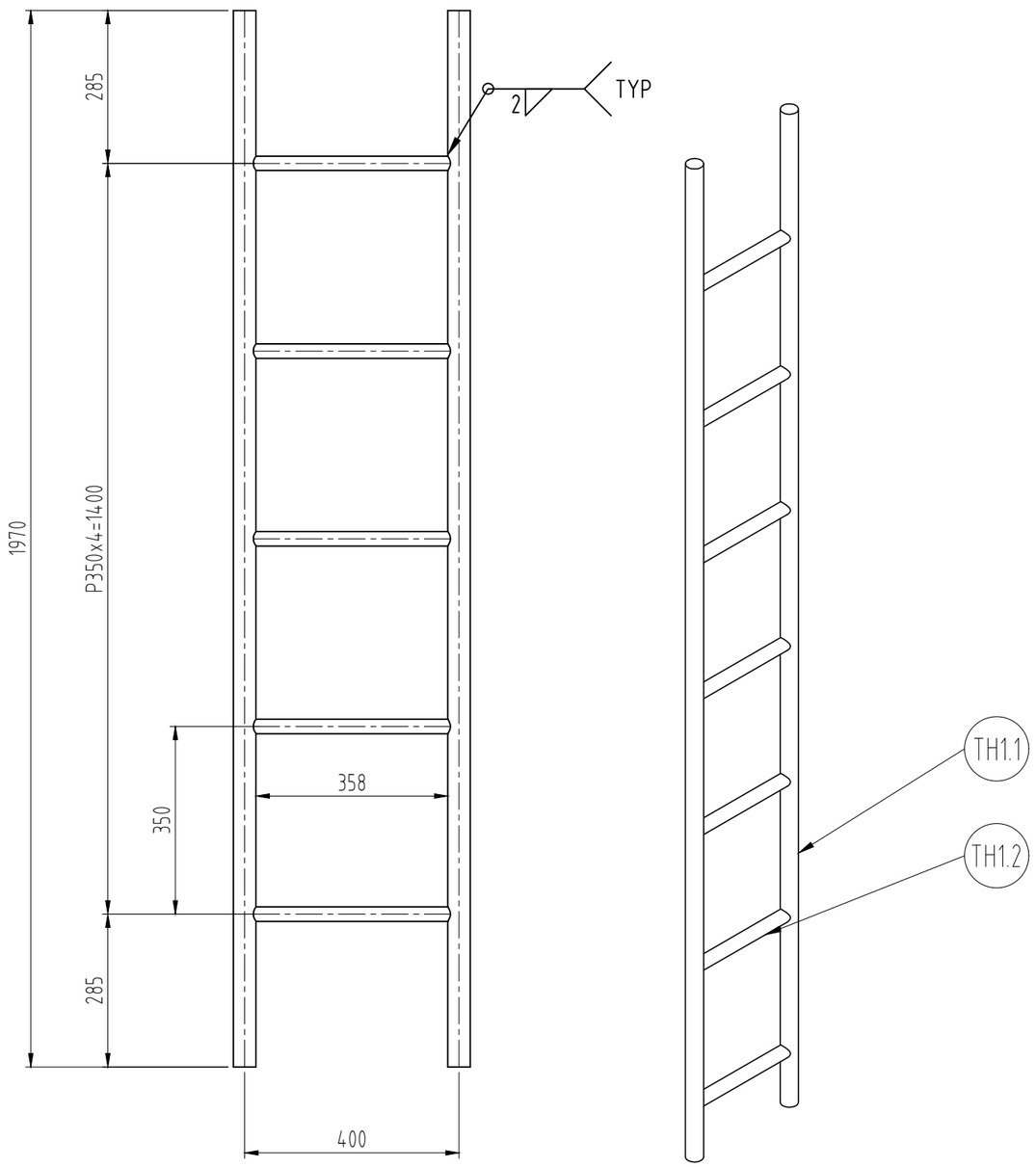
PROJECT JOB CODE: TP2887

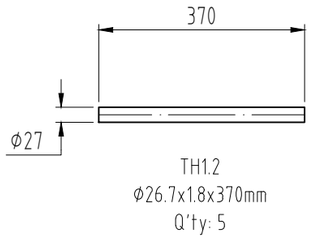
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

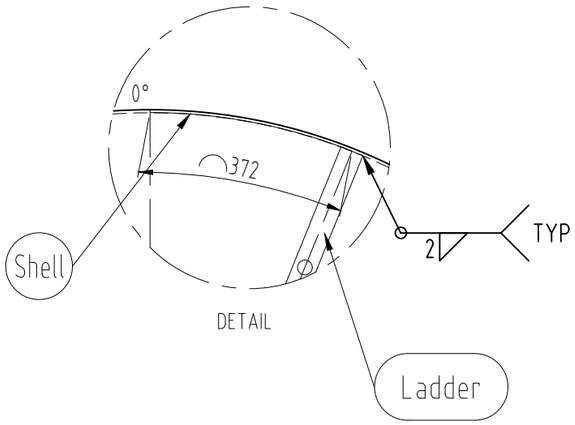
Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.31

LADDER
Q'ty: 1







Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

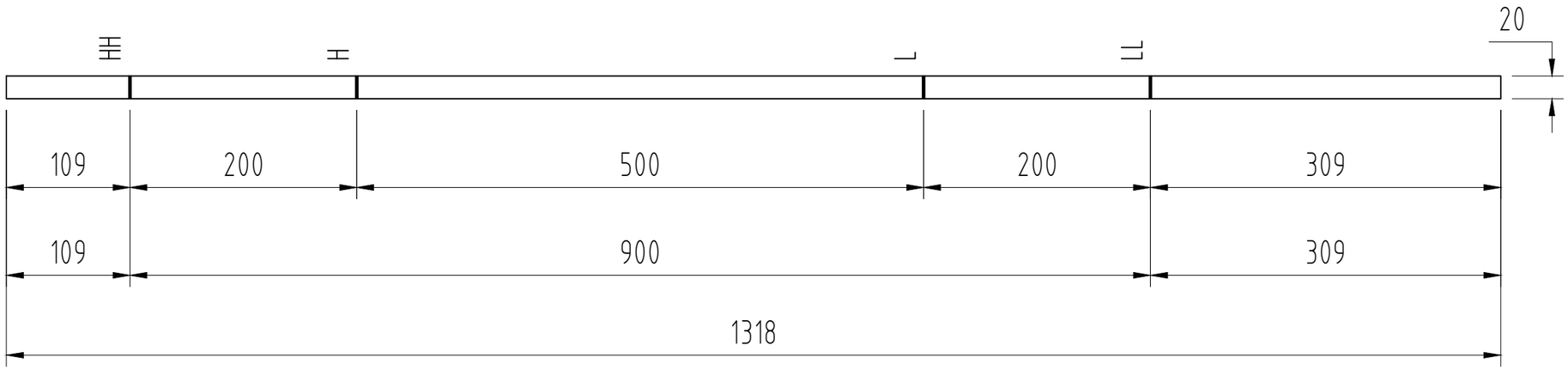
MANUFACTURER: MARTECH BOILER The energy Solutions — MARTECH JOINT STOCK COMPANY

PROJECT JOB CODE: TP2887

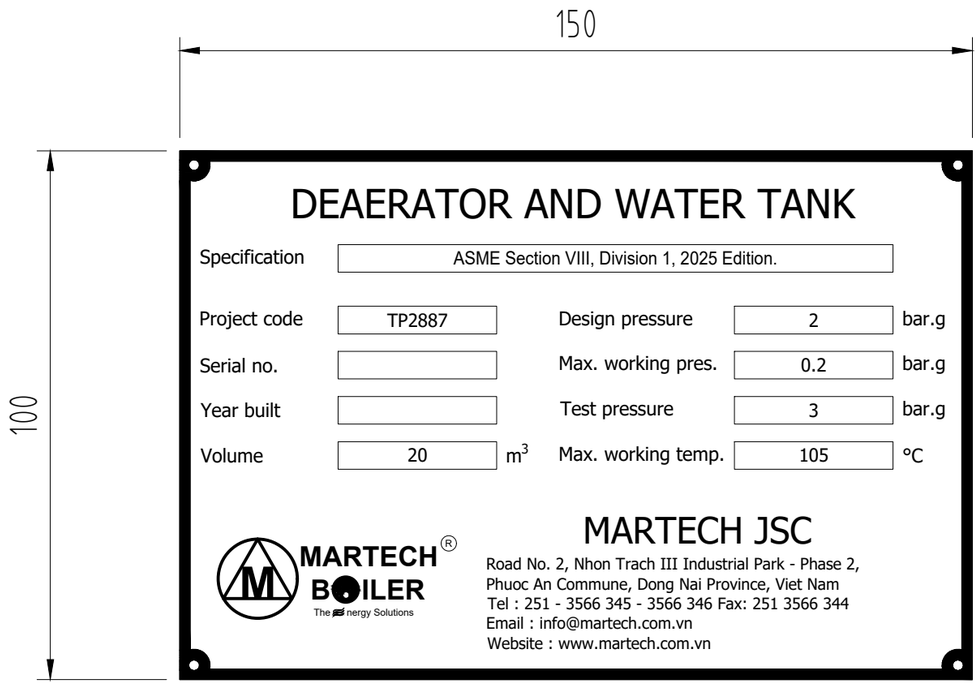
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.32



KT-18(Thước chia mức nước)
SL: 2



NAMEPLATE
Q'ty:1

BẢN VẼ ĐẶT HÀNG

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:
MARTECH BOILER The energy Solutions
MARTECH JOINT STOCK COMPANY

PROJECT JOB CODE: TP2887

TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.33

PLATE LIST

Symbol	Material Decripsion		Specifications (mm)			Q.ty	Area (m2)	Mass (KG)	Paint area
	Type of material	Material	Thickness	Length	Width				
P1.1	Plate steel	SS400	6	1960	490	2	0.7	63.1	0.7
P1.2	Plate steel	SS400	6	2450	1960	2	4.0	376.8	4.0
P2.1	Plate steel	SS400	5	5960	1500	4	8.9	1403.6	8.9
P2.2	Plate steel	SS400	5	245	1500	4	0.4	57.7	0.4
DUONG 1	Plate steel	SS400	5	2050	580	1	1.2	46.7	1.2
DUONG 2	Plate steel	SS400	5	1280	390	1	0.5	19.6	0.5
P3	Plate steel	SS400	6	1550	1550	2	1.8	169.6	1.8
P4	Plate steel	SS400	5	3786	1500	1	5.7	222.9	5.7
P5.1	Plate steel	A36	16	170	160	2	0.0	6.8	0.0
P5.2	Plate steel	SS400	10	200	100	2	0.0	3.1	0.0
P6.1	Plate steel	SS400	8	1160	200	2	0.2	29.1	0.2
P6.2	Plate steel	SS400	6	1763	670	2	0.4	37.7	0.4
P6.3	Plate steel	SS400	4	2030	200	2	0.4	25.5	0.4
P6.4	Plate steel	SS400	6	200	97	4	0.0	3.7	0.0
P6.5	Plate steel	SS400	6	230	97	8	0.0	8.4	0.0
P6.6	Plate steel	SS400	6	373	97	8	0.0	13.6	0.0
P7.1	Plate steel	SS400	10	1350	150	3	0.2	47.7	0.2
P7.2	Plate steel	SS400	10	1711	719	3	0.5	117.8	0.5
P7.3	Plate steel	SS400	4	2338	250	3	0.6	55.1	0.6
P7.4	Plate steel	SS400	6	765	150	6	0.1	32.4	0.1
P7.5	Plate steel	SS400	6	240	70	6	0.0	4.7	0.0
P7.6	Plate steel	SS400	6	265	70	12	0.0	10.5	0.0
P7.7	Plate steel	SS400	6	334	70	12	0.0	13.2	0.0
P8.1	Plate steel	SS400	6	512	384	3	0.2	27.6	0.2
P8.2	Plate steel	SS400	6	990	591	3	0.6	82.7	0.6
P9.1	Plate steel	SS400	6	700	357	2	0.2	23.5	0.2
P9.2	Plate steel	SS400	6	700	349	2	0.2	23.0	0.2
P10	Plate steel	SS400	4	396	120	1	0.0	1.5	0.0
P11.1	Plate steel	SS400	12	203	120	4	0.0	9.2	0.0
P11.3	Plate steel	SS400	12	200	200	4	0.0	15.1	0.0
P12.1	Plate steel	SS400	6	220	100	2	0.0	2.1	0.0
P12.2	Plate steel	SS400	12	229	130	2	0.0	5.6	0.0
P13.1	Plate steel	SS400	6	50	10	4	0.0	0.1	0.0
P13.2	Plate steel	SS400	6	134	134	1	0.0	0.8	0.0
P14.1	Plate steel	SS400	6	239	119	1	0.0	1.3	0.0
P14.2	Plate steel	SS400	6	117	109	2	0.0	1.2	0.0
P14.3	Plate steel	SS400	6	239	239	1	0.1	2.7	0.1
P15.1	Plate steel	SS400	6	134	134	1	0.0	0.8	0.0
P15.2	Plate steel	SS400	6	134	60	1	0.0	0.4	0.0
P15.3	Plate steel	SS400	6	64	60	2	0.0	0.4	0.0
P16.1	Plate steel	SS400	6	330	330	1	0.1	5.1	0.1
P16.2	Plate steel	SS400	10	202	202	1	0.0	3.2	0.0
P17	Plate steel	SS400	6	102	102	1	0.0	0.5	0.0
P18.1	Plate steel	A240 304	6	139	91	2	0.0	1.2	0.0
P18.2	Plate steel	A240 304	6	30	5	4	0.0	0.0	0.0
P18.3	Plate steel	A240 304	16	40	40	1	0.0	0.2	0.0
P18.4	Plate steel	A240 304	16	36	36	1	0.0	0.2	0.0
P18.5	Plate steel	A240 304	16	73	73	1	0.0	0.7	0.0
P19.1	Plate steel	A240 304	3	644	1312	1	0.8	19.9	0.0
P19.2	Plate steel	A240 304	3	644	1312	1	0.8	19.9	0.0
P19.3	Plate steel	A240 304	3	544	1312	2	0.7	33.6	0.0
P19.4	Plate steel	A240 304	3	644	544	1	0.4	8.3	0.0
P19.5	Plate steel	A240 304	3	470	300	1	0.1	3.3	0.0
P19.6	Plate steel	A240 304	6	458	500	1	0.2	10.8	0.0
P20	Plate steel	A240 304	10	200	90	10	0.0	14.1	0.0
P21.1	Plate steel	A240 304	1	350	132	32	0.0	11.6	0.0
P21.2	Plate steel	A240 304	1	551	69	128	0.0	38.2	0.0
P22	Plate steel	SS400	6	434	446	1	0.2	9.1	0.2
KT10	Plate steel	SS400	4	30	25	4	0.0	0.1	0.0
KT8.1	Plate steel	A240 304	2	1340	113	2	0.3	4.8	0.0
KT8.2	Plate steel	A240 304	2	1340	92	1	0.1	1.9	0.0
KT9.1	Plate steel	A240 304	2	113	88	2	0.0	0.3	0.0
KT9.2	Plate steel	A240 304	2	113	23	4	0.0	0.2	0.0
KT14.1	Plate steel	A240 304	2	85	30	2	0.0	0.1	0.0
KT14.2	Plate steel	A240 304	2	85	61	2	0.0	0.2	0.0
KT15.1	Plate steel	A240 304	2	1432	30	2	0.1	1.3	0.0
KT15.2	Plate steel	A240 304	2	1432	25	4	0.1	2.2	0.0
MH1.1	Plate steel	SS400	6	1417	210	1	0.3	14.0	0.3
MH1.4	Plate steel	SS400	6	567	571	1	0.3	15.2	0.3
MH1.5a	Plate steel	SS400	10	149	178	2	0.0	4.2	0.0
MH1.5b	Plate steel	SS400	10	140	178	2	0.0	3.9	0.0
MH1.6	Plate steel	SS400	10	170	30	2	0.0	0.8	0.0
MH2.1	Plate steel	SS400	6	1258	200	1	0.3	11.9	0.3
MH2.2	Plate steel	SS400	6	511	538	1	0.3	12.9	0.3
MH2.3	Plate steel	SS400	10	130	90	1	0.0	0.9	0.0
MH2.4	Plate steel	SS400	6	42	42	1	0.0	0.1	0.0
TOTAL						353.0	32.9	3222.4	28.9

TUBE LIST

Symbol	Material Decripsion		Specifications (mm)			Q.ty	Mass (KG)	Paint area (M2)
	Type of material	Material	OD	Thickness	Length			
O1	Pipe	(S)A106 Gr. B	114.3	6.02	214	1	3.44	0.07
O2	Pipe	P235GH-TC1	60.3	3.91	146	1	0.79	0.03
O3	Pipe	(S)A106 Gr. B	26.7	2.87	170	1	0.29	0.01
O4	Pipe	(S)A106 Gr. B	26.7	2.87	170	1	0.29	0.01
O6.1	Pipe	(S)A106 Gr. B	114.3	6.02	254	1	4.08	0.09
O6.2	Pipe	(S)A106 Gr. B	114.3	6.02	224	1	3.60	0.08
O7.1	Pipe	P235GH-TC1	219.1	8.18	258	1	10.98	0.17
O7.2	Pipe	P235GH-TC1	219.1	8.18	152	1	6.47	0.10
O8	Pipe	(S)A106 Gr. B	33.4	3.38	146	1	0.37	0.01
O9.1	Pipe	P235GH-TC1	219.1	8.18	1500	1	63.82	0.99
O9.2	Pipe	P235GH-TC1	219.1	8.18	4960	1	211.04	3.29
O10.1	Pipe	(S)A106 Gr. B	114.3	6.02	1394	1	22.41	0.47
O10.2	Pipe	(S)A106 Gr. B	114.3	6.02	303	1	4.87	0.10
O10.3	Pipe	(S)A106 Gr. B	114.3	6.02	4125	1	66.31	1.40
O11	Pipe	(S)A106 Gr. B	323.9	10.31	180	1	14.35	0.18
O12.1	Pipe	(S)A106 Gr. B	33.4	3.38	415	1	1.04	0.04
O12.2	Pipe	(S)A106 Gr. B	26.7	2.87	310	1	0.52	0.02
O13	Pipe	(S)A106 Gr. B	42.2	3.56	170	1	0.58	0.02
O14.1	Pipe	(S)A106 Gr. B	88.9	5.49	344	1	3.88	0.09
O14.2	Pipe	(S)A106 Gr. B	88.9	5.49	190	1	2.15	0.05
O15	Pipe	(S)A106 Gr. B	33.4	3.38	610	1	1.53	0.06
O16.1	Pipe	(S)A106 Gr. B	33.4	3.38	972	1	2.43	0.09
O16.2	Pipe	(S)A106 Gr. B	42.2	3.56	130	1	0.44	0.02
O17.1	Pipe	(S)A106 Gr. B	26.7	2.87	249	1	0.42	0.02
O17.2	Pipe	(S)A106 Gr. B	26.7	2.87	249	1	0.42	0.02
O18.1	Pipe	A312 TP304	76.1	5.16	20	1	0.18	0.00
O18.2	Pipe	A312 TP304	76.1	5.16	65	1	0.59	0.00
O19	Pipe	A312 TP304	21.3	2.77	50	4	0.25	0.00
OKT1	Pipe	(S)A106 Gr. B	114.3	6.02	1578	1	25.37	0.54
OKT2.1	Pipe	(S)A106 Gr. B	26.7	2.87	112	1	0.19	0.01
OKT2.2	Pipe	(S)A106 Gr. B	26.7	2.87	112	1	0.19	0.01
TH1.1	Welded pipe	JIS G3131 SPHC	42.2	1.8	1970	2	7.07	0.50
TH1.2	Welded pipe	JIS G3131 SPHC	26.7	1.8	370	5	2.04	0.14
TOLTAL						41.00	462.40	8.63

BẢNG KÊ VẬT TƯ LINH KIỆN

STT	Mã VT	Loại VT	Đơn vị	Xưởng	Công trình	Tổng	Ghi chú
1	VC019267	Mặt bích Type 01.B1 ASTM A516, DN100-PN16	Cái	1	0	1	N1
2	VC000594	Roong mặt bích SW DN100 PN10/16, 316L FG, 316L IR, CS OR 4.5mmT	Cái	0	1	1	
3	VC001389	Bulong M16x70mm + tán cường lực + lông đền/35K 8.8 mạ kẽm	Bộ	0	8	8	
4	VC019279	Mặt bích Type 01.B1 ASTM A516, DN50-PN16	Cái	1	0	1	N2
5	VC000585	Roong mặt bích SW DN50 PN10/16/25/40, 316 FG, 316 IR, CS OR 4.5mmT	Cái	0	1	1	
6	VC001389	Bulong M16x70mm + tán cường lực + lông đền/35K 8.8 mạ kẽm	Bộ	0	4	4	
7	VC006409	Măng xông ren suốt A-105, G 1/2 x 30mm, Sch40	Cái	1	0	1	N3
8	VC006409	Măng xông ren suốt A-105, G 1/2 x 30mm, Sch40	Cái	1	0	1	N4
9	VC019295	Mặt bích Type 05.B1 ASTM A516, DN450-PN16	Cái	1	0	1	N5(Manhole Bồn Nước)
10	VC020000	Mặt bích Type 01.B1 ASTM A516, DN450-PN16	Cái	1	0	1	
11	VC000611	Roong mặt bích SW DN450 PN16, 316L FG, 316L IR, CS OR 4.5mmT	Cái	1	0	1	
12	VC001471	Bulong M27x130mm + tán cường lực + lông đền phe, 35K 8.8 mạ kẽm	Bộ	20	0	20	
13	VC010719	Bulong M12x50mm + tán + lông đền, 35K 8.8 mạ kẽm	Bộ	2	0	2	
14	VC019267	Mặt bích Type 01.B1 ASTM A516, DN100-PN16	Cái	2	0	2	N6
15	VC000594	Roong mặt bích SW DN100 PN10/16, 316L FG, 316L IR, CS OR 4.5mmT	Cái	0	1	1	
16	VC001389	Bulong M16x70mm + tán cường lực + lông đền/35K 8.8 mạ kẽm	Bộ	0	8	8	
17	VC019272	Mặt bích Type 01.B1 ASTM A516, DN200-PN16	Cái	2	0	2	N7
18	VC000602	Roong mặt bích SW DN200 PN16, 316 FG, 316 IR, CS OR 4.5mmT	Cái	0	1	1	
19	VC001446	Bulong M20x80mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	Bộ	0	12	12	
20	VC019273	Mặt bích Type 01.B1 ASTM A516, DN25-PN16	Cái	1	0	1	N8
21	VC000578	Roong mặt bích SW DN25 PN10/16/25/40, 316 FG, 316 IR, CS OR 4.5mmT	Cái	0	1	1	
22	VC001413	Bulong M12x60mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	Bộ	0	4	4	
23	VC019272	Mặt bích Type 01.B1 ASTM A516, DN200-PN16	Cái	1	0	1	N9
24	VC000602	Roong mặt bích SW DN200 PN16, 316 FG, 316 IR, CS OR 4.5mmT	Cái	0	1	1	
25	VC001446	Bulong M20x80mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	Bộ	0	12	12	
26	VC000077	Co đúc 90º LR A-234 WPB, DN200 Sch40 BW	Cái	1	0	1	
27	VC006377	Cùm ống chữ U + tán + long đền, Ø219 x M8, CT3	Bộ	3	0	3	
28	VC019267	Mặt bích Type 01.B1 ASTM A516, DN100-PN16	Cái	1	0	1	N10
29	VC000594	Roong mặt bích SW DN100 PN10/16, 316L FG, 316L IR, CS OR 4.5mmT	Cái	0	1	1	
30	VC001389	Bulong M16x70mm + tán cường lực + lông đền/35K 8.8 mạ kẽm	Bộ	0	8	8	
31	VC005918	Co đúc 90º LR A-234 WPB, DN100 Sch40 BW	Cái	1	0	1	
32	VC000110	Co đúc 45º LR A-234 WPB, DN100 Sch40 BW	Cái	1	0	1	
33	VC006376	Cùm ống chữ U + tán + long đền, Ø114 x M8, CT3	Bộ	3	0	3	
34	VC019275	Mặt bích Type 01.B1 ASTM A516, DN300-PN16	Cái	1	0	1	N11
35	VC005623	Roong mặt bích SW DN300 PN16, 304L FG, 304L IR, CS OR 4.5mmT	Cái	0	1	1	
36	VC001461	Bulong M24x100mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	Bộ	0	12	12	
37	VC019273	Mặt bích Type 01.B1 ASTM A516, DN25-PN16	Cái	1	0	1	N12
38	VC000578	Roong mặt bích SW DN25 PN10/16/25/40, 316 FG, 316 IR, CS OR 4.5mmT	Cái	0	1	1	
39	VC001413	Bulong M12x60mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	Bộ	0	4	4	
40	VC019276	Mặt bích Type 01.B1 A516, DN32-PN16	Cái	1	0	1	N13
41	VC005343	Roong mặt bích SW DN32 PN10/16/25/40, 316L FG, 316L IR, CS OR 4.5mmT	Cái	0	1	1	
42	VC001408	Bulong M16x60mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	Bộ	0	4	4	
43	VC019281	Mặt bích Type 01.B1 ASTM A516, DN80-PN16	Cái	1	0	1	N14
44	VC005349	Roong mặt bích SW DN80 PN10/16/25/40, 316L FG, 316L IR, CS OR 4.5mmT	Cái	0	1	1	
45	VC001389	Bulong M16x70mm + tán cường lực + lông đền/35K 8.8 mạ kẽm	Bộ	0	8	8	
46	VC000380	Bầu giảm đức đồng tâm A-234 WPB, DN80 x 65 Sch40 BW	Cái	1	0	1	
47	VC019273	Mặt bích Type 01.B1 ASTM A516, DN25-PN16	Cái	1	0	1	N15
48	VC000578	Roong mặt bích SW DN25 PN10/16/25/40, 316 FG, 316 IR, CS OR 4.5mmT	Cái	0	1	1	
49	VC001413	Bulong M12x60mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	Bộ	0	4	4	
50	VC000274	Mặt bích Type 01 CT3/SS400, DN400 PN16	Cái	1	0	1	N16(Manhole BKK)
51	VC005492	Mặt bích Type 05 CT3/SS400, DN400 PN16	Cái	1	0	1	
52	VC000609	Roong mặt bích SW DN400 PN16, 316 FG, 316 IR, CS OR 4.5mmT	Cái	1	0	1	
53	VC001470	Bulong M27x110mm + tán cường lực + lông đền phe, 35K 8.8 mạ kẽm	Bộ	16	0	16	
54	VC010727	Tán M14 + lông đền, SC20 4.6 xi	Bộ	2	0	2	
55	VC019285	Mặt bích Type 01.B1 A516, DN20-PN16	Cái	2	0	2	N17
56	VC005584	Roong mặt bích SW DN20 PN10/16/25/40, 316L FG, 316L IR, CS OR 4.5mmT	Cái	0	2	2	
57	VC001413	Bulong M12x60mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	Bộ	0	8	8	
58	VC001388	Bulong M16x50mm + tán cường lực + lông đền/35K 8.8 mạ kẽm	Bộ	0	10	10	Lắp chân BKK+Bồn nước
59	VC001342	Bulong inox M12x40mm + tán + lông đền, SUS304	Bộ	2	0	2	Bộ SPRAY
60	VC014144	Tán M12, SUS304	Con	2	0	2	
61	VC003540	Cao su tấm, dày 3mm	m2	0.1	0	0.1	
62	VC015730	Lò xo nén Ø3xØ30, p= 9.5mm, L= 40mm, SU304	Cái	1	0	1	
63	VC013548	Tán M14, SUS304	Con	12	0	12	Bộ treo TRAY
64	VC002968	Van cầu lắp bích DN25-PN16, KSB	Cái	1	0	1	Cụm lắp PG, PT
65	VC019289	Mặt bích Type 05.B1 ASTM A516, DN25-PN16	Cái	1	0	1	
66	VC000578	Roong mặt bích SW DN25 PN10/16/25/40, 316 FG, 316 IR, CS OR 4.5mmT	Cái	1	0	1	
67	VC001413	Bulong M12x60mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	Bộ	4	0	4	
68	VC009371	Măng xông ren suốt SUS 304, G 1/2 x 30mm, Sch40	Cái	4	0	4	
69	VC000492	Tê đúc A-234 WPB, DN20 Sch40 BW	Cái	1	0	1	
70	VC006464	Siphons SUS 304, 1/2 NPT	Cái	1	0	1	
71	VC006299	Cà rá đồng, 1/2 NPT x 1/4 NPT	Cái	1	0	1	
72	VC000787	Teplong cuộn	Cuộn	2	2	4	
73	VC001406	Bulong M10x50mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	Bộ	24	0	24	KHUNG SP STT
74	VC015735	Bảng tên inox, nền inox, chữ đen, 0.5x150x100mm	Cái	1	0	1	BẢNG TÊN
75	VC010552	Vít đầu dù đuôi cá M4.2x12mm	Bộ	4	0	4	
76	VC019267	Mặt bích Type 01.B1 ASTM A516, DN100-PN16	Cái	1	0	1	Ống thủy tối
77	VC019283	Mặt bích Type 05.B1 ASTM A516, DN100-PN16	Cái	1	0	1	
78	VC000594	Roong mặt bích SW DN100 PN10/16, 316L FG, 316L IR, CS OR 4.5mmT	Cái	1	0	1	
79	VC001389	Bulong M16x70mm + tán cường lực + lông đền/35K 8.8 mạ kẽm	Bộ	8	0	8	
80	VC019285	Mặt bích Type 01.B1 A516, DN20-PN16	Cái	2	0	2	
81	VC005584	Roong mặt bích SW DN20 PN10/16/25/40, 316L FG, 316L IR, CS OR 4.5mmT	Cái	0	2	2	
82	VC001413	Bulong M12x60mm + tán cường lực + lông đền, 35K 8.8 mạ kẽm	Bộ	0	8	8	
83	VC006408	Măng xông ren suốt A-105, M18x25	Cái	4	0	4	
84	VC006409	Măng xông ren suốt A-105, G 1/2 x 30mm, Sch40	Cái	2	0	2	
85	VC003051	Van kính thủy sáng lắp ren Ø16	Bộ	2	0	2	
86	VC002811	Kính thủy bồn nước Ø16	Bộ	1	0	1	
87	VC009342	Vít đầu dù đuôi cá M4.2x12mm	Con	20	0	20	
88	VC004182	Bulong M6x20mm + tán + long đền, 35K 8.8 mạ kẽm	Bộ	4	0	4	
89	VC000382	Bầu giảm đức đồng tâm A-234 WPB, DN100 x 25 Sch40 BW	Cái	1	0	1	

SHAPE STEEL LIST

Symbol	Material Decripsion			Q.ty	Mass (kg)	Paint area (M2)
	Type of material	Material	Length (mm)			
P11.2	Thép hình H125x125x6.5x9x12000mm/SS400	SS400	710	4	67.6	2.1
V1.1	Thép hình V63x63x6x6000mm CT38/SS400/AGS 400	SS400	526	3	9.0	0.4
V1.2	Thép hình V63x63x6x6000mm CT38/SS400/AGS 400	SS400	195	6	6.7	0.3
V2.1	Thép hình V50x50x5x6000 mm, SUS304	SUS304	1059	4	19.06	-
V2.2	Thép hình V50x50x5x6000 mm, SUS304	SUS304	650	4	11.70	-
V2.3	Thép hình V50x50x5x6000 mm, SUS304	SUS304	266	2	2.39	-
V2.4	Thép hình V30x30x3x6000mm, SUS304	SUS304	1150	2	3.15	-
V3.1	Thép hình V50x50x5x6000 mm, SUS304	SUS304	644	2	5.80	-
V3.2	Thép hình V50x50x5x6000 mm, SUS304	SUS304	644	2	5.80	-
PP2	Ty ren M12	SUS304	110	1	0.10	-
T1	Ty ren M14	SUS304	404	4	1.99	-
L1	Láp JIS G4051 S45C Ø16x6000mm	S45C	260	1	0.4	0.0
L2.1	Láp JIS G4051 S45C Ø16x6000mm	S45C	260	1	0.4	0.0
L2.2	Láp S45C Ø14x6000mm	S45C	250	1	0.4	0.0
L2.3	Láp S45C Ø14x6000mm	S45C	350	1	0.6	0.0
L3.1	Láp ASTM A276 304/304L, Ø10x6000mm	SUS304	250	2	0.32	-
L3.2	Láp ASTM A276 304/304L, Ø10x6000mm	SUS304	200	1	0.13	-
L4	Láp SUS304 Ø14x6000mm	SUS304	25	8	0.25	-
TOLTAL				49.00	135.70	2.90

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:

MARTECH BOILER® The energy Solutions

MARTECH JOINT STOCK COMPANY

PROJECT JOB CODE: TP2887

TITLE: DETAIL DRAWING

DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.34

STT	Loai vat lieu	O19			
		4	Ldinhhinh	Lduthua	Lsudung
		50	(mm)	(mm)	(mm)
1	Ống đúc ASTM A312 TP304, Ø21.3x2.77x6000mm	4	6000	5800	200
Tong		4	6000	5800	200

So luong thanh vat lieu Ống đúc ASTM A312 TP304, Ø21.3x2.77x6000mm su dung : 1 (thanh)
% khong su dung : 96.67%

STT	Loai vat lieu	O3	O4	O12.2	O17.1	O17.2	OKT2.1	OKT2.2			
		1	1	1	1	1	1	1	Ldinhhinh	Lduthua	Lsudung
		170	170	310	249	249	112	112	(mm)	(mm)	(mm)
1	Ống đúc ASTM A106 Gr.B, Ø26.7x2.87x6000 mm	1	1	1	1	1	1	1	6000	4628	1372
Tong		1	1	1	1	1	1	1	6000	4628	1372

So luong thanh vat lieu Ống đúc ASTM A106 Gr.B, Ø26.7x2.87x6000 mm su dung : 1 (thanh)
% khong su dung : 77.13%

STT	Loai vat lieu	TH1.2			
		5	Ldinhhinh	Lduthua	Lsudung
		370	(mm)	(mm)	(mm)
1	Ống hàn JIS G3131 SPHC, Ø26.7x1.8x6000mm	5	6000	4150	1850
Tong		5	6000	4150	1850

So luong thanh vat lieu Ống hàn JIS G3131 SPHC, Ø26.7x1.8x6000mm su dung : 1 (thanh)
% khong su dung : 69.17%

STT	Loai vat lieu	O8	O12.1	O15	O16.1			
		1	1	1	1	Ldinhhinh	Lduthua	Lsudung
		146	415	610	972	(mm)	(mm)	(mm)
1	Ống đúc ASME SA-106 Gr.B, Ø33.4x3.38x11800mm	1	1	1	1	11800	9657	2143
Tong		1	1	1	1	11800	9657	2143

So luong thanh vat lieu Ống đúc ASME SA-106 Gr.B, Ø33.4x3.38x11800mm su dung : 1 (thanh)
% khong su dung : 81.85%

STT	Loai vat lieu	O13	O16.2			
		1	1	Ldinhhinh	Lduthua	Lsudung
		170	130	(mm)	(mm)	(mm)
1	Ống đúc ASTM A106 Gr.B, Ø42.2x3.56x6000 mm	1	1	6000	5700	300
Tong		1	1	6000	5700	300

So luong thanh vat lieu Ống đúc ASTM A106 Gr.B, Ø42.2x3.56x6000 mm su dung : 1 (thanh)
% khong su dung : 95.02%

STT	Loai vat lieu	TH1.1			
		2	Ldinhhinh	Lduthua	Lsudung
		1970	(mm)	(mm)	(mm)
1	Ống hàn JIS G3131 SPHC, Ø42.2x1.8x6000mm	2	6000	2060	3940
Tong		2	6000	2060	3940

So luong thanh vat lieu Ống hàn JIS G3131 SPHC, Ø42.2x1.8x6000mm su dung : 1 (thanh)
% khong su dung : 34%

STT	Loai vat lieu	O2			
		1	Ldinhhinh	Lduthua	Lsudung
		146	(mm)	(mm)	(mm)
1	Ống đúc DIN EN 10216-2 P235GH-TC1, Ø60.3x3.91x11800mm	1	11800	11654	146
Tong		1	11800	11654	146

So luong thanh vat lieu Ống đúc DIN EN 10216-2 P235GH-TC1, Ø60.3x3.91x11800mm su dung : 1 (thanh)
% khong su dung : 98.76%

STT	Loai vat lieu	O18.1	O18.2			
		1	1	Ldinhhinh	Lduthua	Lsudung
		20	65	(mm)	(mm)	(mm)
1	Ống đúc ASTM A312 TP304, Ø76.1x5.16x6000mm	1	1	6000	5915	85
Tong		1	1	6000	5915	85

So luong thanh vat lieu Ống đúc ASTM A312 TP304, Ø76.1x5.16x6000mm su dung : 1 (thanh)
% khong su dung : 98.58%

STT	Loai vat lieu	O14.1	O14.2			
		1	1	Ldinhhinh	Lduthua	Lsudung
		344	190	(mm)	(mm)	(mm)
1	Ống đúc ASTM A106 Gr.B, Ø88.9x5.49x6000 mm	1	1	6000	5466	534
Tong		1	1	6000	5466	534

So luong thanh vat lieu Ống đúc ASTM A106 Gr.B, Ø88.9x5.49x6000 mm su dung : 1 (thanh)
% khong su dung : 91.1%

STT	Loai vat lieu	O1	O6.1	O6.2	O10.1	O10.2	O10.3	OKT1			
		1	1	1	1	1	1	1	Ldinhhinh	Lduthua	Lsudung
		214	254	224	1394	303	4125	1578	(mm)	(mm)	(mm)
1	Ống đúc ASTM A106 Gr.B, Ø114.3x6.02x6000mm		1				1	1	6000	43	5957
2	Ống đúc ASTM A106 Gr.B, Ø114.3x6.02x6000mm	1		1	1	1			6000	3865	2135
Tong		1	1	1	1	1	1	1	12000	3908	8092

So luong thanh vat lieu Ống đúc ASTM A106 Gr.B, Ø114.3x6.02x6000mm su dung : 2 (thanh)
% khong su dung : 32.57%

STT	Loai vat lieu	O7.1	O7.2	O9.1	O9.2			
		1	1	1	1	Ldinhhinh	Lduthua	Lsudung
		258	152	1500	4960	(mm)	(mm)	(mm)
1	Ống đúc DIN EN 10216-2 P235GH-TC1, Ø219.1x8.18x11800mm	1	1	1	1	11800	4930	6870
Tong		1	1	1	1	11800	4930	6870

So luong thanh vat lieu Ống đúc DIN EN 10216-2 P235GH-TC1, Ø219.1x8.18x11800mm su dung : 1 (thanh)
% khong su dung : 41.78%

STT	Loai vat lieu	L3.1	L3.2			
		2	1	Ldinhhinh	Lduthua	Lsudung
		250	200	(mm)	(mm)	(mm)
1	Láp ASTM A276 304/304L, Ø10x6000mm	2	1	6000	5300	700
Tong		2	1	6000	5300	700

So luong thanh vat lieu Láp ASTM A276 304/304L, Ø10x6000mm su dung : 1 (thanh)
% khong su dung : 88.33%

STT	Loai vat lieu	L1	L2.1			
		1	1	Ldinhhinh	Lduthua	Lsudung
		260	260	(mm)	(mm)	(mm)
1	Láp JIS G4051 S45C Ø16x6000mm	1	1	6000	5480	520
Tong		1	1	6000	5480	520

So luong thanh vat lieu Láp JIS G4051 S45C Ø16x6000mm su dung : 1 (thanh)
% khong su dung : 91.33%

STT	Loai vat lieu	L4			
		8	Ldinhhinh	Lduthua	Lsudung
		25	(mm)	(mm)	(mm)
1	Láp SUS304 Ø14x6000mm	8	6000	5800	200
Tong		8	6000	5800	200

So luong thanh vat lieu Láp SUS304 Ø14x6000mm su dung : 1 (thanh)
% khong su dung : 96.67%

STT	Loai vat lieu	PP2			
		1	Ldinhhinh	Lduthua	Lsudung
		110	(mm)	(mm)	(mm)
1	Ty ren M12, SUS304	1	6000	5890	110
Tong		1	6000	5890	110

So luong thanh vat lieu Ty ren M12, SUS304 su dung : 1 (thanh)
% khong su dung : 98.17%

STT	Loai vat lieu	T1			
		4	Ldinhhinh	Lduthua	Lsudung
		404	(mm)	(mm)	(mm)
1	Ty ren M14, SUS304	4	6000	4384	1616
Tong		4	6000	4384	1616

So luong thanh vat lieu Ty ren M14, SUS304 su dung : 1 (thanh)
% khong su dung : 74%

STT	Loai vat lieu	P11.2			
		4	Ldinhhinh	Lduthua	Lsudung
		710	(mm)	(mm)	(mm)
1	Thép hình H125x125x6.5x9x12000mm/SS400	4	6000	3160	2840
Tong		4	6000	3160	2840

So luong thanh vat lieu Thép hình H125x125x6.5x9x12000mm/SS400 su dung : 1 (thanh)
% khong su dung : 52.67%

STT	Loai vat lieu	V2.4			
		2	Ldinhhinh	Lduthua	Lsudung
		1150	(mm)	(mm)	(mm)
1	Thép hình V30x30x3x6000mm, SUS304	2	6000	3700	2300
Tong		2	6000	3700	2300

So luong thanh vat lieu Thép hình V30x30x3x6000mm, SUS304 su dung : 1 (thanh)
% khong su dung : 61.67%

STT	Loai vat lieu	V2.1	V2.2	V2.3	V3.1	V3.2			
		4	4	2	2	2	Ldinhhinh	Lduthua	Lsudung
		1059	650	266	644	644	(mm)	(mm)	(mm)
1	Thép hình V50x50x5x6000 mm, SUS304	4	2	1			6000	198	5802
2	Thép hình V50x50x5x6000 mm, SUS304		2	1	2	2	6000	1858	4142
Tong		4	4	2	2	2	12000	2056	9944

So luong thanh vat lieu Thép hình V50x50x5x6000 mm, SUS304 su dung : 2 (thanh)
% khong su dung : 17.13%

STT	Loai vat lieu	V1.1	V1.2			
		3	6	Ldinhhinh	Lduthua	Lsudung
		526	195	(mm)	(mm)	(mm)
1	Thép hình V63x63x6x6000mm CT38/SS400/AGS 40	3	6	6000	3252	2748
Tong		3	6	6000	3252	2748

So luong thanh vat lieu Thép hình V63x63x6x6000mm CT38/SS400/AGS 400 su dung : 1 (thanh)
% khong su dung : 54.2%

STT	Loai vat lieu	O11			
		1	Ldinhhinh	Lduthua	Lsudung
		180	(mm)	(mm)	(mm)
1	Ống đúc ASTM A106 Gr.B, Ø323.9x10.31x6000mm	1	6000	5820	180
Tong		1	6000	5820	180

So luong thanh vat lieu Ống đúc ASTM A106 Gr.B, Ø323.9x10.31x6000mm su dung : 1 (thanh)
% khong su dung : 97%

STT	Loai vat lieu	L2.2	L2.3			
		1	1	Ldinhhinh	Lduthua	Lsudung
		250	350	(mm)	(mm)	(mm)
1	Láp S45C Ø14x6000mm	1	1	6000	5400	600
Tong		1	1	6000	5400	600

So luong thanh vat lieu Láp S45C Ø14x6000mm su dung : 1 (thanh)
% khong su dung : 90%

MÔ TẢ	Đ/v TÍNH	SỐ LƯỢNG	GHI CHÚ
KHỐI LƯỢNG	Kg	600	Bao gồm thép ống và thép hình
DIỆN TÍCH SƠN LÓT	m2	12	Bao gồm thép ống và thép hình

MÔ TẢ	BẮN BI	SƠN LÓT	GHI CHÚ
THỰC HIỆN	YES	YES	Trừ Inox

MÔ TẢ	MÁY	BÚT XÓA	GHI CHÚ
SỐ LƯỢNG MARKING	90	NO	.....

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

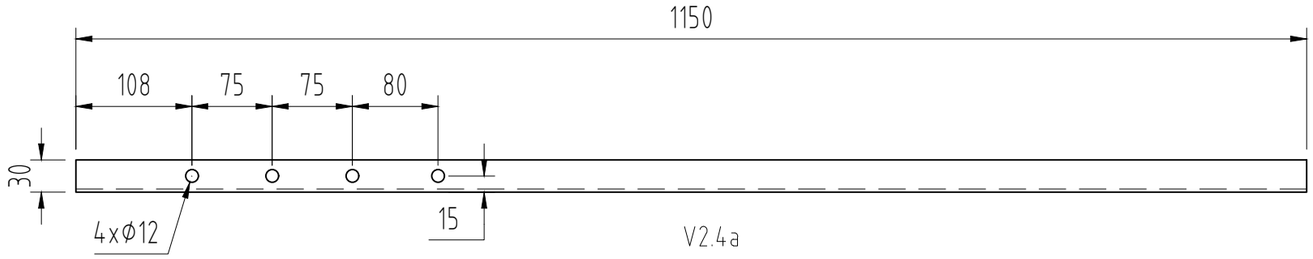
MANUFACTURER:
MARTECH BOILER The Energy Solutions
MARTECH JOINT STOCK COMPANY

PROJECT JOB CODE: TP2887

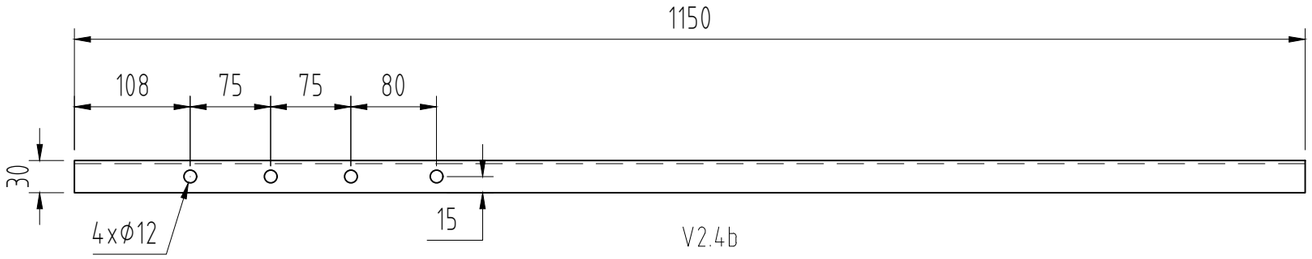
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

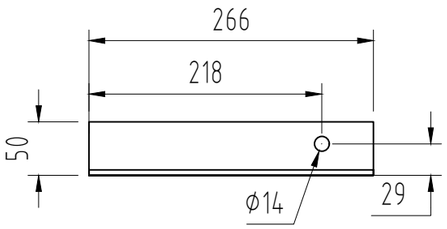
DRAWING NO.: 2887.0100.MDWG.35



V2.4a
SUS304 V30x30x3-1150mm
Q'ty:01

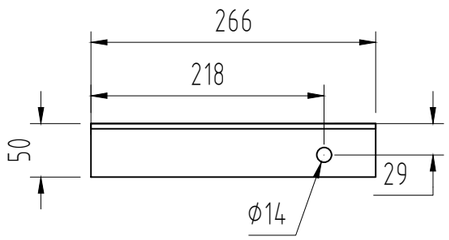


V2.4b
SUS304 V30x30x3-1150mm
Q'ty:01

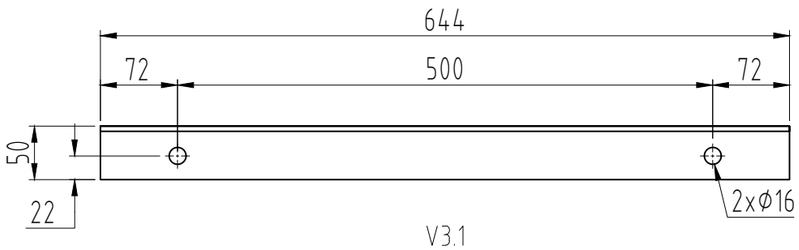


V2.3a
SUS304 V50x50x5-266mm
Q'ty:01

V2.1
SUS304 V50x50x5-1059mm
Q'ty:04



V2.3b
SUS304 V50x50x5-266mm
Q'ty:01



V3.1
SUS304-V50x50x5-644mm
Q'ty:2

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

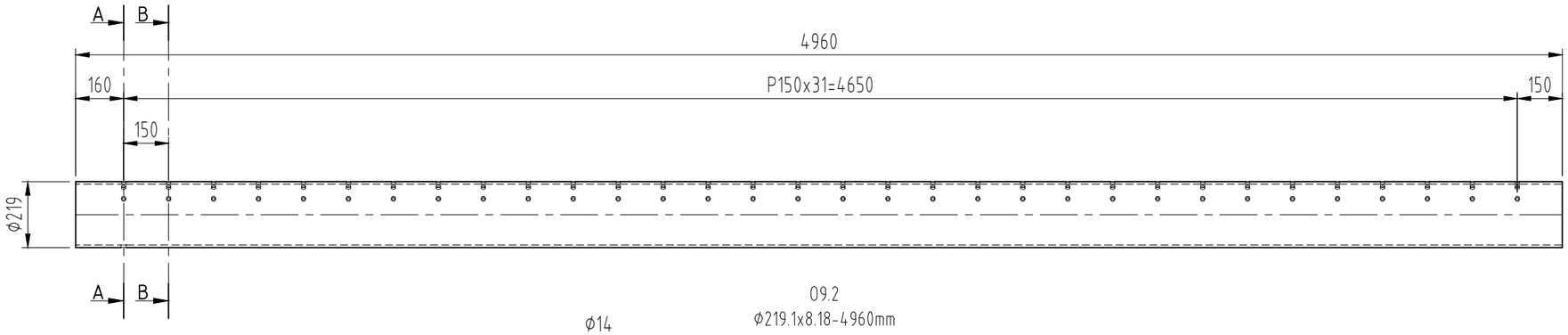
MANUFACTURER:
MARTECH BOILER The energy Solutions
MARTECH JOINT STOCK COMPANY

PROJECT JOB CODE:
TP2887

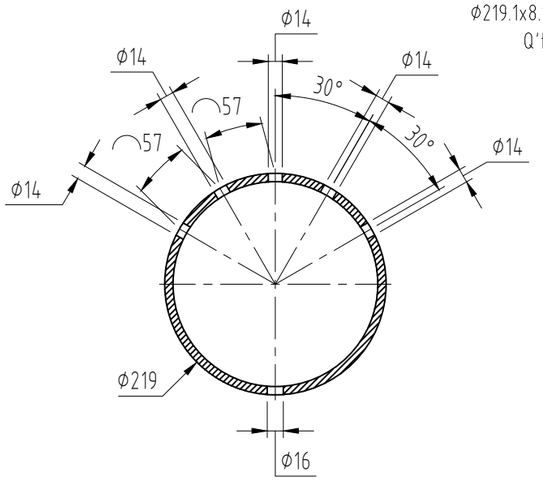
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

DRAWING NO.:
2887.0100.MDWG.36

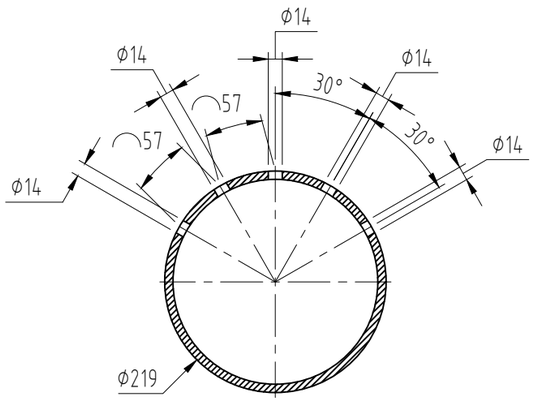


09.2
Φ219.1x8.18-4960mm
Q'ty:01



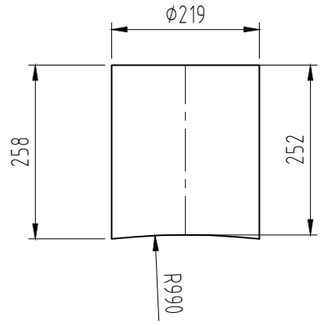
SECTION A-A
SC:3/1

Khoan 160 lỗ Ø14mm,
1 lỗ Ø16 xả cặn

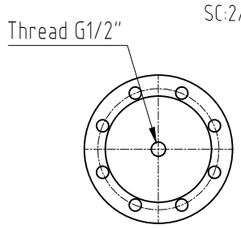


SECTION B-B
SC:3/1

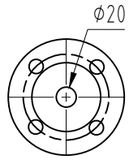
BẢN VẼ GIA CÔNG



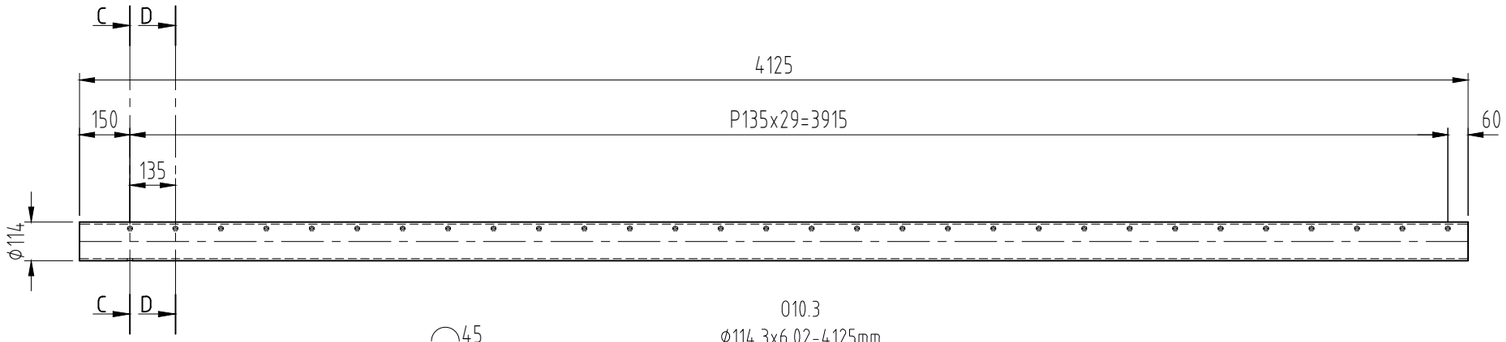
07.1
Φ219.1x8.18-258mm
SC:2/1-Q'ty:01



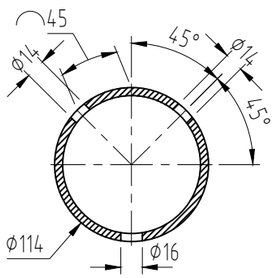
Flange type05, DN100-PN16
SC:2/1-Q'ty:01



Flange type05, DN25-PN16
SC: 3/1-Q'ty:01

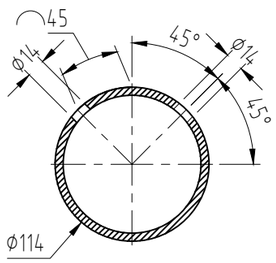


010.3
Φ114.3x6.02-4125mm
Q'ty:01



SECTION C-C
SC:4/1

Khoan 60 lỗ Ø14mm,
1 lỗ Ø16 xả cặn



SECTION D-D
SC:4/1

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

MANUFACTURER:
MARTECH BOILER
The energy Solutions
MARTECH JOINT STOCK COMPANY

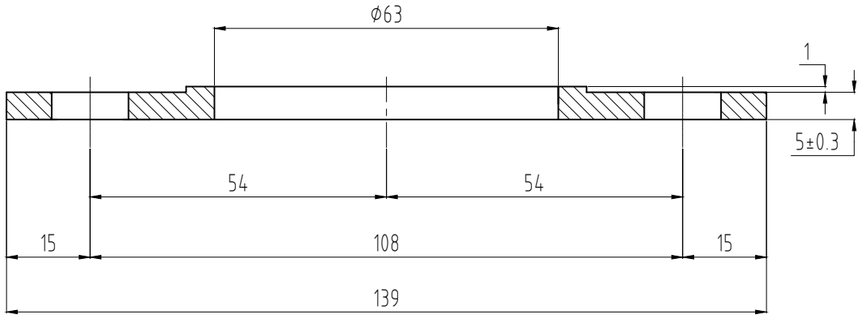
PROJECT JOB CODE: TP2887

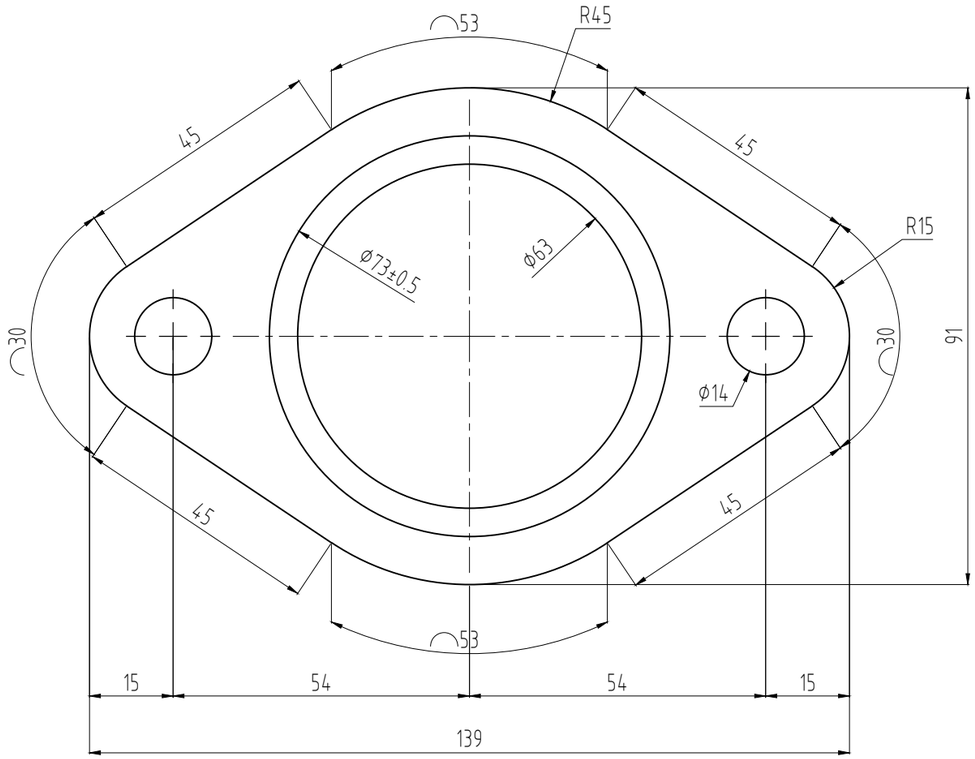
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

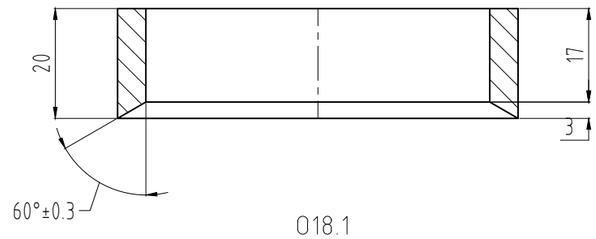
DRAWING NO.: 2887.0100.MDWG.37

BẢN VẼ GIA CÔNG

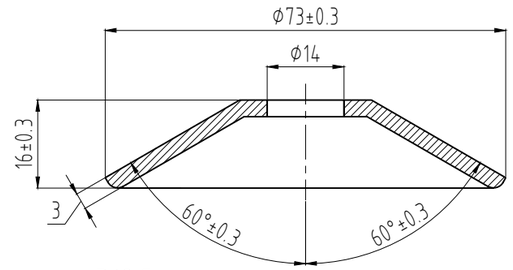




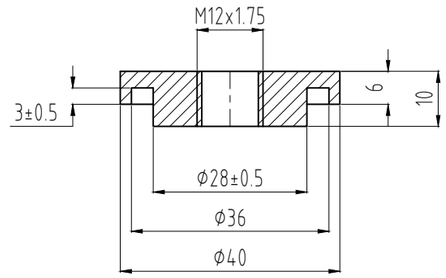
P18.1
SUS304-6mm
Q'ty:2



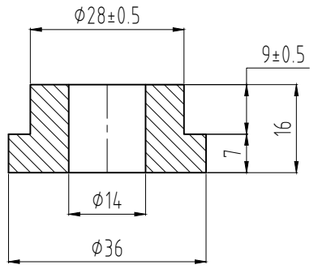
O18.1
A312 TP304,Ø76x5.16mm
SL:01



P18.5
SUS304-16mm
Q'ty:01



P18.3
SUS304-16mm
Q'ty:01



P18.4
SUS304-16mm
Q'ty:01

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

CLIENT:

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MARTECH JOINT STOCK COMPANY

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DEAERATOR AND WATER TANK

Scale	Size	Rev.
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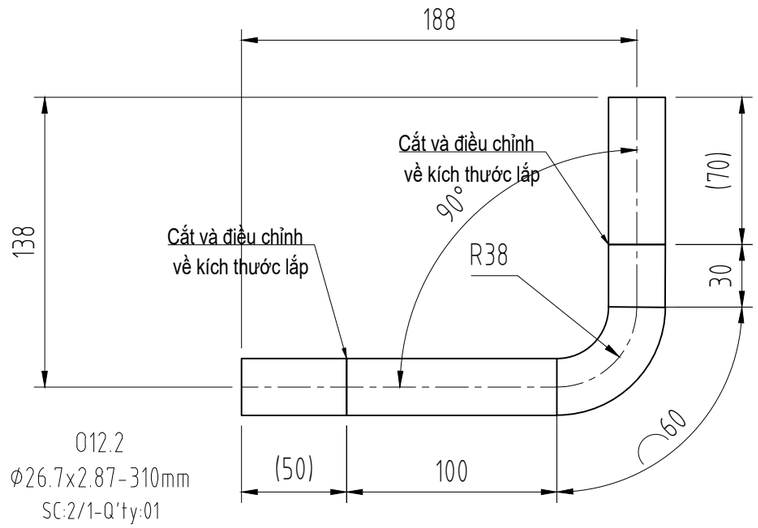
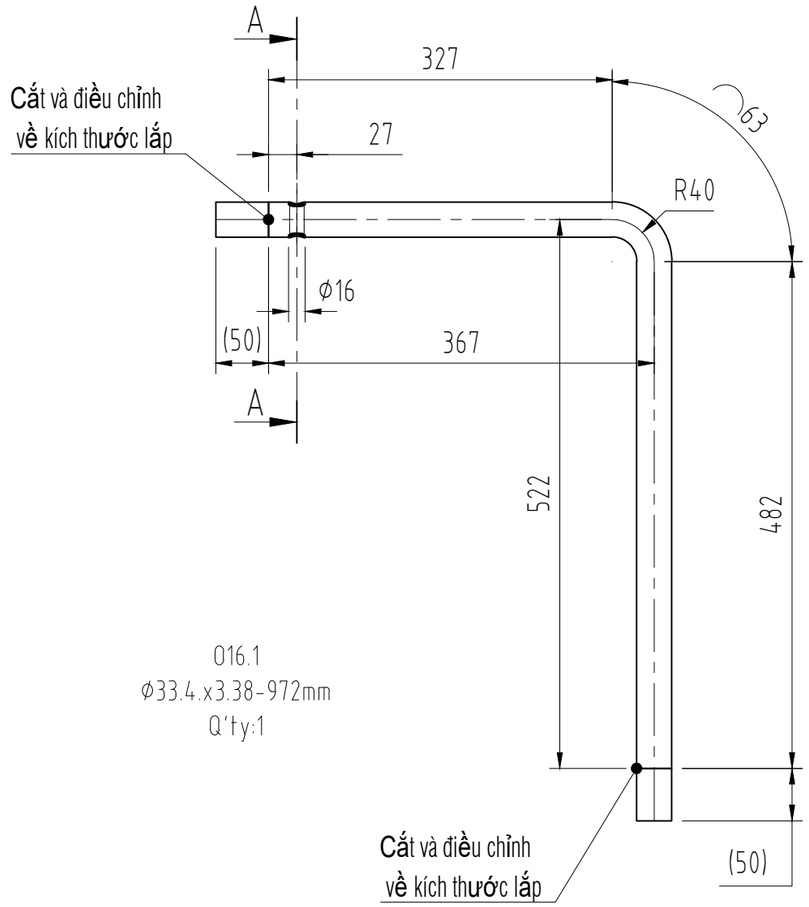
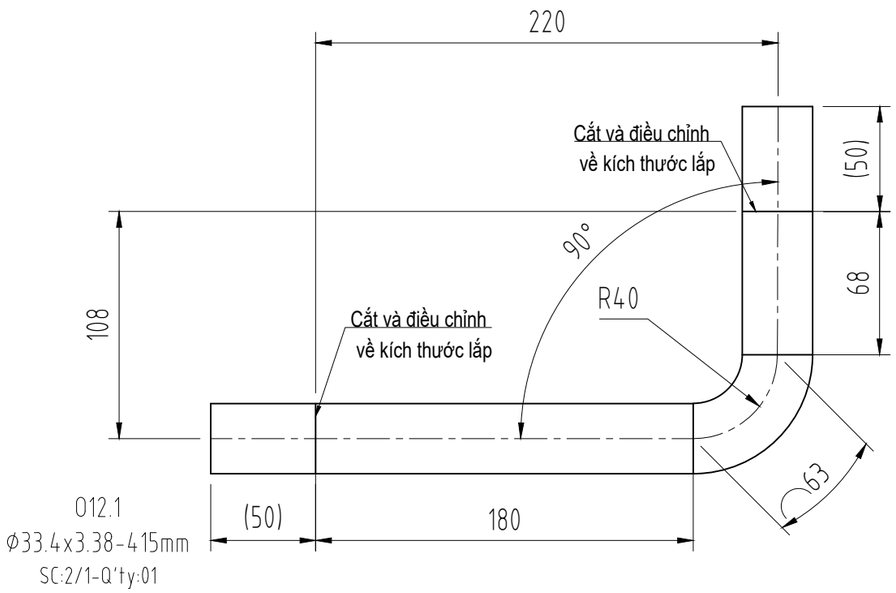
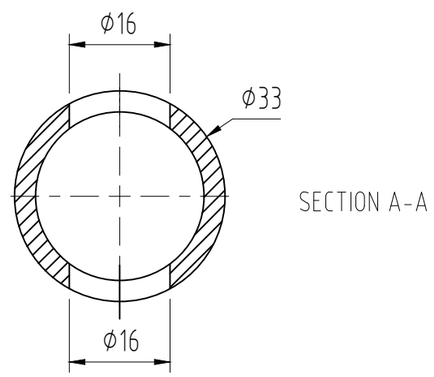
DRAWING NO.: 2887.0100.MDWG.38

BẢN VẼ GIA CÔNG









Rev.	Date	Description	Drawn	Checked	Approved
3					
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0	30.01.26	For approval	N.M.Vuong	D.M.Tuan	L.H.Phuong

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MANUFACTURER: MARTECH BOILER The energy Solutions — MARTECH JOINT STOCK COMPANY

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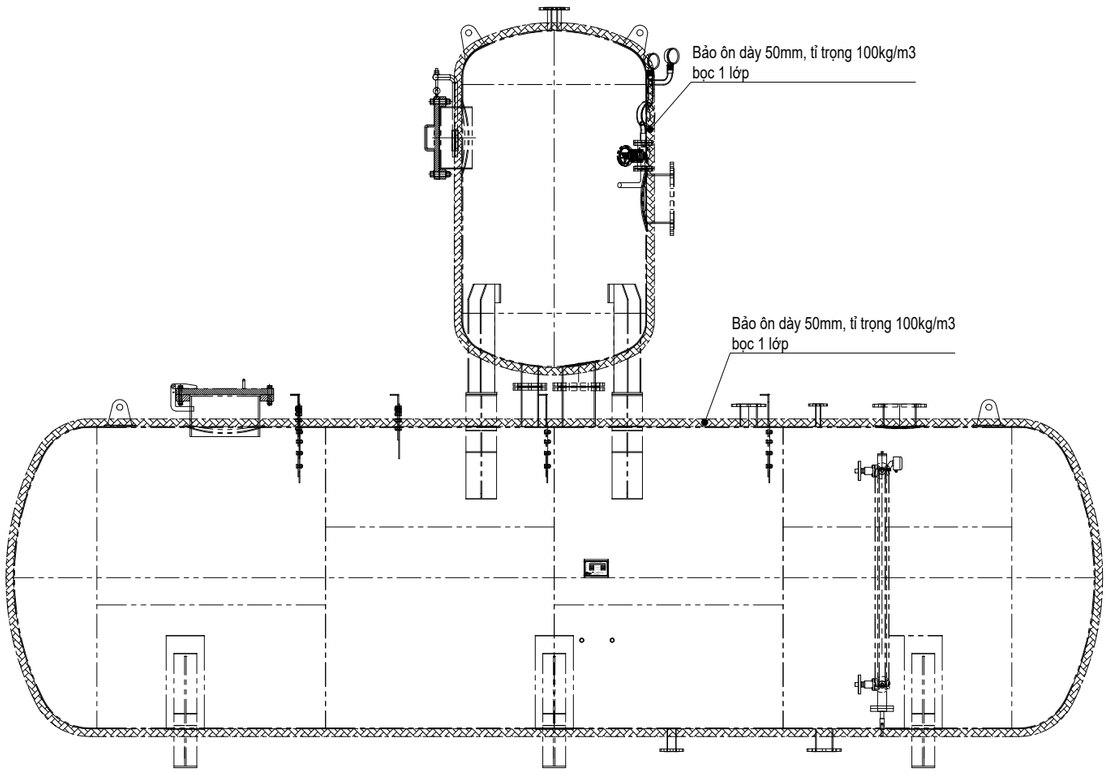
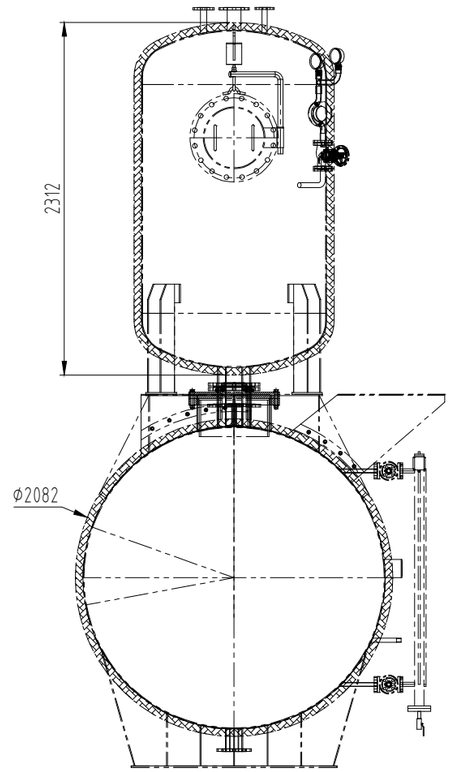
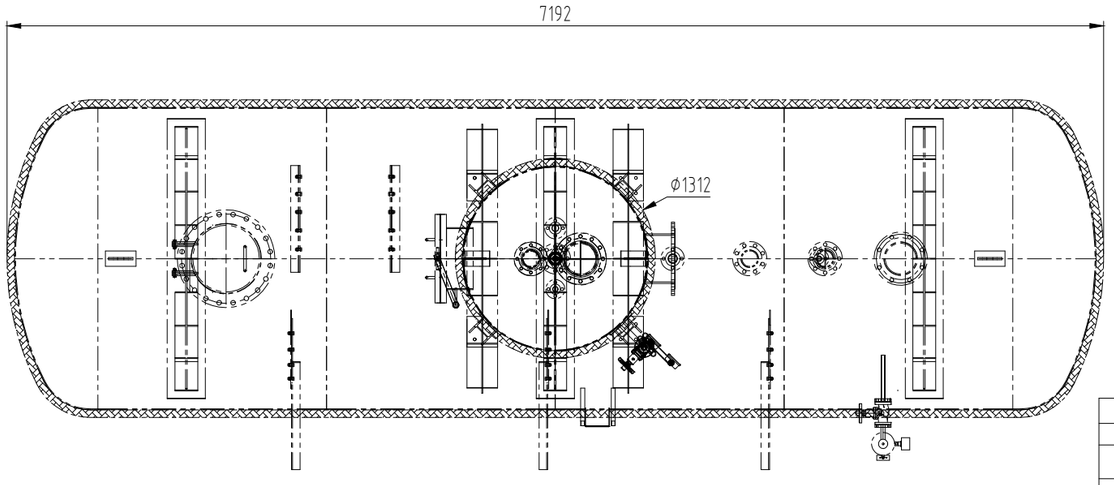
TITLE: DETAIL DRAWING
DEAERATOR AND WATER TANK

Scale	Size	Rev.
1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.39







YÊU CẦU KỸ THUẬT

- Bảo ôn dày 50mm(bọc 1 lớp), tỷ trọng 100kg/m^3 cho thân, chõm bồn nước và bình khử khí.
- Bảo ôn phải đúng tỉ trọng, đúng số lớp, siết chặc không có khe hở và phải được buộc chặc (khe hở >7mm được xem là lớn, phải nhét bảo ôn vụn vào).
- Bảo ôn thi công còn phải nguyên nhãn mác, bao bì.
- Làm cổ áo tại các nozzle, tai cẩu, part chờ thang thao tác.
- Hai tấm tole vỏ liền kề chồng mí nhỏ nhất 50mm, tại chỏm cầu chồng mí 100mm.

Diện tích bọc bảo ôn:

- Bồn nước: 49.5 m2.
- Bộ khử khí: 10.5 m2.

THÔNG TIN BẢO ÔN

STT	VẬT TƯ	ĐVT	SỐ LƯỢNG	KHỐI LƯỢNG (kg)
1	Bảo ôn vàng 6 tấm 600x1200mm, dày 50mm, tỉ trọng 100kg/m3	Kiện	15	324.0
2	Tôn vỏ	m	52	244.9
3	Vít đầu dù đuôi cá M4.2x12mm	Con	1800	
4	Kẽm buộc	Kg	10	10
Tổng				

Rev.	Date	Description	Drawn	Checked	Approved
3					
2					
1					
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1:1	A3	00

DRAWING NO.: 2887.0100.MDWG.40